



# SE46-LOST AND Found Management System

Topics in Computer Science II (COMSATS University Islamabad)



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# **Lost and Found Management System**

**Final Year Project**

**Session 2017-2021**

**A 4th Year Student**

A project submitted in partial fulfilment of the  
COMSATS University Degree  
of  
BS Software Engineering (CUI)



Department of Computer Science  
COMSATS University Islamabad, Lahore Campus

16 June 2021

## Project Detail

|                              |                                                                                                                                      |                |                              |                        |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------|------------------------|
| Type (Nature of project)     | [ <input checked="" type="checkbox"/> ] Development      [ <input type="checkbox"/> ] Research      [ <input type="checkbox"/> ] R&D |                |                              |                        |
| Area of specialization       | Web technologies and android                                                                                                         |                |                              |                        |
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\*The candidates confirm that the work submitted is their own and appropriate credit has been given where reference has been made to work of others

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## Table of Contents

|        |                                       |    |
|--------|---------------------------------------|----|
| 1      | Abstract.....                         | 13 |
| 2      | CHAPTER 01 .....                      | 14 |
| 2.1    | INTRODUCTION .....                    | 14 |
| 2.2    | OBJECTIVES .....                      | 14 |
| 2.3    | PROBLEM STATEMENT .....               | 15 |
| 2.4    | ASSUMPTION AND LIMITATION .....       | 15 |
| 2.5    | PROJECT SCOPE .....                   | 15 |
| 3      | CHAPTER 02 REQUIREMENT ANALYSIS.....  | 16 |
| 3.1    | LITERATURE REVIEW .....               | 16 |
| 3.2    | STACKHOLDER’S LIST .....              | 16 |
| 3.3    | REQUIREMENT ELICITATION.....          | 17 |
| 3.3.1  | FUNCTIONAL REQUIREMENTS .....         | 17 |
| 3.3.2  | NON-FUNCTIONAL REQUIREMENTS .....     | 21 |
| 3.3.3  | REQUIREMENT TRACEABILITY MATRIC:..... | 24 |
| 3.4    | USE CASE DESCRIPTION .....            | 26 |
| 3.4.1  | UC:001.....                           | 26 |
| 3.4.2  | UC:002.....                           | 27 |
| 3.4.3  | UC:003.....                           | 28 |
| 3.4.4  | UC:004.....                           | 29 |
| 3.4.5  | UC:005.....                           | 30 |
| 3.4.6  | UC:006.....                           | 31 |
| 3.4.7  | UC:007.....                           | 32 |
| 3.4.8  | UC:008.....                           | 33 |
| 3.4.9  | UC:009.....                           | 34 |
| 3.4.10 | UC:010.....                           | 35 |
| 3.4.11 | UC:011 .....                          | 36 |

|        |                                             |    |
|--------|---------------------------------------------|----|
| 3.4.12 | UC:012.....                                 | 37 |
| 3.4.13 | UC:013.....                                 | 38 |
| 3.4.14 | UC:014.....                                 | 39 |
| 3.4.15 | UC:015.....                                 | 40 |
| 3.4.16 | UC:016.....                                 | 41 |
| 3.4.17 | UC:017.....                                 | 42 |
| 3.5    | USE CASE DIAGRAM .....                      | 43 |
| 3.6    | SOFTWARE DEVELOPMENT LIFECYCLE (SDLC) ..... | 47 |
| 4      | CHAPTER 03 SYSTEM DESIGN.....               | 48 |
| 4.1    | WORK BREAKDOWN STRUCTURE:.....              | 48 |
| 4.2    | ACTIVITY DIAGRAM .....                      | 49 |
| 4.2.1  | CREATE ACCOUNT.....                         | 49 |
| 4.2.2  | LOGIN.....                                  | 50 |
| 4.2.3  | REPORT LOST ITEM .....                      | 51 |
| 4.2.4  | REPORT FOUND ITEM .....                     | 52 |
| 4.2.5  | VIEW FOUND ITEMS .....                      | 53 |
| 4.2.6  | VIEW LOST ITEMS .....                       | 54 |
| 4.2.7  | CHAT .....                                  | 55 |
| 4.2.8  | FEEDBACK .....                              | 56 |
| 4.2.9  | UPDATE ACCOUNT INFO .....                   | 57 |
| 4.2.10 | DEACTIVATE ACCOUNT .....                    | 58 |
| 4.2.11 | VIEW STATUS .....                           | 59 |
| 4.2.12 | CONTACT DETAILS .....                       | 60 |
| 4.2.13 | View history .....                          | 61 |
| 4.2.14 | View home page.....                         | 62 |
| 4.2.15 | Search by category .....                    | 63 |
| 4.2.16 | View statistics .....                       | 64 |

|            |                                          |           |
|------------|------------------------------------------|-----------|
| 4.2.17     | User guidance .....                      | 65        |
| <b>4.3</b> | <b>SEQUENCE DIAGRAM .....</b>            | <b>66</b> |
| 4.3.1      | Register new user: .....                 | 66        |
| 4.3.2      | Login: .....                             | 67        |
| 4.3.3      | Report lost item: .....                  | 67        |
| 4.3.4      | Report found item: .....                 | 68        |
| 4.3.5      | View status: .....                       | 69        |
| 4.3.6      | View Contact details: .....              | 70        |
| 4.3.7      | Feedback: .....                          | 71        |
| 4.3.8      | Chat: .....                              | 72        |
| 4.3.9      | View homepage: .....                     | 73        |
| 4.3.10     | View lost items: .....                   | 74        |
| 4.3.11     | View found items: .....                  | 75        |
| 4.3.12     | Update account info: .....               | 76        |
| 4.3.13     | Deactivate account: .....                | 77        |
| 4.3.14     | User guidance: .....                     | 78        |
| 4.3.15     | View history: .....                      | 79        |
| 4.3.16     | Search by Category .....                 | 80        |
| 4.3.17     | View Statistics .....                    | 81        |
| <b>4.4</b> | <b>SOFTWARE ARCHITECTURE .....</b>       | <b>82</b> |
| <b>4.5</b> | <b>CLASS DIAGRAM .....</b>               | <b>83</b> |
| <b>4.6</b> | <b>ENTITY RELATIONSHIP DIAGRAM .....</b> | <b>84</b> |
| <b>4.7</b> | <b>COLLABORATION DIAGRAM .....</b>       | <b>85</b> |
| 4.7.1      | login .....                              | 85        |
| 4.7.2      | REGISTER .....                           | 86        |
| 4.7.3      | REPORT FOUND ITEM .....                  | 87        |
| 4.7.4      | REPORT LOST ITEM .....                   | 88        |
| 4.7.5      | VIEW FOUND ITEM .....                    | 89        |

|        |                                  |     |
|--------|----------------------------------|-----|
| 4.7.6  | VIEW LOST ITEM .....             | 90  |
| 4.7.7  | VIEW ITEMS.....                  | 91  |
| 4.7.8  | CHAT .....                       | 92  |
| 4.7.9  | CONTACT DETAILS .....            | 93  |
| 4.7.10 | VIEW STATUS .....                | 94  |
| 4.7.11 | DEACTIVATE ACCOUNT .....         | 95  |
| 4.7.12 | GUIDANCE .....                   | 96  |
| 4.7.13 | HIGHLIGHTS .....                 | 97  |
| 4.7.14 | UPDATE ACCOUNT.....              | 98  |
| 4.7.15 | FEEDBACK .....                   | 99  |
| 4.7.16 | View Statistics:.....            | 100 |
| 4.7.17 | View History: .....              | 101 |
| 4.7.18 | Search with Category:.....       | 102 |
| 4.8    | NETWORK DIAGRAM .....            | 103 |
| 5      | CHAPTER 04 SYSTEM TESTING .....  | 104 |
| 5.1    | TEST CASES .....                 | 104 |
| 5.1.1  | TC01: REGISTER NEW USER .....    | 104 |
| 5.1.2  | TC02: LOGIN.....                 | 105 |
| 5.1.3  | TC03: REPORT LOST ITEM .....     | 106 |
| 5.1.4  | TC04: REPORT FOUND ITEM.....     | 107 |
| 5.1.5  | TC05: VIEW STATUS .....          | 108 |
| 5.1.6  | TC06: VIEW CONTACT DETAILS ..... | 109 |
| 5.1.7  | TC07: FEEDBACK.....              | 110 |
| 5.1.8  | TC08: CHAT .....                 | 111 |
| 5.1.9  | TC09: VIEW HOME PAGE .....       | 112 |
| 5.1.10 | TC10: VIEW LOST ITEM .....       | 113 |
| 5.1.11 | TC11: VIEW FOUND ITEMS .....     | 114 |

|        |                                          |     |
|--------|------------------------------------------|-----|
| 5.1.12 | TC12: UPDATE ACCOUNT INFO .....          | 115 |
| 5.1.13 | TC13: DEACTIVATE ACCOUNT .....           | 116 |
| 5.1.14 | TC14: USER GUIDANCE .....                | 117 |
| 5.1.15 | TC15: VIEW HISTORY .....                 | 118 |
| 5.1.16 | TC16: SEARCH BY CATEGORY .....           | 119 |
| 5.1.17 | TC17: VIEW STATISTICS.....               | 120 |
| 6      | CHAPTER CONCLUSION .....                 | 121 |
| 6.1    | Problems Faced and Lessons Learned:..... | 121 |
| 6.2    | Project Summary: .....                   | 121 |
| 6.3    | Future Work:.....                        | 121 |
| 6.4    | Web Prototypes: .....                    | 123 |
| 6.4.1  | Sign up screen.....                      | 123 |
| 6.4.2  | Login screen: .....                      | 123 |
| 6.4.3  | Home screen .....                        | 124 |
| 6.4.4  | Report lost item screen.....             | 125 |
| 6.4.5  | Report found item screen: .....          | 126 |
| 6.4.6  | Lost item screen: .....                  | 127 |
| 6.4.7  | Found item screen: .....                 | 128 |
| 6.4.8  | View status screen: .....                | 129 |
| 5.5    | Mobile Prototypes.....                   | 130 |
| 5.5.1  | View Home Page .....                     | 130 |
| 5.5.2  | Menu Screen.....                         | 131 |
| 5.5.3  | User Details Screen .....                | 132 |
| 5.5.4  | Item details screen.....                 | 133 |
| 5.5.5  | My Lost items Screen .....               | 134 |
| 5.5.6  | My found items Screen .....              | 135 |
| 5.5.7  | All Lost items screen .....              | 136 |

|                                   |     |
|-----------------------------------|-----|
| 5.5.8 All found items screen..... | 137 |
| 5.5.9 About us screen .....       | 138 |

## LIST OF TABLES:

|                                                             |     |
|-------------------------------------------------------------|-----|
| Table 1 Requirement Traceability Matric .....               | 25  |
| Table 2 Use case Description for Register New User .....    | 26  |
| Table 3 Use case Description for Login .....                | 27  |
| Table 4 Use case Description for Report lost item .....     | 28  |
| Table 5 case Description Use for Register found item .....  | 29  |
| Table 6 Use case Description for View status .....          | 30  |
| Table 7 Use case Description for View Contact Details ..... | 31  |
| Table 8 Use Case Description for feedback .....             | 32  |
| Table 9 Use case Description for Chat .....                 | 33  |
| Table 10 Use case Description for View home page .....      | 34  |
| Table 11 Use case Description for View lost item .....      | 35  |
| Table 12 Use case Description for View found items .....    | 36  |
| Table 13 Use case Description for Update account info ..... | 37  |
| Table 14 Use case Description for Deactivate Account .....  | 38  |
| Table 15 Use case Description for User guidance .....       | 39  |
| Table 16 Use case Description for View item .....           | 40  |
| Table 17 Use case description for search by category .....  | 41  |
| Table 18 use case description for view statistics .....     | 42  |
| Table 19 Test case for register user .....                  | 104 |
| Table 20 Test case for login .....                          | 105 |
| Table 21 Test case for Report lost item .....               | 106 |
| Table 22 Test case for Report found item .....              | 107 |
| Table 23 Test case for View status .....                    | 108 |
| Table 24 Test case for View contact details .....           | 109 |
| Table 25 Test case for Feedback .....                       | 110 |
| Table 26 Test case for chat .....                           | 111 |
| Table 27 Test case for View home page .....                 | 112 |
| Table 28 Test case for View lost items .....                | 113 |
| Table 29 Test case for View found items .....               | 114 |
| Table 30 Test case for Update Account info .....            | 115 |
| Table 31 Test case for Deactivate account .....             | 116 |
| Table 32 Test case for User guidance .....                  | 117 |
| Table 33 Test case for View history .....                   | 118 |

|                                                 |     |
|-------------------------------------------------|-----|
| Table 34 test case for search by category ..... | 119 |
| Table 35 testcase for view statistics .....     | 120 |

## LIST OF FIGURES:

|                                                           |    |
|-----------------------------------------------------------|----|
| Figure 1 Use case Diagram for Whole Project.....          | 43 |
| Figure 2 Use case Diagram for Finder .....                | 44 |
| Figure 3 Use case Diagram for Owner .....                 | 45 |
| Figure 4 Use case Diagram for Admin.....                  | 46 |
| Figure 5 system development lifecycle .....               | 47 |
| Figure 6 Work breakdown structure(WBS).....               | 48 |
| Figure 7 Activity Diagram for Create Account.....         | 49 |
| Figure 8 Activity Diagram for login.....                  | 50 |
| Figure 9 Activity Diagram for Register lost item.....     | 51 |
| Figure 10 Activity Diagram for Register found item .....  | 52 |
| Figure 11 Activity diagram for view found item.....       | 53 |
| Figure 12 Activity diagram for view lost items.....       | 54 |
| Figure 13 Activity diagram for chat .....                 | 55 |
| Figure 14 Activity diagram for feedback .....             | 56 |
| Figure 15 Activity diagram for update account info .....  | 57 |
| Figure 16 Activity diagram for Deactivate account .....   | 58 |
| Figure 17 Activity diagram for View status .....          | 59 |
| Figure 18 Activity diagram for Contact details .....      | 60 |
| Figure 19 view history.....                               | 61 |
| Figure 20 view homepage .....                             | 62 |
| Figure 21search by category.....                          | 63 |
| Figure 22view statistics .....                            | 64 |
| Figure 23user guidance.....                               | 65 |
| Figure 24 Sequence diagram for Register new user .....    | 66 |
| Figure 25 Sequence diagram for login .....                | 67 |
| Figure 26 Sequence diagram for Report lost item.....      | 67 |
| Figure 27 Sequence diagram for Report found item .....    | 68 |
| Figure 28 Sequence diagram for view status.....           | 69 |
| Figure 29 Sequence Diagram for view Contact details ..... | 70 |
| Figure 30 Sequence diagram for Feedback .....             | 71 |

|                                                               |     |
|---------------------------------------------------------------|-----|
| Figure 31 Sequence diagram for chat.....                      | 72  |
| Figure 32 Sequence diagram for view homepage .....            | 73  |
| Figure 33 Sequence diagram for view lost items .....          | 74  |
| Figure 34 Sequence diagrams for view found items .....        | 75  |
| Figure 35 Sequence diagram for update account info .....      | 76  |
| Figure 36 Sequence diagram for Deactivate account .....       | 77  |
| Figure 37 Sequence diagram for user guidance.....             | 78  |
| Figure 38 Sequence diagram for view history.....              | 79  |
| Figure 39 sequence diagram of search by category.....         | 80  |
| Figure 40 sequence diagram of view statistics .....           | 81  |
| Figure 41 Software Architecture .....                         | 82  |
| Figure 43 Class diagram.....                                  | 83  |
| Figure 44Entity relationship diagram.....                     | 84  |
| Figure 45 Collaboration diagram for login.....                | 85  |
| Figure 46 Collaboration diagram for registration.....         | 86  |
| Figure 47 collaboration diagram for report found item.....    | 87  |
| Figure 48 Collaboration diagram for report lost item.....     | 88  |
| Figure 49 Collaboration diagram for View found items .....    | 89  |
| Figure 50 Collaboration diagram for View lost items.....      | 90  |
| Figure 51 Collaboration diagram for View items.....           | 91  |
| Figure 52 Collaboration diagram for Chat .....                | 92  |
| Figure 53 Collaboration diagram for Contact details .....     | 93  |
| Figure 54 Collaboration diagram for View status .....         | 94  |
| Figure 55 Collaboration diagram for deactivate account .....  | 95  |
| Figure 56 Collaboration diagram for Guidance.....             | 96  |
| Figure 57 Collaboration diagrams for highlights .....         | 97  |
| Figure 58 Collaboration diagram for update account info ..... | 98  |
| Figure 59 Collaboration diagram for Feedback.....             | 99  |
| Figure 60 collaboration diagram for view statistics. ....     | 100 |
| Figure 61 collaboration diagram for view history. ....        | 101 |
| Figure 62 collaboration diagram for search by category. ....  | 102 |
| Figure 63 Network Diagram.....                                | 103 |
| Figure 64 sign up prototype.....                              | 123 |
| Figure 65 login prototype .....                               | 123 |
| Figure 66 home prototype .....                                | 124 |

|                                                         |     |
|---------------------------------------------------------|-----|
| Figure 67 Report lost item prototype.....               | 125 |
| Figure 68 Report found item prototype .....             | 126 |
| Figure 69 Lost item prototype .....                     | 127 |
| Figure 70 Found item prototype .....                    | 128 |
| Figure 71 View status prototype.....                    | 129 |
| Figure 72 Menu screen mobile prototype.....             | 131 |
| Figure 73 User Details Screen Mobile prototype .....    | 132 |
| Figure 74 Item details screen Mobile prototype.....     | 133 |
| Figure 75 My Lost items Screen mobile prototype .....   | 134 |
| Figure 76 my found items Screen mobile prototype .....  | 135 |
| Figure 77 All Lost items screen mobile prototype .....  | 136 |
| Figure 78 all found items screen mobile prototype ..... | 137 |
| Figure 79 about us screen mobile prototype.....         | 138 |

# 1 Abstract

Lost and Found Management System is a web as well as android application. As web technology and android are well-known and commonly used nowadays, so everyone has an easy access to them no matter where they are. This application is developed especially for the students, faculty members and visitors of university to find their lost items. It has an attractive GUI (graphical user interface) which will be friendly to every kind of user. The focus of the application is on the person who has lost valuable item. Our system is keeping track of lost items.

## **2 CHAPTER 01**

### **2.1 INTRODUCTION**

These days many of us lose our valuable items but no such proper platform is available for returning these lost items. So, it is mandatory to develop a system that can help to overcome these problems. This project is helpful because there is no proper tracking of lost item in our country. As this system's functionality is described here, whenever a person will lose his item, first of all he will get register on the website and after successful login a form will appear for describing his lost item after filling and adding location of lost item his ad of lost item will go live. Highlights of that item will also appear on main page. Similarly, when finder will found lost item he will also get register on the website and after login he will provide information regarding the item and location of that lost item and when mislayer will see his lost item he will then contact to the finder.

Anyone can see lost items on the website but in order to claim his item he/she have to register on the site or on the mobile app. Security breaches are increasing day by day so we have provided two step verification for every user for better security. The mislayer will get immediate status of his lost item.

Lost and found management system, manage and track the lost and found items. This website is developed for the students, faculty members and visitors of the universities yet anyone is able to use it. Many times we come across multiple incidents someone has lost his/her valuable item and someone has found a lost item. Usually the people who find something submit it to the in-charge of the department and these items are placed in waiting for the person who lost it, but sometimes person who lost his item is not aware about this. The in-charge of the department wants to give back the items to their owners but due to unavailability of suitable platform, lost items are not returned. Due to this reason, the students, employees, as well as visitors face a lot of difficulties to find their missing items. For these kinds of issues, we have developed a web and mobile application of lost and found management system.

### **2.2 OBJECTIVES**

Our goal is to return the owners their lost items. As of now there is very little development of such type so we have developed an application that ease out the whole process of finding the lost items. We have made a web based and android application because these platforms are very common and easily accessible these days.

We have developed this application to provide a very basic and easy to use, user interface so that every person can easily use the application. We have provided Maps (geographical location) service so that the owners and the finder can get the exact location where the lost item is located. A chat feature is also provided so that the owners and finder can interact regarding the lost object. Every lost item will be shown in the highlights on the main app page.

## 2.3 PROBLEM STATEMENT

These days many of us lose our valuable items but no such proper platform is available for returning these lost items. So, it is mandatory to develop a system that can help to overcome these problems. This project is helpful because there is no proper tracking of lost item in our country.

## 2.4 ASSUMPTION AND LIMITATION

### ASSUMPTIONS:

- There will be many users who will use this app or website therefore it is not possible to train every user that's why we have made a user friendly interface that help every user to train them.
- This project will be able to use on smart phones, tablets, desktops, laptops.
- At the project meeting the Stakeholders of the project are available.

### LIMITATIONS:

- This project is able to work only on desktop system and android devices. User will not be able to use it on IOS.
- Some of the functionalities in the android application are limited.
- Internet connection is compulsory.
- Above 4000 user can't use the app or software at a time.

## 2.5 PROJECT SCOPE

Nowadays most of us go somewhere to visit place and sometimes we forget our valuable item at that place. This is very stressful for every owner. There are many platforms but sometimes they neither submit lost items nor return item to its exact owner. Some platforms are paid which are not suitable for some users. Our project is based on web as well as on android. Both owner and finder will register themselves.

User needs registration before performing any operation. After registration and login owner can search his item and can also see all lost items. We have provided form to finder for providing information regarding the lost item, he will also give the geographical location of that particular place from where he found the item similarly owner will also give the location where he lost the item. Both owner and finder can chat with each other for verification. Owner will mark item found in his profile after he has found his lost item.

## 3 CHAPTER 02 REQUIREMENT ANALYSIS

### 3.1 LITERATURE REVIEW

(Thomas Campbell:2003). [1] In this development a system is designed which restores an item to its owner. The item includes a tag that is attached to the object which includes pre assigned identification number associated of the owner and also a toll free number. This pre identification number is then sent to the information centre from finder. The information centre contacts with the owner. After that owner receives information about his lost item from the information centre. The major drawback of this system is the third medium used for returning the lost item that is information centre and sometimes their server may can go down which will lead more problems for the mislayer.

(Tom Price:2008). [2] This is automated lost item and found system named as LINR that avoids language barriers by providing a short description of the item entered by the finder which is stored in a database in any lost item notification message communicated to the owner concerning the item. The lost item notification message also tells language used by the reporting party when placing the lost item report, and may also include an indication of the geographic location where the lost item report originated. There are many drawbacks in this system. The major drawback of this system is when owner will submit its information about the lost item after submitting he/she does not know the status of his item as this system avoids language barriers because all type of lost information is handled by the system.

(Harkamal Kaur:2017). [3] This is a simple web based client server site which was made for students to of university to easily found there lost item. The main features of this application include an authentication, registration and login system, a lost and found listings page where the user can enter the information about the lost and found items. Due to some security issues, the users cannot delete any items because some users may delete the items just to mess around. This privilege is given to the admin only. The application also allows the users to update the listed items if needed. The users can update the contact information and change the password. This system has no proper GUI. Owner and finder cannot directly chat with each. The major negative point is both owner and finder cannot add their geographical location.

### 3.2 STACKHOLDER'S LIST

| Name      | Role                         |
|-----------|------------------------------|
| 1. Admin  | Manage the working of system |
| 2. Owner  | Report lost item             |
| 3. Finder | Report found item            |

### 3.3 REQUIREMENT ELICITATION

#### 3.3.1 FUNCTIONAL REQUIREMENTS

##### 3.3.1.1 Register new user:

| Req. No. | Functional Requirement                                                 |
|----------|------------------------------------------------------------------------|
| FR01-01  | The owner or finder can register him/herself into the system.          |
| FR01-02  | The owner or finder is able to enter his/her first name.               |
| FR01-03  | The owner or finder is able to enter his/her last name.                |
| FR01-04  | The owner or finder is able to enter a valid email address.            |
| FR01-05  | The owner or finder is able to set a password                          |
| FR01-06  | The system allows the user to set up two step verification if desired. |
| FR01-07  | The user is able to add his phone number.                              |

##### 3.3.1.2 Login:

| Req. No. | Functional Requirement                                                                                         |
|----------|----------------------------------------------------------------------------------------------------------------|
| FR02-01  | If the owner or finder is registered with the system, then he/she can log in with the system.                  |
| FR02-02  | The owner or finder is able to enter registered email address.                                                 |
| FR02-03  | The owner or finder is able to enter password.                                                                 |
| FR02-04  | If the owner or finder has given the correct email and password, then they are able to press the login button. |
| FR02-05  | The admin is able to login into the system.                                                                    |

##### 3.3.1.3 Report lost item:

| Req. No. | Functional Requirement                                                            |
|----------|-----------------------------------------------------------------------------------|
| FR03-01  | The owner is able to report his/her lost item.                                    |
| FR03-02  | The owner is able to enter his/her name.                                          |
| FR03-03  | The owner is able to enter his contact email.                                     |
| FR03-04  | The owner is able to enter the date on which his/her item was lost.               |
| FR03-05  | The owner is able to enter the geographical location where his/her item was lost. |
| FR03-06  | The owner is able to enter the category of his/her lost item.                     |
| FR03-07  | The owner is able to enter the brand of his/her lost item.                        |

|         |                                                                                                                                 |
|---------|---------------------------------------------------------------------------------------------------------------------------------|
| FR03-08 | The owner is able to enter the colour of his/her lost item.                                                                     |
| FR03-09 | The owner is able to enter the brief description of his/her lost item.                                                          |
| FR03-10 | After entering all the details, the owner is able to press the submit button.                                                   |
| FR03-11 | After submitting the form, the system generates a reference id of that particular item.                                         |
| FR03-12 | When user(owner) add all the information regarding item, an automated mail is sent to the provided email with reference number. |

#### 3.3.1.4 Report found item:

| Req. No. | Functional Requirement                                                                                                           |
|----------|----------------------------------------------------------------------------------------------------------------------------------|
| FR04-01  | The finder is able to report his/her lost item.                                                                                  |
| FR04-02  | The finder is able to enter his/her name.                                                                                        |
| FR04-03  | The finder is able to enter his contact email.                                                                                   |
| FR04-04  | The finder is able to enter the date on which item was found.                                                                    |
| FR04-05  | The finder is able to enter the geographical location where item was found.                                                      |
| FR04-06  | The finder is able to enter the category of found item.                                                                          |
| FR04-07  | The finder is able to enter the brand of found item.                                                                             |
| FR04-08  | The finder is able to enter the colour of found item.                                                                            |
| FR04-09  | The finder is able to enter the brief description of found item.                                                                 |
| FR04-10  | After entering all the details, the finder is able to press the submit button.                                                   |
| FR04-11  | After entering all the details, the owner is able to press the submit button.                                                    |
| FR04-12  | When user(finder) add all the information regarding item, an automated mail is sent to the provided email with reference number. |

#### 3.3.1.5 View status:

| Req. No. | Functional Requirement                                                                                              |
|----------|---------------------------------------------------------------------------------------------------------------------|
| FR05-01  | The finder or the owner can check the status of lost item by entering the reference number of that particular item. |
| FR05-02  | The admin is able to check the performance of the system.                                                           |
| FR05-03  | The admin is able to block the user, who is doing something wrong like posting abusive content etc.                 |

### 3.3.1.6 View Contact details:

| Req. No. | Functional Requirement                                                                                |
|----------|-------------------------------------------------------------------------------------------------------|
| FR06-01  | The user is able to contact with the LOST&FOUND management team by clicking on the contact us button. |
| FR06-02  | The user is able to get the email address of the LOST&FOUND management team from the contact us page. |

### 3.3.1.7 Feedback:

| Req. No. | Functional Requirement                                                        |
|----------|-------------------------------------------------------------------------------|
| FR07-01  | The user is able to give his/her feedback by clicking on the feedback button. |
| FR07-02  | The user is able report bugs if any.                                          |
| FR07-03  | Admin can see the feedbacks of the users.                                     |

### 3.3.1.8 Chat:

| Req. No. | Functional Requirement                                                                                                                              |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| FR08-01  | The system shall enable the item owner and finder to decide on what location they want to meet.                                                     |
| FR08-02  | The system shall enable the finder to ask for more details of the item to check the validity of the ownership of the lost item from the item owner. |
| FR08-03  | Owner and finder are able to directly communicate with each other                                                                                   |

### 3.3.1.9 View homepage:

| Req. No. | Functional Requirement                                                 |
|----------|------------------------------------------------------------------------|
| FR09-01  | The lost items highlights are displayed on the main page..             |
| FR09-02  | The user (owner/finder) can search the lost item from the search bar.  |
| FR09-03  | The user (owner/finder) can search the found item from the search bar. |
| FR09-04  | The user (owner/finder) is able to report lost item.                   |
| FR09-05  | The user owner/finder) is able to report found item.                   |

#### **3.3.1.10 View lost items:**

| <b>Req. No.</b> | <b>Functional Requirement</b>                                            |
|-----------------|--------------------------------------------------------------------------|
| FR10-01         | The user(owner/finder) is able to see all the lost items after login in. |
| FR10-02         | The user (owner/finder) is able to see details of the lost item.         |
| FR10-03         | The user (owner/finder) is able to see the status of lost item.          |
| FR10-04         | The admin is able to view the lost item list.                            |

#### **3.3.1.11 View found items:**

| <b>Req. No.</b> | <b>Functional Requirement</b>                                             |
|-----------------|---------------------------------------------------------------------------|
| FR11-01         | The user(owner/finder) is able to see all the found items after login in. |
| FR11-02         | The user (owner/finder) is able to see details of the found items.        |
| FR11-03         | The user (owner/finder) is able to see the status of found items.         |
| FR11-04         | The admin is able to view the found item list.                            |

#### **3.3.1.12 Update account info:**

| <b>Req. No.</b> | <b>Functional Requirement</b>                               |
|-----------------|-------------------------------------------------------------|
| FR12-01         | The user(owner/finder) is able to update item details.      |
| FR12-02         | The user(owner/finder) is able to update his email address. |
| FR12-03         | The user(owner/finder) is able to update his name.          |
| FR12-04         | The user(owner/finder) is able to update his phone number.  |

#### **3.3.1.13 Deactivate account:**

| <b>Req. No.</b> | <b>Functional Requirement</b>                          |
|-----------------|--------------------------------------------------------|
| FR13-01         | The user (owner/finder) is able to delete his account. |
| FR13-02         | The user (owner/finder) is able to delete item.        |
| FR13-03         | Admin can block and deactivate the account of user.    |

#### 3.3.1.14 User guidance:

| Req. No. | Functional Requirement                                                                                 |
|----------|--------------------------------------------------------------------------------------------------------|
| FR14-01  | There is a short video clip for “ <b>how to use the system</b> ” and app for the guidance of the user. |

#### 3.3.1.15 View history:

| Req. No. | Functional Requirement                                                        |
|----------|-------------------------------------------------------------------------------|
| FR15-01  | The finder is able to see the returned items history only if he is logged in. |
| FR15-02  | The owner is able to see the received items history only if he is logged in.  |

#### 3.3.1.16 Search by category

| Req. No. | Functional Requirement                                         |
|----------|----------------------------------------------------------------|
| FR16-01  | The user can search the desired item by entering the category. |

#### 3.3.1.17 View statistics

| Req. No. | Functional Requirement                 |
|----------|----------------------------------------|
| FR17-01  | Admin can view the overall statistics. |

### 3.3.2 NON-FUNCTIONAL REQUIREMENTS

#### 3.3.2.1 Performance:

| Req. No. | Non-Functional Requirement                                                                                                         |
|----------|------------------------------------------------------------------------------------------------------------------------------------|
| NFR01-01 | The average page loading time of our application is less than 3 seconds                                                            |
| NFR01-02 | Response time of our application page is less than 1 second                                                                        |
| NFR01-03 | The page being used, supporting 5 thousand users per hour must provide 6 second or less response time in a chrome desktop browser. |

### 3.3.2.2 Usability:

| Req. No. | Non-Functional Requirement                                                                                          |
|----------|---------------------------------------------------------------------------------------------------------------------|
| NFR02-01 | The main actions are completed under 1 minute once the user see the interface                                       |
| NFR02-02 | The system UI is easy to understand that every time the user re-uses the system he/she shall easily get used to it. |
| NFR02-03 | The Error rate of users to submit their Item details at the submission page must not exceed 10 percent.             |

### 3.3.2.3 Security:

| Req. No. | Non-Functional Requirement                                                                   |
|----------|----------------------------------------------------------------------------------------------|
| NFR03-01 | For isolation of information from other users every user would have a two step-verification. |
| NFR03-02 | No other user can access the system functionality that is not present in our database.       |

### 3.3.2.4 Portability:

| Req. No. | Non-Functional Requirement                                                        |
|----------|-----------------------------------------------------------------------------------|
| NFR04-01 | The System is compatible with windows Operating systems.                          |
| NFR04-02 | The system is compatible with devices having chrome browser version 1.31 or above |

### 3.3.2.5 Reliability:

| Req. No. | Non-Functional Requirement                                              |
|----------|-------------------------------------------------------------------------|
| NFR05-01 | The average failure rate of the system is 3 or less than that annually. |
| NFR05-02 | The system runs 7 days a week, 24 hours a day                           |

### 3.3.2.6 Availability:

| Req. No. | Non-Functional Requirement                                                                                    |
|----------|---------------------------------------------------------------------------------------------------------------|
| NFR06-01 | Mean Time to Repair (MTTR): Even if the system fails, the system is recovered back up within an hour or less. |

### 3.3.3 REQUIREMENT TRACEABILITY MATRIC:

| ID | FR ID | DESCRIPTION            | USE CASE # | TEST CASE # | ACTIVITY # | SEQUENCE # |
|----|-------|------------------------|------------|-------------|------------|------------|
| 01 | FR01  | REGISTER NEW USER      | UC001      | TC01        | 001        | 001        |
| 02 | FR02  | LOGIN                  | UC002      | TC02        | 002        | 002        |
| 03 | FR03  | REPORT LOST ITEM       | UC003      | TC03        | 003        | 003        |
| 04 | FR04  | REPORT FOUND ITEM      | UC004      | TC04        | 004        | 004        |
| 05 | FR05  | VIEW STATUS            | UC005      | TC05        | 011        | 005        |
| 06 | FR06  | VIEW CONTACT DETAILS   | UC006      | TC06        | 012        | 006        |
| 07 | FR07  | FEEDBACK               | UC007      | TC07        | 008        | 007        |
| 08 | FR08  | CHAT                   | UC008      | TC08        | 007        | 008        |
| 09 | FR09  | VIEW HOME PAGE         | UC009      | TC09        | 014        | 009        |
| 10 | FR10  | VIEW LOST ITEMS        | UC010      | TC10        | 006        | 010        |
| 11 | FR11  | VIEW FOUND ITEMS       | UC011      | TC11        | 005        | 011        |
| 12 | FR12  | UPDATE ACCOUNT INFO    | UC012      | TC12        | 009        | 012        |
| 13 | FR13  | DEACTIVATE ACCOUNT     | UC013      | TC13        | 010        | 013        |
| 14 | FR14  | USER GUIDANCE          | UC014      | TC14        | 017        | 014        |
| 15 | FR15  | VIEW HISTORY           | UC015      | TC15        | 013        | 015        |
| 16 | FR16  | VIEW ITEMS BY CATEGORY | UC016      | TC16        | 015        | 016        |

|    |      |                    |       |      |     |     |
|----|------|--------------------|-------|------|-----|-----|
| 17 | FR15 | VIEW<br>STATISTICS | UC017 | TC17 | 016 | 017 |
|----|------|--------------------|-------|------|-----|-----|

*Table 1 Requirement Traceability Matrix*

### 3.4 USE CASE DESCRIPTION

#### 3.4.1 UC:001

|                                                                                                  |                                                                                                                             |                                                                             |                   |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------------|
| <b>Use case ID:</b>                                                                              | 001                                                                                                                         | <b>Use case Name:</b>                                                       | Register new user |
| <b>Priority</b>                                                                                  | 01                                                                                                                          |                                                                             |                   |
| <b>Actor</b>                                                                                     | Finder, Owner                                                                                                               |                                                                             |                   |
| <b>Use Case Summary:</b>                                                                         | It allows the user (finder, owner) to register himself with the system by providing the information like name, valid email. |                                                                             |                   |
| <b>Pre-conditions</b>                                                                            | The email should be valid.                                                                                                  |                                                                             |                   |
| <b>Normal Follow of Paths</b>                                                                    |                                                                                                                             | <b>Alternative Path</b>                                                     |                   |
| 1. The user clicks on sign up button.                                                            |                                                                                                                             |                                                                             |                   |
| 2. The user gives the information like name, email, and password.                                |                                                                                                                             |                                                                             |                   |
| 3. A verification mail is sent to the provided email.                                            |                                                                                                                             | If user does not click on sign up then verification email will not be sent. |                   |
| <b>Exception:</b>                                                                                |                                                                                                                             |                                                                             |                   |
| 1. The system do not allow the user to add email which is already registered with in the system. |                                                                                                                             |                                                                             |                   |
| <b>Post-conditions:</b>                                                                          |                                                                                                                             |                                                                             |                   |
| <b>Step no #</b>                                                                                 | <b>Description</b>                                                                                                          |                                                                             |                   |
| 1.                                                                                               | The user (finder, owner) will be successfully registered.                                                                   |                                                                             |                   |

*Table 2 Use case Description for Register New User*

### 3.4.2 UC:002

|                                                                                                       |                                                                                                                                  |                  |       |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------|-------|
| Use case ID:                                                                                          | 002                                                                                                                              | Use case Name:   | Login |
| Priority                                                                                              | 02                                                                                                                               |                  |       |
| Actor                                                                                                 | Finder, Owner                                                                                                                    |                  |       |
| Use Case Summary:                                                                                     | It allows the user to login himself into the system using the email and password which they provide at the time of registration. |                  |       |
| Pre-conditions                                                                                        | User should be registered with the system.                                                                                       |                  |       |
| Normal Follow of Paths                                                                                |                                                                                                                                  | Alternative Path |       |
| 1. The user click on the login button.                                                                |                                                                                                                                  |                  |       |
| 2. The user provides the email and password.                                                          |                                                                                                                                  |                  |       |
| Exception:                                                                                            |                                                                                                                                  |                  |       |
| 1. If the email or password is not valid then the user will not be able to logged in with the system. |                                                                                                                                  |                  |       |
| Post-conditions:                                                                                      |                                                                                                                                  |                  |       |
| Step no #                                                                                             | Description                                                                                                                      |                  |       |
| 1.                                                                                                    | The user (finder, owner will be successfully logged in.                                                                          |                  |       |

*Table 3 Use case Description for Login*

### 3.4.3 UC:003

|                                                                                                                     |                                                                                      |                                                                           |                  |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------|
| Use case ID:                                                                                                        | 003                                                                                  | Use case Name:                                                            | Report Lost Item |
| Priority                                                                                                            | 01                                                                                   |                                                                           |                  |
| Actor                                                                                                               | Owner                                                                                |                                                                           |                  |
| Use Case Summary:                                                                                                   | It allows the owner to report his lost item in to the system.                        |                                                                           |                  |
| Pre-conditions                                                                                                      | Owner should be registered and logged in with the system before reporting lost item. |                                                                           |                  |
| Normal Follow of Paths                                                                                              |                                                                                      | Alternative Path                                                          |                  |
| 1. The owner clicks on the report lost item button.                                                                 |                                                                                      |                                                                           |                  |
| 2. The owner provides the information like name, email, lost item description, location, colour, and brand.         |                                                                                      |                                                                           |                  |
| 3. After entering all the information the owner will press the submit button.                                       |                                                                                      |                                                                           |                  |
| 4. After pressing the submit button an automated mail is sent to the provided email with a unique reference number. |                                                                                      | If the user does not press submit button then lost item is be registered. |                  |
| Exception:                                                                                                          |                                                                                      |                                                                           |                  |
| If the item is already registered, the system does not allow the owner to report.                                   |                                                                                      |                                                                           |                  |
| Post-conditions:                                                                                                    |                                                                                      |                                                                           |                  |
| Step no #                                                                                                           | Description                                                                          |                                                                           |                  |
| 1.                                                                                                                  | Item will be successfully reported and the database will be updated.                 |                                                                           |                  |

*Table 4 Use case Description for Report lost item*

### 3.4.4 UC:004

|                                                                                                                     |                                                                                        |                                                                                  |                   |
|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------------|
| Use case ID:                                                                                                        | 004                                                                                    | Use case Name:                                                                   | Report Found Item |
| Priority                                                                                                            | 01                                                                                     |                                                                                  |                   |
| Actor                                                                                                               | Finder                                                                                 |                                                                                  |                   |
| Use Case Summary:                                                                                                   | It allows the finder to report the found item in to the system.                        |                                                                                  |                   |
| Pre-conditions                                                                                                      | Finder should be registered and logged in with the system before reporting found item. |                                                                                  |                   |
| Normal Follow of Paths                                                                                              |                                                                                        | Alternative Path                                                                 |                   |
| 1. The finder clicks on the report found item button.                                                               |                                                                                        |                                                                                  |                   |
| 2. The finder provides the information like name, email, lost item description, location, colour, and brand.        |                                                                                        |                                                                                  |                   |
| 3. After entering all the information the finder press the submit button.                                           |                                                                                        |                                                                                  |                   |
| 4. After pressing the submit button an automated mail is sent to the provided email with a unique reference number. |                                                                                        | If the user does not press submit button then found item will not be registered. |                   |
| Exception:                                                                                                          |                                                                                        |                                                                                  |                   |
| If the item is already reported, the system do not allow the finder to report.                                      |                                                                                        |                                                                                  |                   |
| Post-conditions:                                                                                                    |                                                                                        |                                                                                  |                   |
| Step no #                                                                                                           | Description                                                                            |                                                                                  |                   |
| 1.                                                                                                                  | Item will be successfully reported and the database will be updated.                   |                                                                                  |                   |

*Table 5 Use case Description for Register found item*

### 3.4.5 UC:005

|                                                                            |                                                                 |                                                                        |             |
|----------------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------|-------------|
| Use case ID:                                                               | 005                                                             | Use case Name:                                                         | View Status |
| Priority                                                                   | 03                                                              |                                                                        |             |
| Actor                                                                      | Finder, Owner                                                   |                                                                        |             |
| Use Case Summary:                                                          | It allows the user (finder, owner) to check the status of item. |                                                                        |             |
| Pre-conditions                                                             | The item should be reported in to the system.                   |                                                                        |             |
| Normal Follow of Paths                                                     |                                                                 | Alternative Path                                                       |             |
| 1. The user will click on the view status button.                          |                                                                 |                                                                        |             |
| 2. The user will enter the reference number of the item.                   |                                                                 |                                                                        |             |
| 3. The user will press the enter button.                                   |                                                                 | The user will not be able to see the status if the session is expired. |             |
| Exception:                                                                 |                                                                 |                                                                        |             |
| If the reference number is not valid, the system will not show any status. |                                                                 |                                                                        |             |
| Post-conditions:                                                           |                                                                 |                                                                        |             |
| Step no #                                                                  | Description                                                     |                                                                        |             |
| 1.                                                                         | The status of the item will be displayed on the screen.         |                                                                        |             |

*Table 6 Use case Description for View status*

### 3.4.6 UC:006

|                                                                                                  |                                                                                                                                                                                                   |                  |                      |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------|
| Use case ID:                                                                                     | 006                                                                                                                                                                                               | Use case Name:   | View Contact Details |
| Priority                                                                                         | 04                                                                                                                                                                                                |                  |                      |
| Actor                                                                                            | Finder, Owner                                                                                                                                                                                     |                  |                      |
| Use Case Summary:                                                                                | In case the user feels to contact the management team of Lost and Found management system, a contact detail section would be provided so that the user can contact them for their own assistance. |                  |                      |
| Pre-conditions                                                                                   | The user must be login into the system to view the contact details.                                                                                                                               |                  |                      |
| Normal Follow of Paths                                                                           |                                                                                                                                                                                                   | Alternative Path |                      |
| 1. The user (Owner) logs into the system by providing email and password.                        |                                                                                                                                                                                                   |                  |                      |
| 2. The user scroll to the contact us Section on the main page of the application.                |                                                                                                                                                                                                   |                  |                      |
| 3. The user clicks on the required contact button to get the contact information.                |                                                                                                                                                                                                   |                  |                      |
| Exception:                                                                                       |                                                                                                                                                                                                   |                  |                      |
| The system do not allow the user to see contact details if he/she is not logged into the system. |                                                                                                                                                                                                   |                  |                      |
| Post-conditions:                                                                                 |                                                                                                                                                                                                   |                  |                      |
| Step no #                                                                                        | Description                                                                                                                                                                                       |                  |                      |
| 1.                                                                                               | The user will see the contact information.                                                                                                                                                        |                  |                      |

*Table 7 Use case Description for View Contact Details*

### 3.4.7 UC:007

|                                                                                                |                                                                                                                                                                                                     |                  |          |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------|
| Use case ID:                                                                                   | 007                                                                                                                                                                                                 | Use case Name:   | Feedback |
| Priority                                                                                       | 04                                                                                                                                                                                                  |                  |          |
| Actor                                                                                          | Finder, Owner                                                                                                                                                                                       |                  |          |
| Use Case Summary:                                                                              | The user can give his/her feedback regarding the experience that he/she got while using our system. They can also report any bug if they found so that our team can rectify it as soon as possible. |                  |          |
| Pre-conditions                                                                                 | The user must be login into the system to enter the feedback.                                                                                                                                       |                  |          |
| Normal Follow of Paths                                                                         |                                                                                                                                                                                                     | Alternative Path |          |
| 1. The user (Owner) logs into the system by providing email and password.                      |                                                                                                                                                                                                     |                  |          |
| 2. The user search for the feedback button provided on the main page.                          |                                                                                                                                                                                                     |                  |          |
| 3. The user will click on the Feedback button.                                                 |                                                                                                                                                                                                     |                  |          |
| 4. User enter his/her feedback                                                                 |                                                                                                                                                                                                     |                  |          |
| Exception:                                                                                     |                                                                                                                                                                                                     |                  |          |
| The system does not allow the user to insert feedback if he/she is not logged into the system. |                                                                                                                                                                                                     |                  |          |
| The system does not let the user to insert more than 4000 letters in the feedback area.        |                                                                                                                                                                                                     |                  |          |
| Post-conditions:                                                                               |                                                                                                                                                                                                     |                  |          |
| Step no #                                                                                      | Description                                                                                                                                                                                         |                  |          |
| 1.                                                                                             | The user will add a feedback regarding his experience.                                                                                                                                              |                  |          |

*Table 8 Use Case Description for feedback*

### 3.4.8 UC:008

|                                                                                                                        |                                                                                                                                                                                                                                                         |                  |      |
|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------|
| Use case ID:                                                                                                           | 008                                                                                                                                                                                                                                                     | Use case Name:   | Chat |
| Priority                                                                                                               | 02                                                                                                                                                                                                                                                      |                  |      |
| Actor                                                                                                                  | Finder, Owner                                                                                                                                                                                                                                           |                  |      |
| Use Case Summary:                                                                                                      | For the convenience of user, both the user and finder can chat with each other to decide for a mutual place to visit each other and exchange the lost item. It can also initially help the user to communicate with each other regarding the lost item. |                  |      |
| Pre-conditions                                                                                                         | The user must be login into the system and the user must have either uploaded a lost item request or found an item.                                                                                                                                     |                  |      |
| Normal Follow of Paths                                                                                                 |                                                                                                                                                                                                                                                         | Alternative Path |      |
| 1. The user (Owner) login into the system by providing email and password.                                             |                                                                                                                                                                                                                                                         |                  |      |
| 2. The user either posts a lost item request or responds to a person for finding a lost item.                          |                                                                                                                                                                                                                                                         |                  |      |
| 3. The user is having a chat button on the screen that he/she would click to start chat.                               |                                                                                                                                                                                                                                                         |                  |      |
| Exception:                                                                                                             |                                                                                                                                                                                                                                                         |                  |      |
| The system does not allow the user to chat if they have not either posted a lost item or respond to lost item request. |                                                                                                                                                                                                                                                         |                  |      |
| Post-conditions:                                                                                                       |                                                                                                                                                                                                                                                         |                  |      |
| Step no #                                                                                                              | Description                                                                                                                                                                                                                                             |                  |      |
| 1.                                                                                                                     | The user will meet at a mutual place to exchange the lost product.                                                                                                                                                                                      |                  |      |

*Table 9 Use case Description for Chat*

### 3.4.9 UC:009

|                                                                                  |                                                                                                                                                                                 |                  |                |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------|
| Use case ID:                                                                     | 009                                                                                                                                                                             | Use case Name:   | View Home page |
| Priority                                                                         | 04                                                                                                                                                                              |                  |                |
| Actor                                                                            | Finder, Owner                                                                                                                                                                   |                  |                |
| Use Case Summary:                                                                | Homepage would be the place where all the highlights of the lost items would be displayed. The person who has lost his item would type in the search bar or see the highlights. |                  |                |
| Pre-conditions                                                                   | The user must be login into the system to view the Homepage.                                                                                                                    |                  |                |
| Normal Follow of Paths                                                           |                                                                                                                                                                                 | Alternative Path |                |
| 1. The user (Owner) logs into the system by providing email and password.        |                                                                                                                                                                                 |                  |                |
| 2. The lost and found items highlights are displayed on the main screen.         |                                                                                                                                                                                 |                  |                |
| 3. The owner select the search bar to enter details regarding his/her lost item. |                                                                                                                                                                                 |                  |                |
| 4. The finder enters details if he /she have found something.                    |                                                                                                                                                                                 |                  |                |
| 5. Search button is pressed to search for the product.                           |                                                                                                                                                                                 |                  |                |
| Exception:                                                                       |                                                                                                                                                                                 |                  |                |
|                                                                                  |                                                                                                                                                                                 |                  |                |
| Post-conditions:                                                                 |                                                                                                                                                                                 |                  |                |
| Step no #                                                                        | Description                                                                                                                                                                     |                  |                |
| 1.                                                                               | The user will view the home page items.                                                                                                                                         |                  |                |

*Table 10 Use case Description for View home page*

### 3.4.10 UC:010

|                                                                                                     |                                                                                                                                 |                  |                 |
|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| Use case ID:                                                                                        | 010                                                                                                                             | Use case Name:   | View Lost items |
| Priority                                                                                            | 01                                                                                                                              |                  |                 |
| Actor                                                                                               | Finder, Owner, Admin                                                                                                            |                  |                 |
| Use Case Summary:                                                                                   | The owner, admin and finder can see from the highlights that which items are lost. All the lost items can be viewed from there. |                  |                 |
| Pre-conditions                                                                                      | The user must be login into the system to view the details of the lost items.                                                   |                  |                 |
| Normal Follow of Paths                                                                              |                                                                                                                                 | Alternative Path |                 |
| 1. The user (Owner/finder) logs into the system by providing email and password.                    |                                                                                                                                 |                  |                 |
| 2. The user (finder) searches an item.                                                              |                                                                                                                                 |                  |                 |
| 3. The user (owner/finder) clicks on the lost items item button in order to see all the lost items. |                                                                                                                                 |                  |                 |
| 4. The user (owner/finder) clicks on the details to see the lost item details.                      |                                                                                                                                 |                  |                 |
| 5. Admin clicks on the lost items button to see all the received and returned items.                |                                                                                                                                 |                  |                 |
| Exception:                                                                                          |                                                                                                                                 |                  |                 |
| The system does not allow the user to see lost items details if he is not logged into the system.   |                                                                                                                                 |                  |                 |
| Post-conditions:                                                                                    |                                                                                                                                 |                  |                 |
| Step no #                                                                                           | Description                                                                                                                     |                  |                 |
| 1.                                                                                                  | The user will see the lost items.                                                                                               |                  |                 |

*Table 11 Use case Description for View lost item*

### 3.4.11 UC:011

|                                                                                                       |                                                                                                                                                                                                     |                  |                  |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------|
| Use case ID:                                                                                          | 011                                                                                                                                                                                                 | Use case Name:   | View Found Items |
| Priority                                                                                              | 01                                                                                                                                                                                                  |                  |                  |
| Actor                                                                                                 | Owner, Admin                                                                                                                                                                                        |                  |                  |
| Use Case Summary:                                                                                     | The owner and admin will see all the found items reported by finders. They will also be able to see all the details of the lost items for example their category, provided phone number, email etc. |                  |                  |
| Pre-conditions:                                                                                       | The user must be login into the system to view the details of the found items.                                                                                                                      |                  |                  |
| Normal Follow of Paths                                                                                |                                                                                                                                                                                                     | Alternative Path |                  |
| 1. The user (Owner/finder) logs into the system by providing email and password.                      |                                                                                                                                                                                                     |                  |                  |
| 2. The user (Owner/finder) searches an item.                                                          |                                                                                                                                                                                                     |                  |                  |
| 3. The user (Owner/finder) clicks on the found items item button in order to see all the found items. |                                                                                                                                                                                                     |                  |                  |
| 4. The user (Owner/finder) clicks on the details to see the found item details.                       |                                                                                                                                                                                                     |                  |                  |
| 5. The admin clicks on the found items button to see all the received and returned items.             |                                                                                                                                                                                                     |                  |                  |
| Exception:                                                                                            |                                                                                                                                                                                                     |                  |                  |
| The system does not allow the user to see found items details if he is not logged into the system.    |                                                                                                                                                                                                     |                  |                  |
| Post-conditions:                                                                                      |                                                                                                                                                                                                     |                  |                  |
| Step no #                                                                                             | Description                                                                                                                                                                                         |                  |                  |
| 1.                                                                                                    | The user will see the found items.                                                                                                                                                                  |                  |                  |

*Table 12 Use case Description for View found items*

### 3.4.12 UC:012

|                                                                        |                                                                                                                                                                                                                                                |                  |                     |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|
| Use case ID:                                                           | 012                                                                                                                                                                                                                                            | Use case Name:   | Update Account Info |
| Priority                                                               | 2                                                                                                                                                                                                                                              |                  |                     |
| Actor                                                                  | Owner, Admin                                                                                                                                                                                                                                   |                  |                     |
| Use Case Summary:                                                      | Update Account info will provide the user to update his item info. It will also allow the user to update his provided email address, password, phone number, also the location of the item etc.                                                |                  |                     |
| Pre-conditions                                                         | <div>1. The user must be logged into the system.</div> <div>2. Item must be previously added into the system if user wants to update the item info.</div> <div>3. User must be registered before if he wants to update his account info.</div> |                  |                     |
| Normal Follow of Paths                                                 |                                                                                                                                                                                                                                                | Alternative Path |                     |
| 1. The user logs into the system.                                      |                                                                                                                                                                                                                                                |                  |                     |
| 2. The user updates the info which he wants to update.                 |                                                                                                                                                                                                                                                |                  |                     |
| 3. The system responds the user after updating and updates the system. |                                                                                                                                                                                                                                                |                  |                     |
| Exception:                                                             |                                                                                                                                                                                                                                                |                  |                     |
| The system does not allow the user to update another user info.        |                                                                                                                                                                                                                                                |                  |                     |
| Post-conditions:                                                       |                                                                                                                                                                                                                                                |                  |                     |
| Step no #                                                              | Description                                                                                                                                                                                                                                    |                  |                     |
| 1.                                                                     | <div>1. The database will be updated.</div> <div>2. The information will be updated everywhere in the system</div>                                                                                                                             |                  |                     |

*Table 13 Use case Description for Update account info*

### 3.4.13 UC:013

|                                                                                  |                                                                                                                                |                                                  |                    |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------------------|
| Use case ID:                                                                     | 013                                                                                                                            | Use case Name:                                   | Deactivate Account |
| Priority                                                                         | 4                                                                                                                              |                                                  |                    |
| Actor                                                                            | Owner, Finder                                                                                                                  |                                                  |                    |
| Use Case Summary:                                                                | The user will delete his account or any posted items.                                                                          |                                                  |                    |
| Pre-conditions                                                                   | 1. The user must be logged in into the system.<br>2. Only previously user and item registered into the system will be deleted. |                                                  |                    |
| Normal Follow of Paths                                                           |                                                                                                                                | Alternative Path                                 |                    |
| 1. The user logs into the system by clicking on the button.                      |                                                                                                                                |                                                  |                    |
| 2. The user selects any posted item and then directs to the system to delete it. |                                                                                                                                | The item is deleted if user does not confirm it. |                    |
| 3. The user is able to delete his account permanently.                           |                                                                                                                                |                                                  |                    |
| 4. The system respond message to the user after deleting.                        |                                                                                                                                |                                                  |                    |
| Exception:                                                                       |                                                                                                                                |                                                  |                    |
| The system does not allow the user to delete another user info.                  |                                                                                                                                |                                                  |                    |
| Post-conditions:                                                                 |                                                                                                                                |                                                  |                    |
| Step no #                                                                        | Description                                                                                                                    |                                                  |                    |
| 1.                                                                               | Account info will be successfully deleted and the database will be updated.                                                    |                                                  |                    |

*Table 14 Use case Description for Deactivate Account*

#### 3.4.14 UC:014

|                                                                  |                                                            |                  |               |
|------------------------------------------------------------------|------------------------------------------------------------|------------------|---------------|
| Use case ID:                                                     | 014                                                        | Use case Name:   | User Guidance |
| Priority                                                         | 4                                                          |                  |               |
| Actor                                                            | Owner, Finder                                              |                  |               |
| Use Case Summary:                                                | User will see a short video clip of how to use the system. |                  |               |
| Pre-conditions                                                   | Website must be opened.                                    |                  |               |
| Normal Follow of Paths                                           |                                                            | Alternative Path |               |
| 1. The user clicks on the tutorial to see how to use the system. |                                                            |                  |               |
| Exception:                                                       |                                                            |                  |               |
|                                                                  |                                                            |                  |               |
| Post-conditions:                                                 |                                                            |                  |               |
| Step no #                                                        | Description                                                |                  |               |
| 1.                                                               | The user will be able to know how to use the system.       |                  |               |

*Table 15 Use case Description for User guidance*

### 3.4.15 UC:015

|                                                                             |                                                                               |                                                                            |              |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------|
| Use case ID:                                                                | 015                                                                           | Use case Name:                                                             | View History |
| Priority                                                                    | 04                                                                            |                                                                            |              |
| Actor                                                                       | Owner, Finder                                                                 |                                                                            |              |
| Use Case Summary:                                                           | Both finder and owner will be able to see returned and received items history |                                                                            |              |
| Pre-conditions                                                              | Website must be opened in order to see the history of lost and found items.   |                                                                            |              |
| Normal Follow of Paths                                                      |                                                                               | Alternative Path                                                           |              |
| 1. The user clicks on the accounts button.                                  |                                                                               | If the user is not logged in, then he/she will be redirected to login page |              |
| 2. The user sees the returned and received items history.                   |                                                                               |                                                                            |              |
| Exception:                                                                  |                                                                               |                                                                            |              |
| User is not able to see any other users received and returned items history |                                                                               |                                                                            |              |
| Post-conditions:                                                            |                                                                               |                                                                            |              |
| Step no #                                                                   | Description                                                                   |                                                                            |              |
| 1.                                                                          | The user will successfully see his items history.                             |                                                                            |              |

*Table 16 Use case Description for View item*

### 3.4.16 UC:016

|                                                                  |                                                                                                        |                                                                                 |                    |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------|
| Use case ID:                                                     | 016                                                                                                    | Use case Name:                                                                  | Search by category |
| Priority                                                         | 04                                                                                                     |                                                                                 |                    |
| Actor                                                            | Owner, Finder                                                                                          |                                                                                 |                    |
| Use Case Summary:                                                | Both finder and owner will be able to search the desired item by selecting their desired item category |                                                                                 |                    |
| Pre-conditions                                                   | User must be logged-in in order to search the item by category                                         |                                                                                 |                    |
| Normal Follow of Paths                                           |                                                                                                        | Alternative Path                                                                |                    |
| 1. The user clicks on the search by category dropdown.           |                                                                                                        |                                                                                 |                    |
| 2. The user sees all the selected items in the selected category |                                                                                                        | If an item category that is not present is error then no result would be found. |                    |
| Exception:                                                       |                                                                                                        |                                                                                 |                    |
| User is not able to see any other item from other categories.    |                                                                                                        |                                                                                 |                    |
| Post-conditions:                                                 |                                                                                                        |                                                                                 |                    |
| Step no #                                                        | Description                                                                                            |                                                                                 |                    |
| 1.                                                               | The user will successfully see all the items in that category.                                         |                                                                                 |                    |

*Table 17 Use case description for search by category*

### 3.4.17 UC:017

|                                                         |                                                                                           |                                                                                            |                 |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------|
| Use case ID:                                            | 017                                                                                       | Use case Name:                                                                             | View statistics |
| Priority                                                | 04                                                                                        |                                                                                            |                 |
| Actor                                                   | Admin                                                                                     |                                                                                            |                 |
| Use Case Summary:                                       | The Admin is able to view the statistics of all the items that are registered or returned |                                                                                            |                 |
| Pre-conditions                                          | Admin must be logged-in in order to view the statistics                                   |                                                                                            |                 |
| Normal Follow of Paths                                  |                                                                                           | Alternative Path                                                                           |                 |
| 1. The admin logs into the system with valid inputs     |                                                                                           | If the input information provided is not correct , the admin is be able to view statistics |                 |
| 2. The admin can view the statistics on admin home page |                                                                                           |                                                                                            |                 |
| Exception:                                              |                                                                                           |                                                                                            |                 |
|                                                         |                                                                                           |                                                                                            |                 |
| Post-conditions:                                        |                                                                                           |                                                                                            |                 |
| Step no #                                               | Description                                                                               |                                                                                            |                 |
| 1.                                                      | The admin will successfully view the statistics.                                          |                                                                                            |                 |

*Table 18 use case description for view statistics.*

### 3.5 USE CASE DIAGRAM

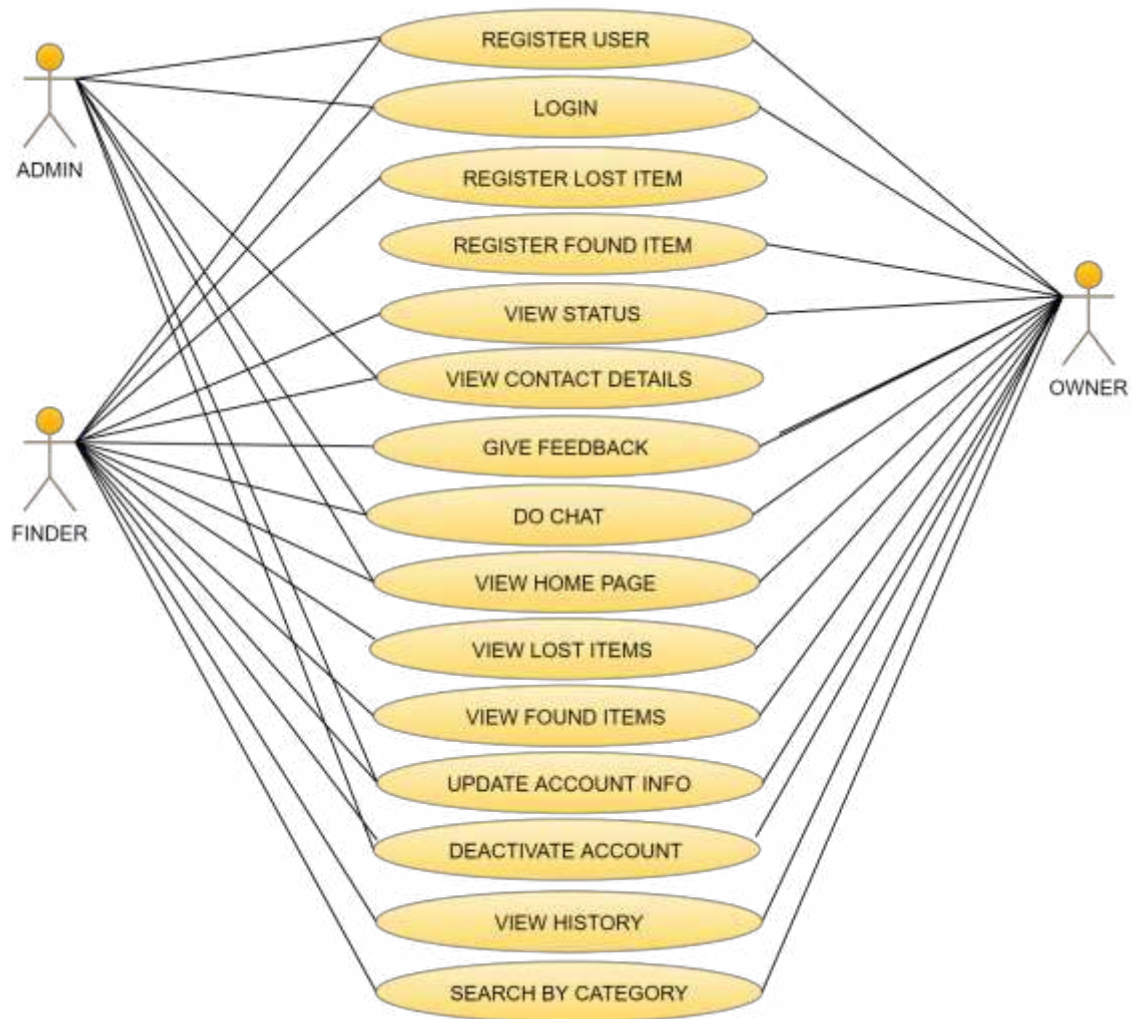


Figure 1 Use case Diagram for Whole Project

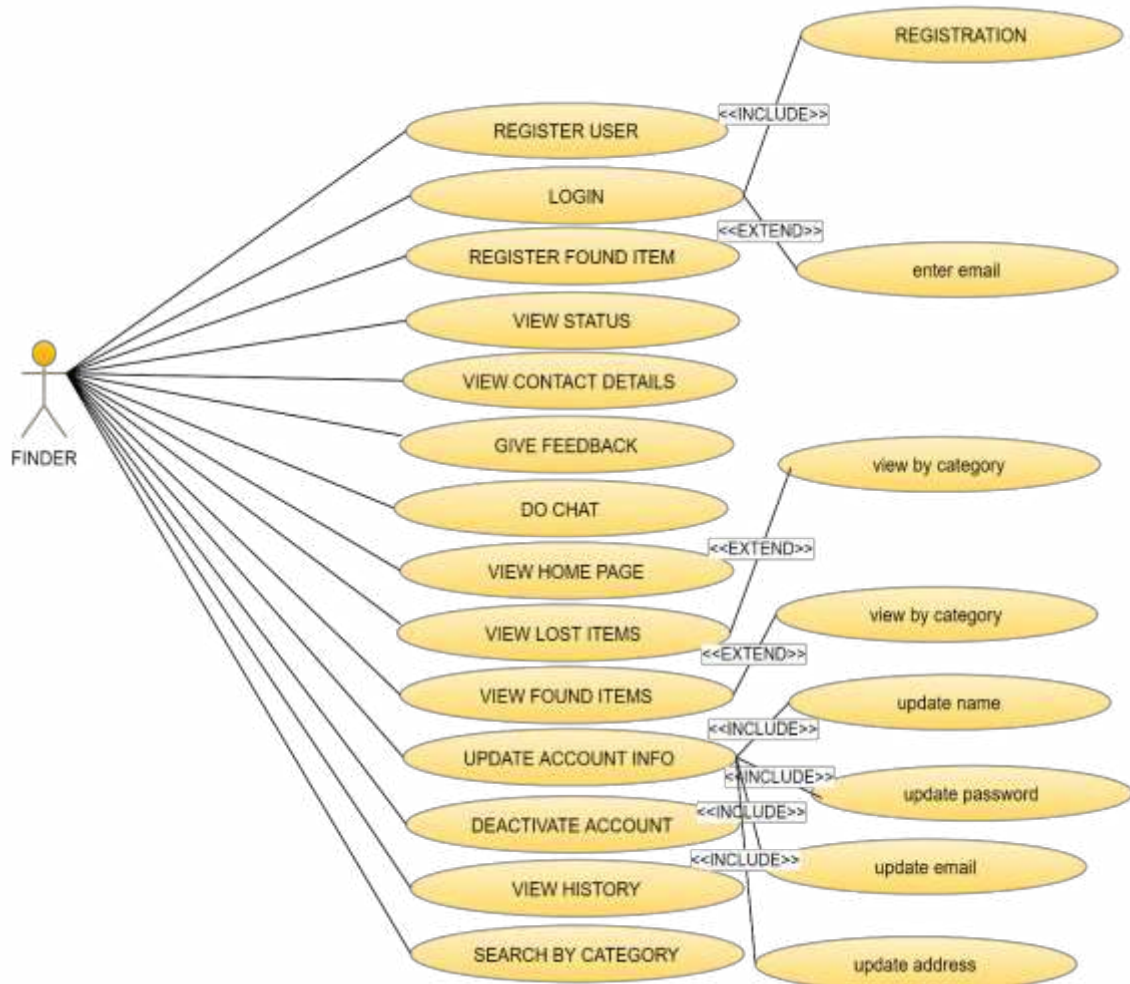


Figure 2 Use case Diagram for Finder

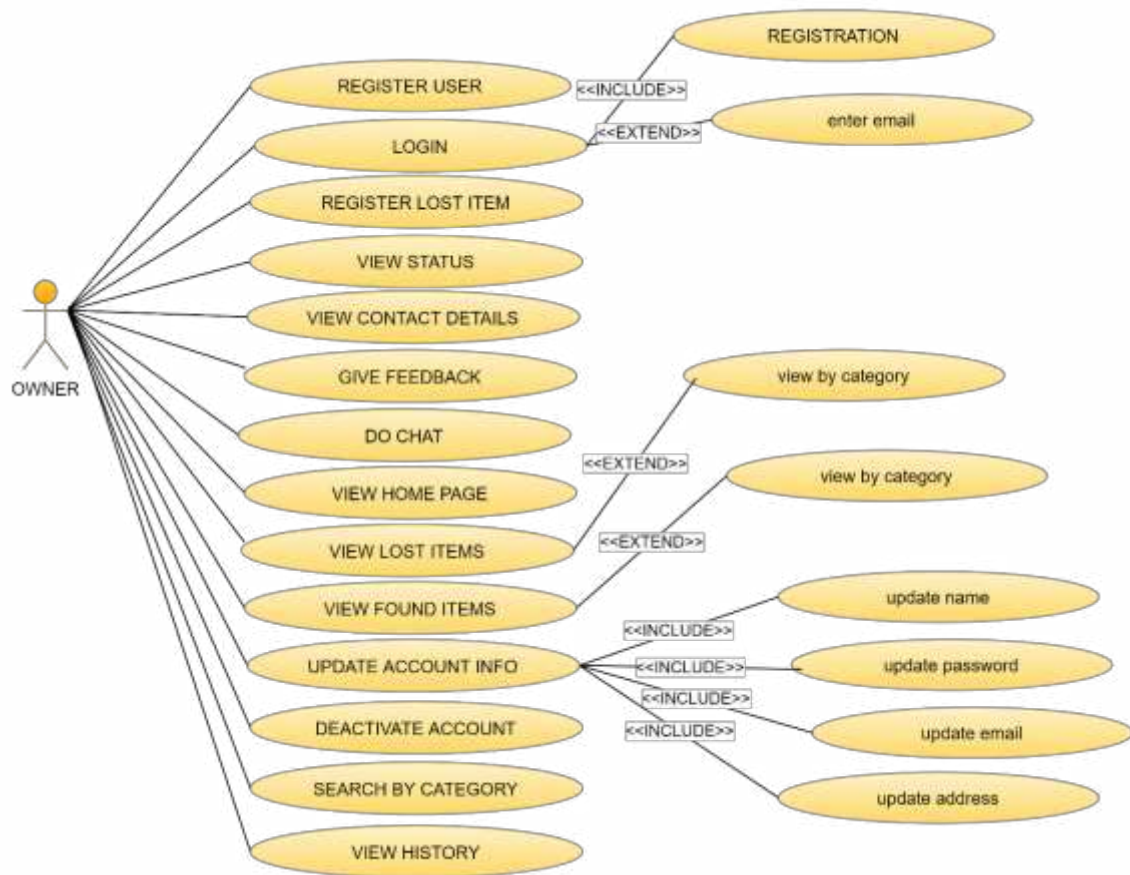


Figure 3 Use case Diagram for Owner

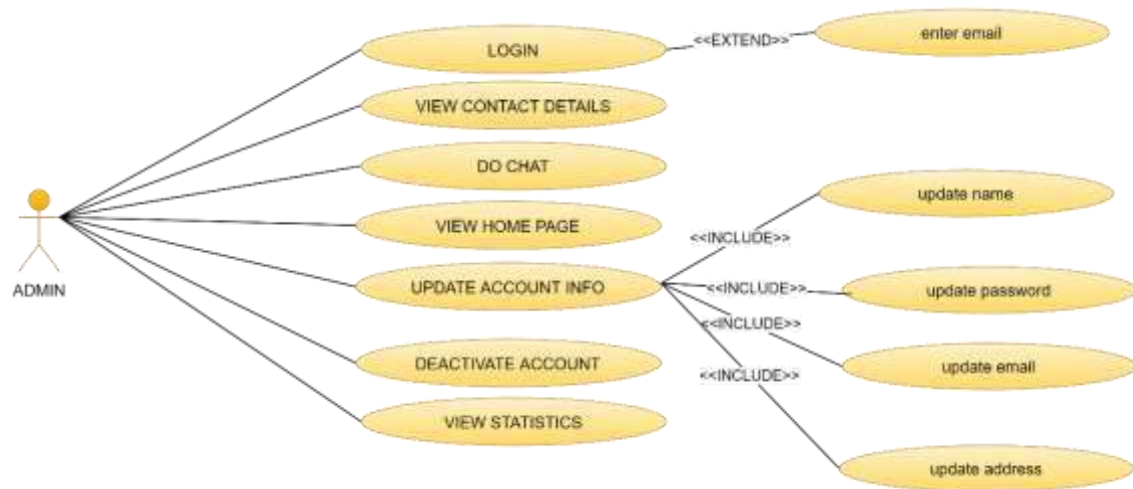
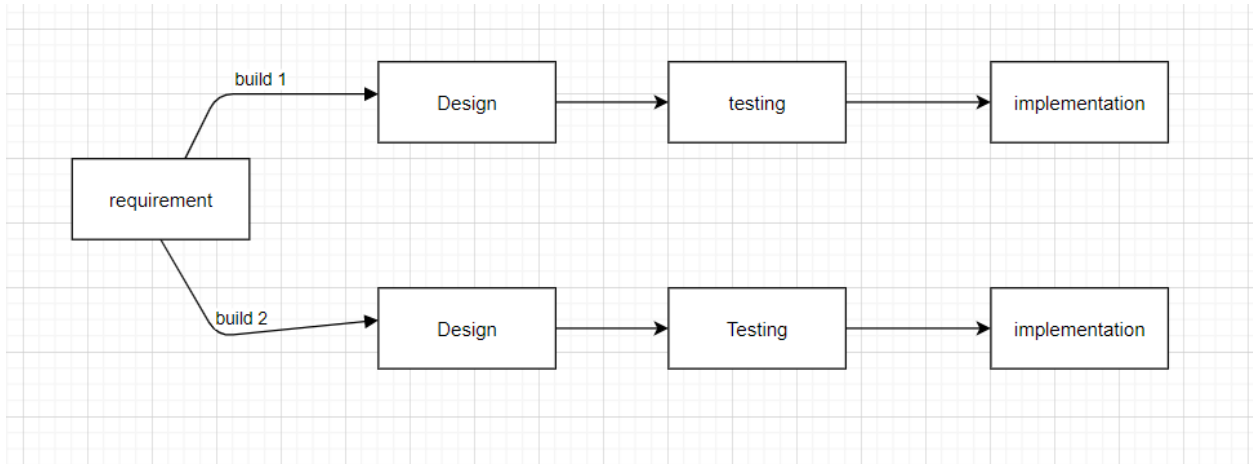


Figure 4 Use case Diagram for Admin

### 3.6 SOFTWARE DEVELOPMENT LIFECYCLE (SDLC)

We are using iterative model to develop our software application. Iterative model is convenient for using in the development of applications where step by step development is conducted. As our work is distributed in three group members so iterative model is used so we can divide our tasks in chunks and work on them step by step for easy and proper implementation. Iterative model also has the flexibility of accommodating any changes if they are made in future. The stages in iterative method are:



*Figure 5 system development lifecycle*

## 4 CHAPTER 03 SYSTEM DESIGN

### 4.1 WORK BREAKDOWN STRUCTURE:

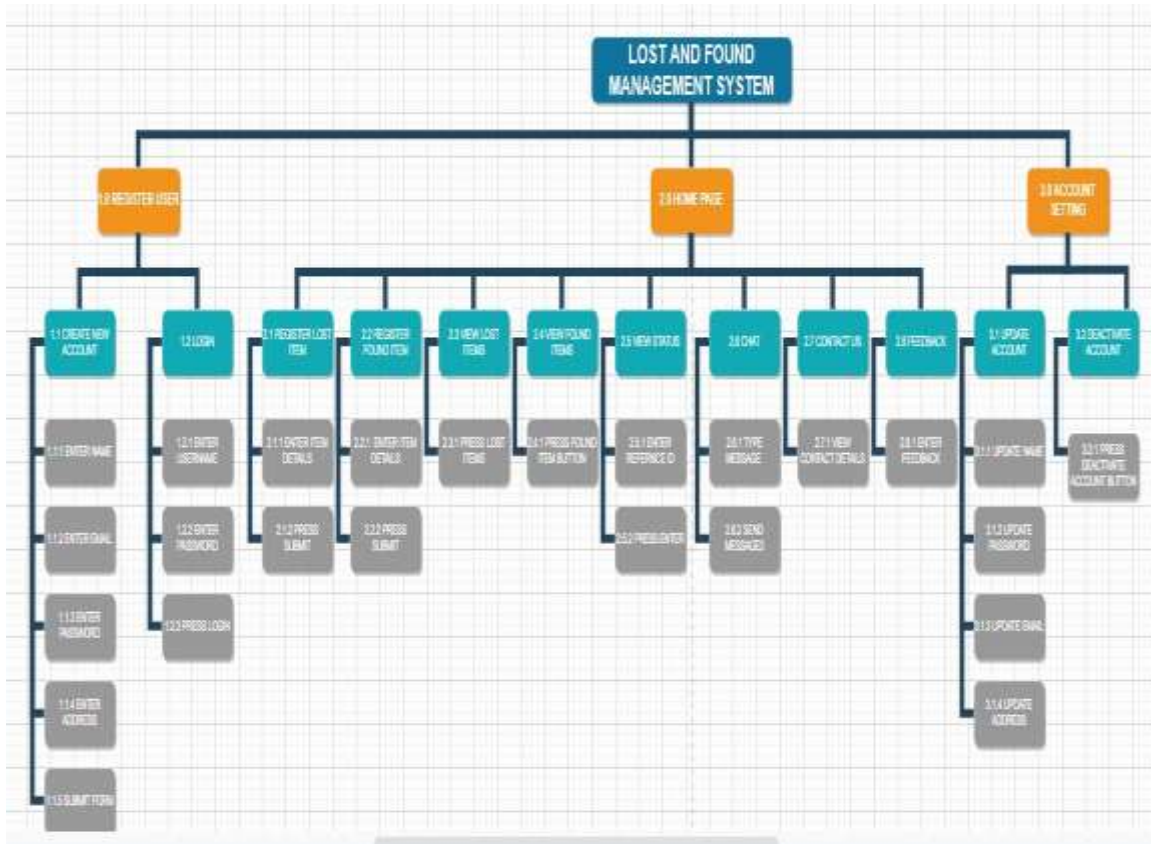


Figure 6 Work breakdown structure(WBS)

## 4.2 ACTIVITY DIAGRAM

### 4.2.1 CREATE ACCOUNT

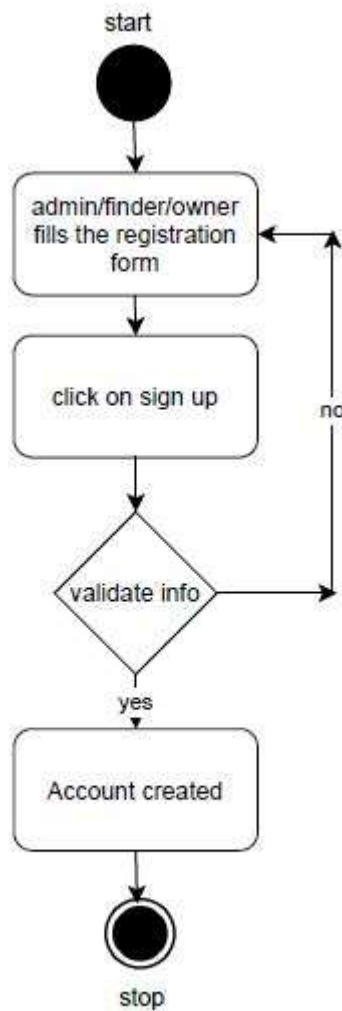


Figure 7 Activity Diagram for Create Account

## 4.2.2 LOGIN

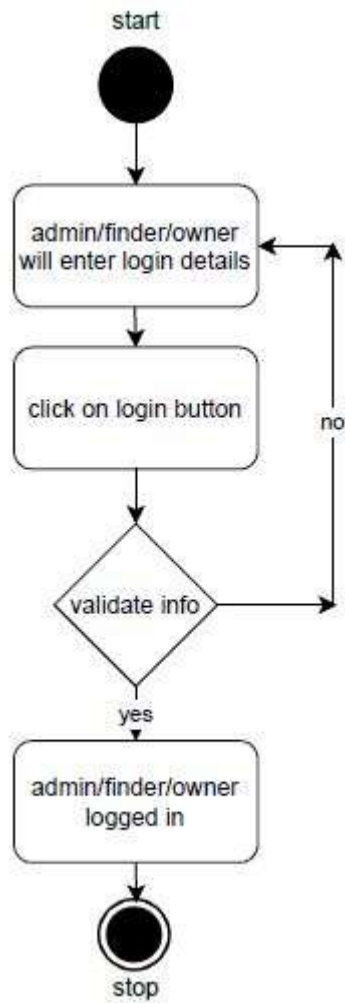


Figure 8 Activity Diagram for login

### 4.2.3 REPORT LOST ITEM

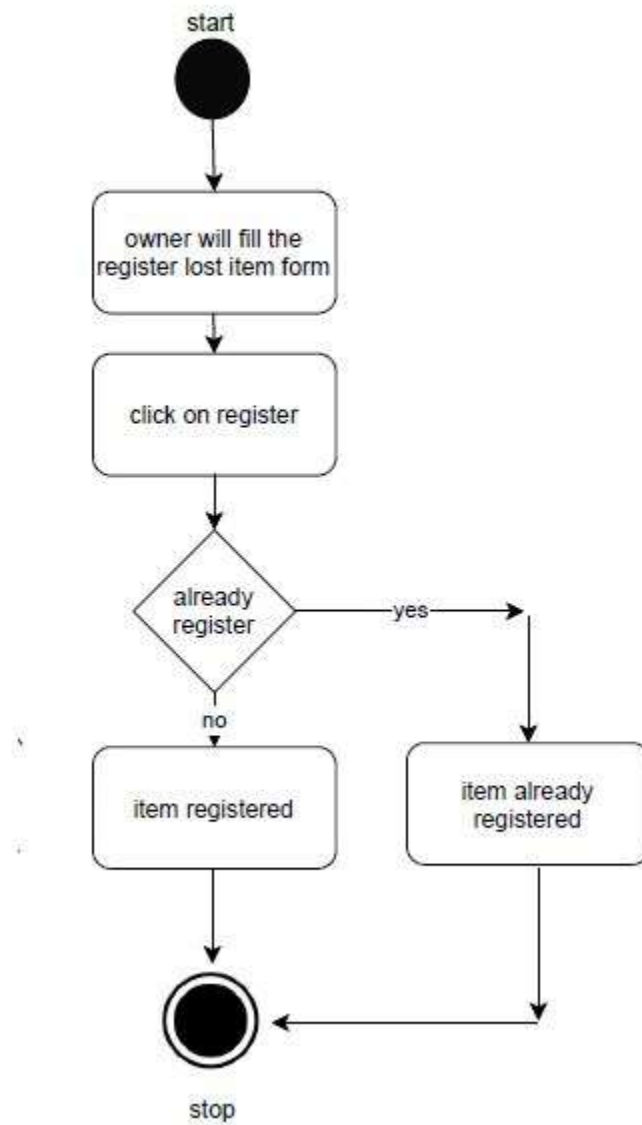


Figure 9 Activity Diagram for Register lost item

#### 4.2.4 REPORT FOUND ITEM

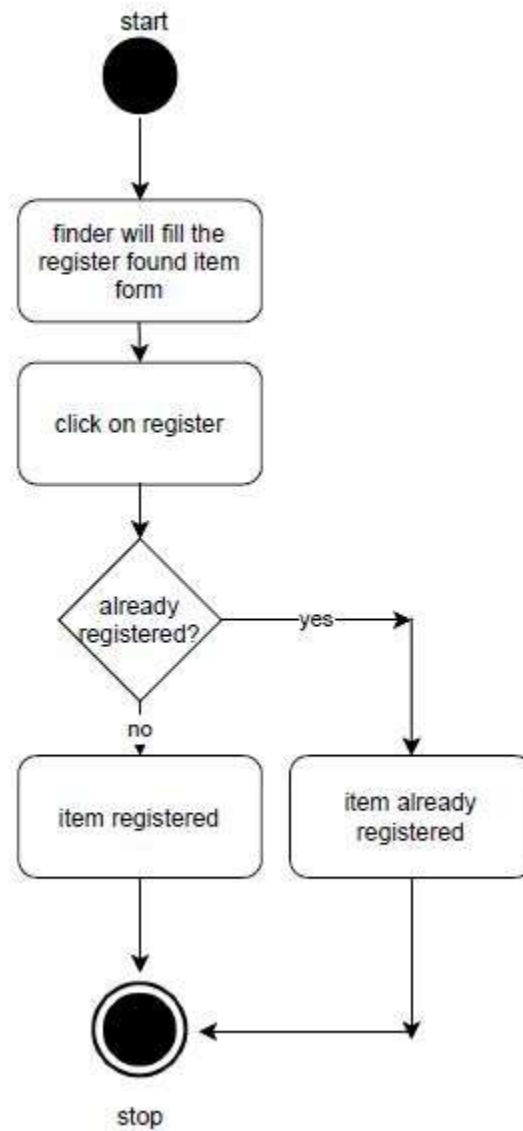


Figure 10 Activity Diagram for Register found item

#### 4.2.5 VIEW FOUND ITEMS



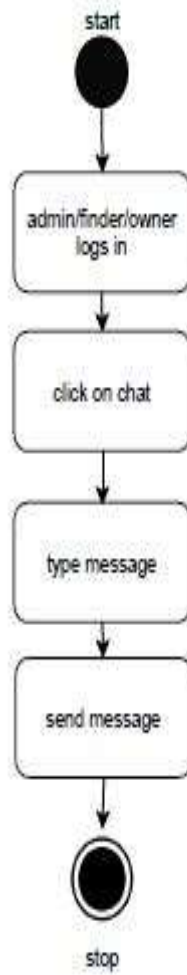
Figure 11 Activity diagram for view found item

#### 4.2.6 VIEW LOST ITEMS



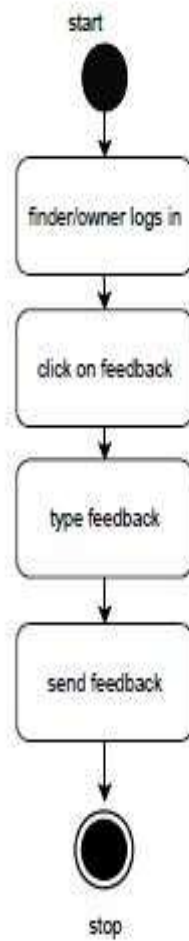
*Figure 12 Activity diagram for view lost items*

#### 4.2.7 CHAT



*Figure 13 Activity diagram for chat*

#### 4.2.8 FEEDBACK



*Figure 14 Activity diagram for feedback*

#### 4.2.9 UPDATE ACCOUNT INFO

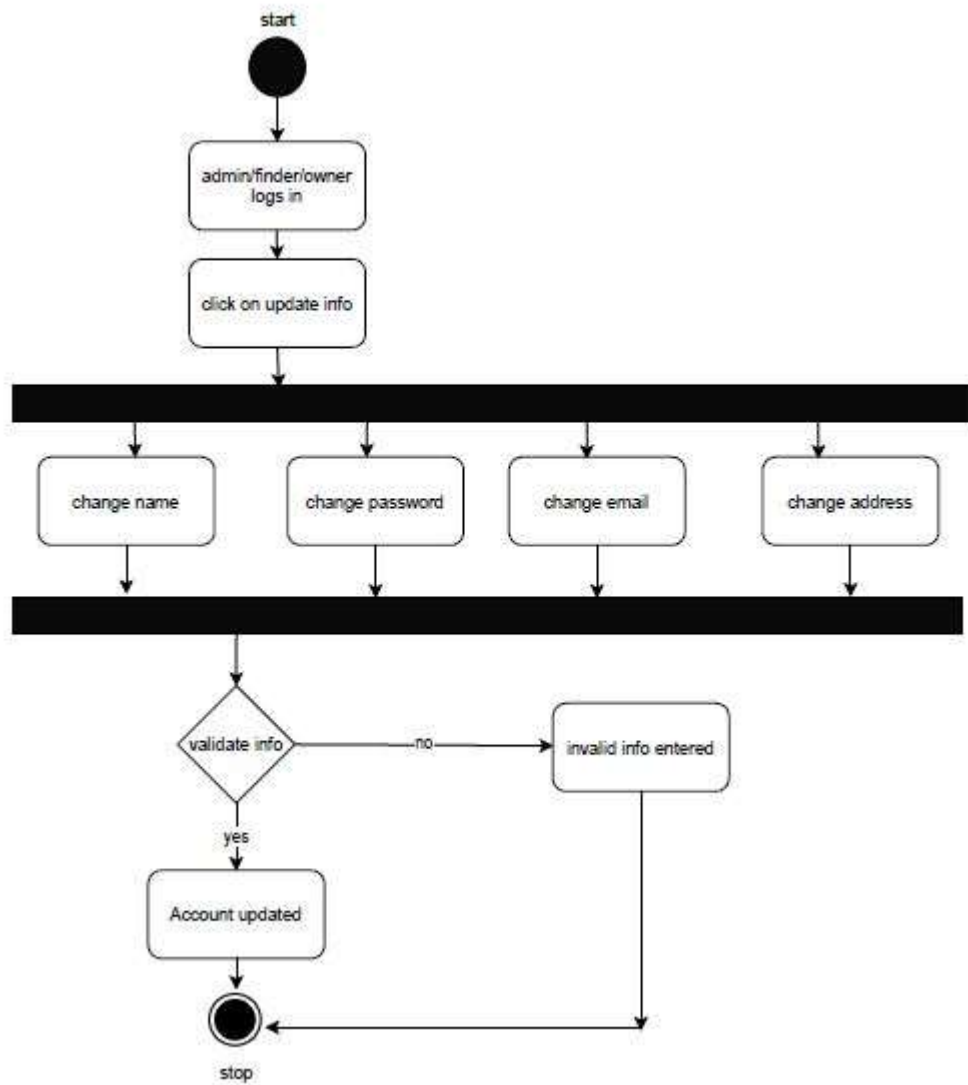


Figure 15 Activity diagram for update account info

#### 4.2.10 DEACTIVATE ACCOUNT



*Figure 16 Activity diagram for Deactivate account*

#### 4.2.11 VIEW STATUS

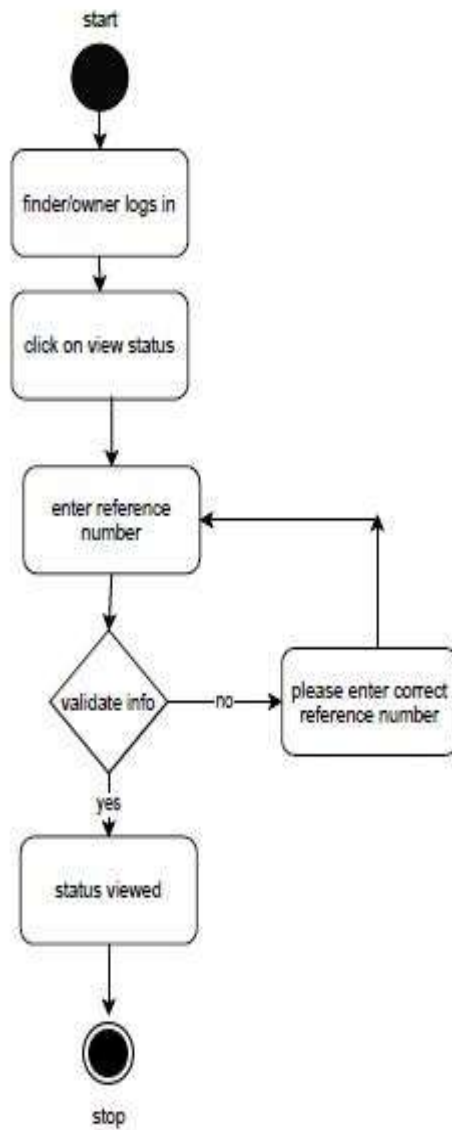


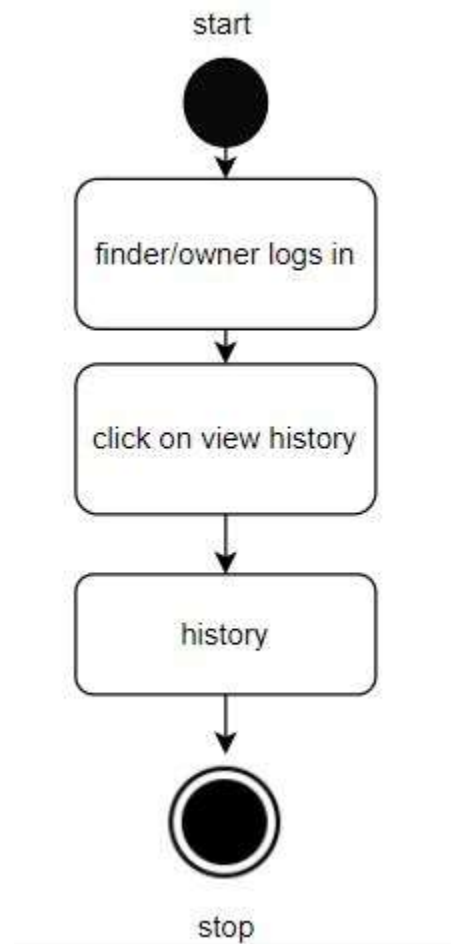
Figure 17 Activity diagram for View status

#### 4.2.12 CONTACT DETAILS



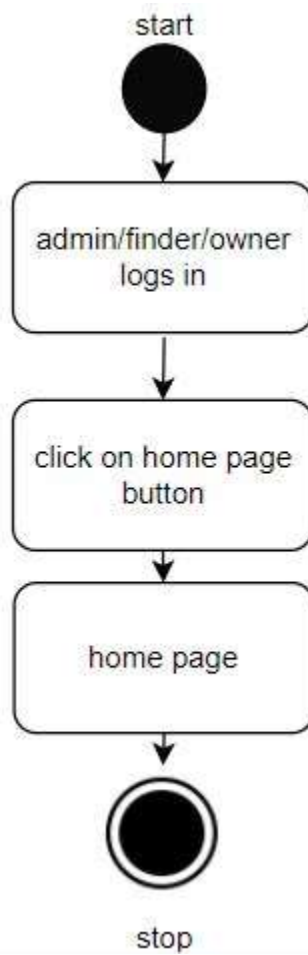
*Figure 18 Activity diagram for Contact details*

#### 4.2.13 View history



*Figure 19 view history*

#### 4.2.14 View home page



*Figure 20 view homepage*

#### 4.2.15 Search by category

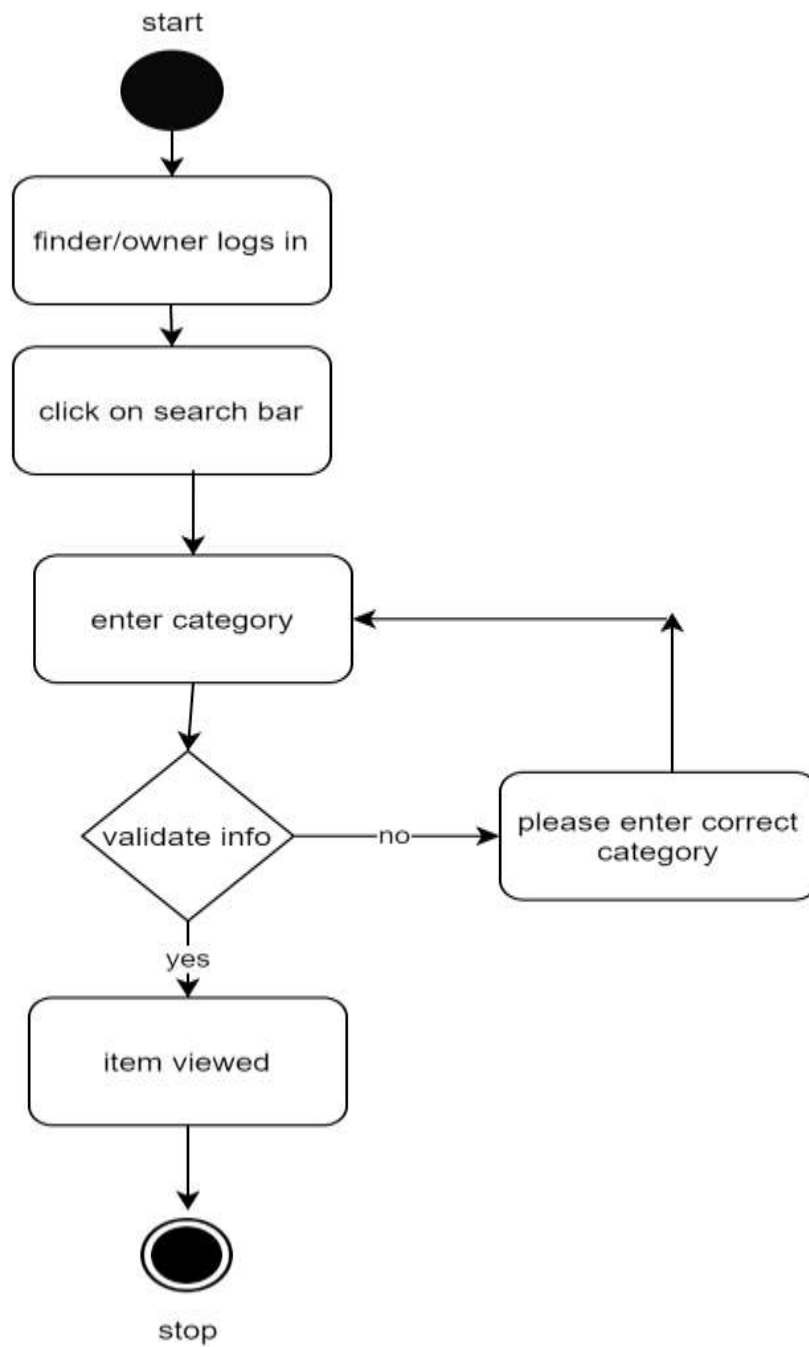
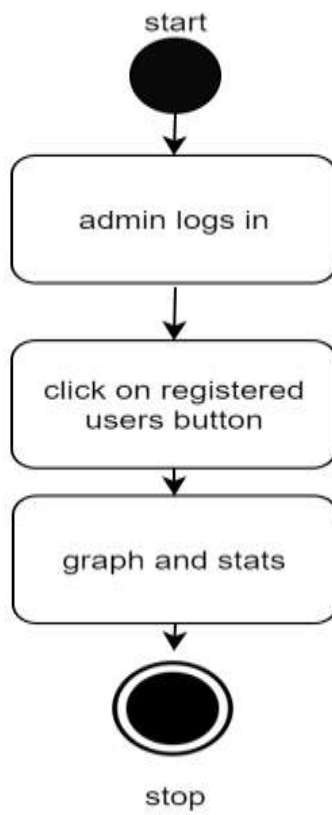


Figure 21search by category

#### 4.2.16 View statistics



*Figure 22view statistics*

#### 4.2.17 User guidance

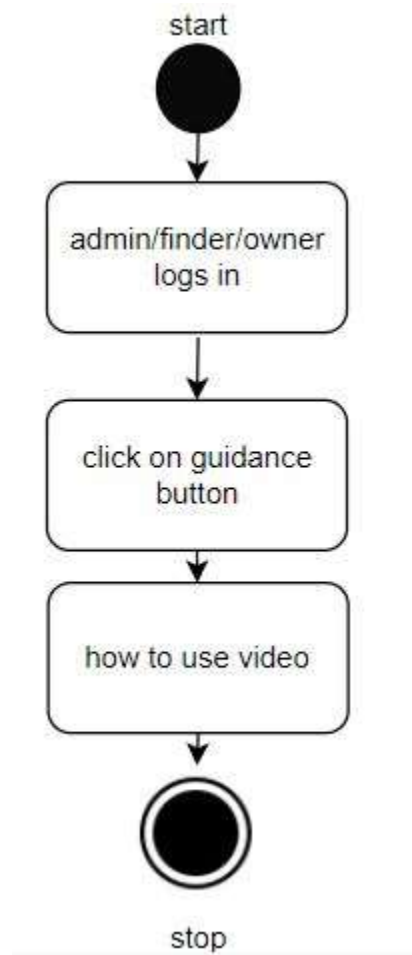


Figure 23user guidance

## 4.3 SEQUENCE DIAGRAM

### 4.3.1 Register new user:

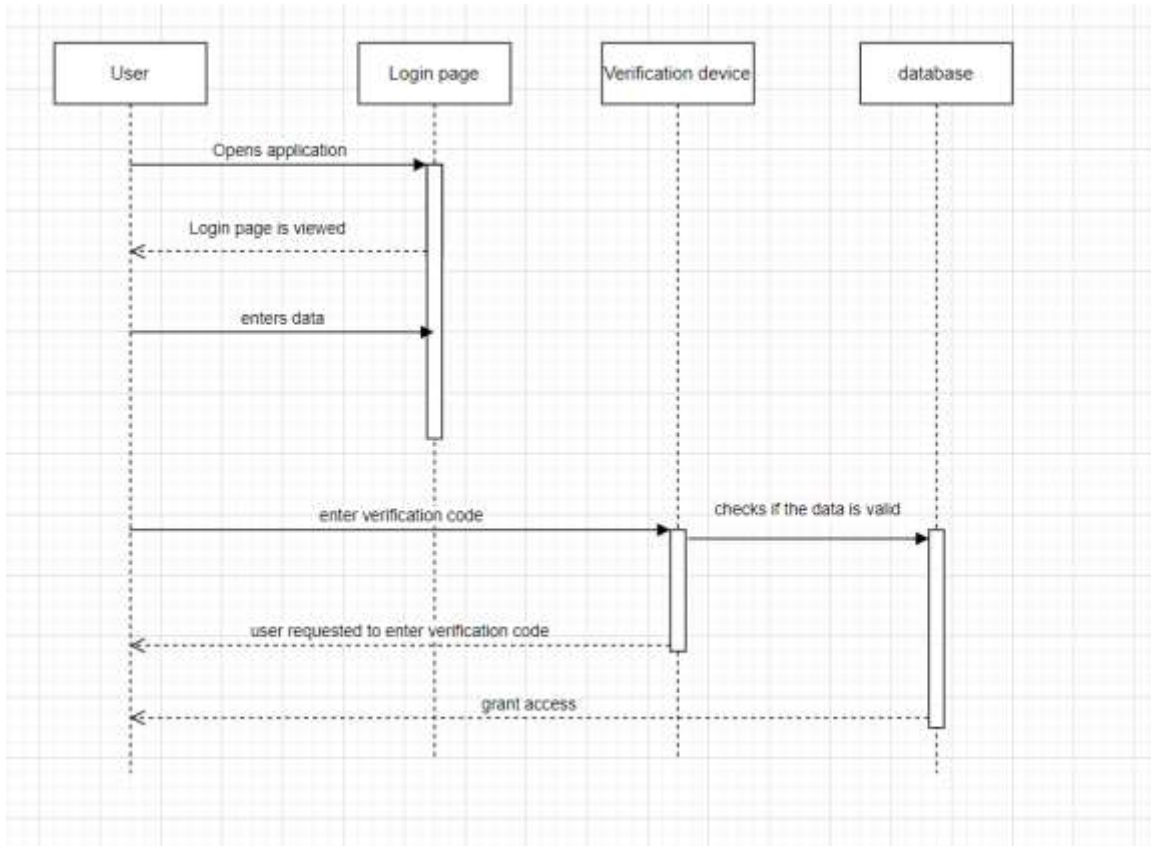


Figure 24 Sequence diagram for Register new user

### 4.3.2 Login:

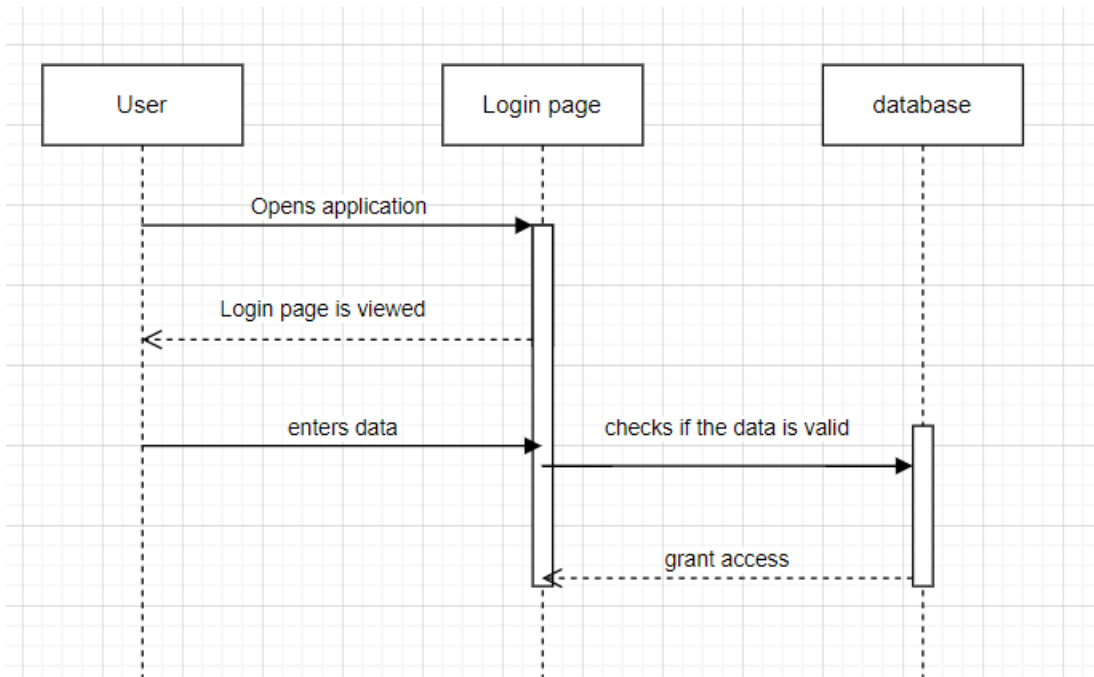


Figure 25 Sequence diagram for login

### 4.3.3 Report lost item:

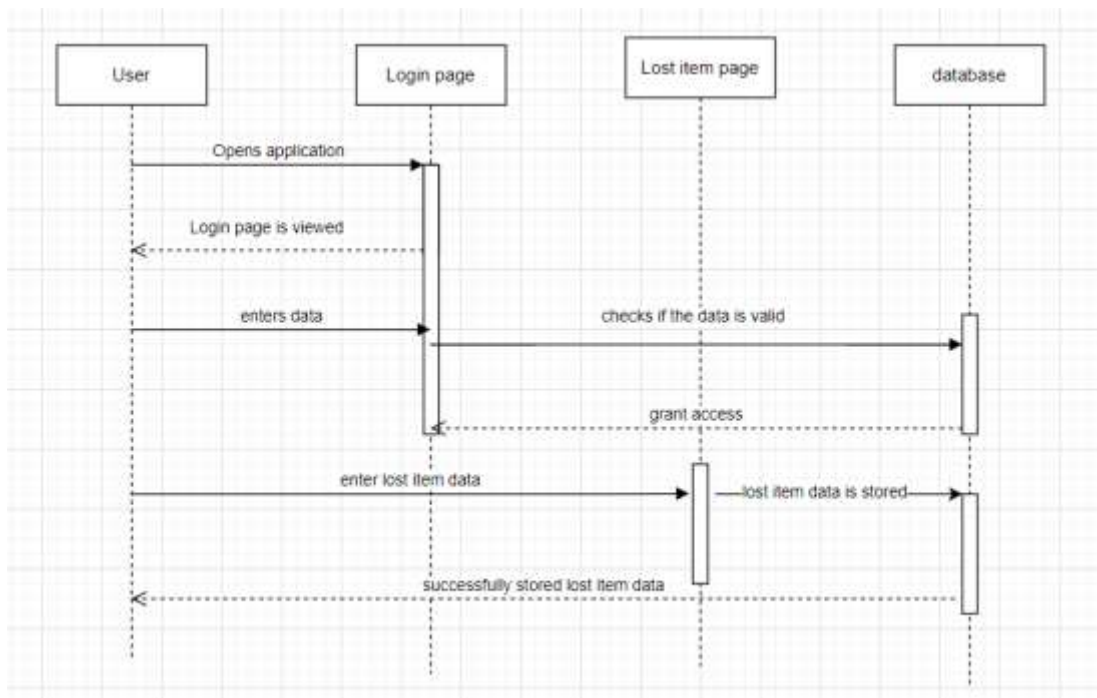


Figure 26 Sequence diagram for Report lost item

#### 4.3.4 Report found item:

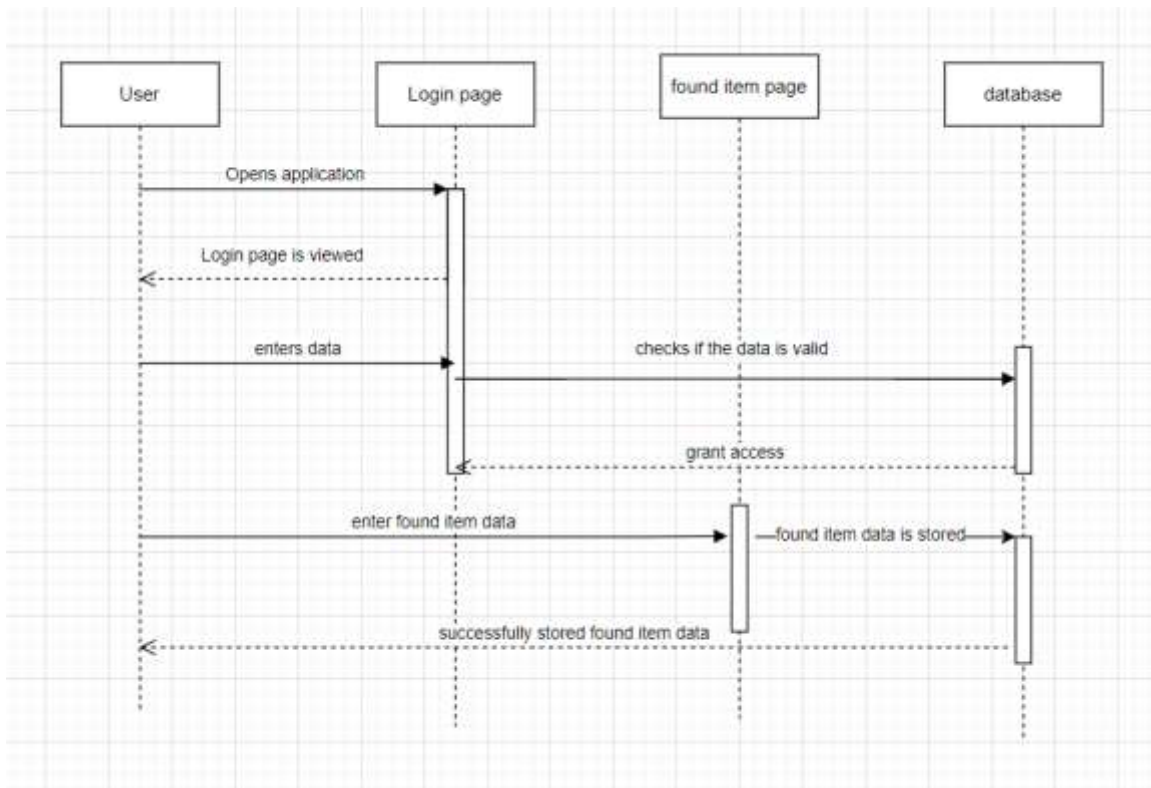


Figure 27 Sequence diagram for Report found item

#### 4.3.5 View status:

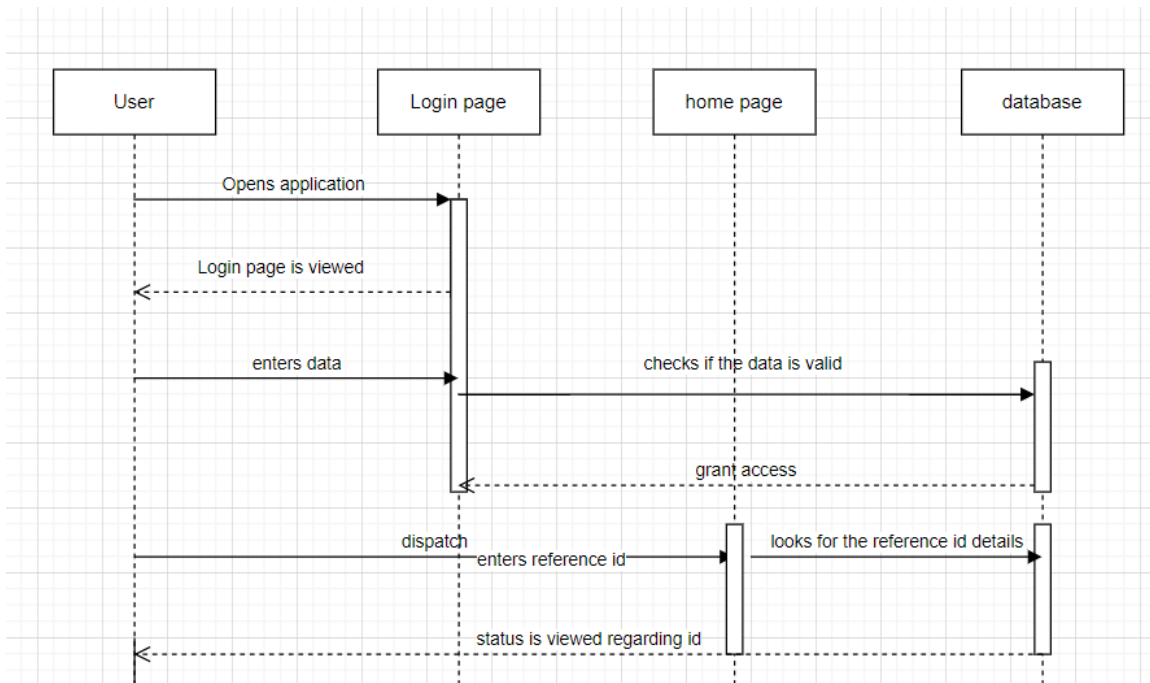


Figure 28 Sequence diagram for view status

#### 4.3.6 View Contact details:

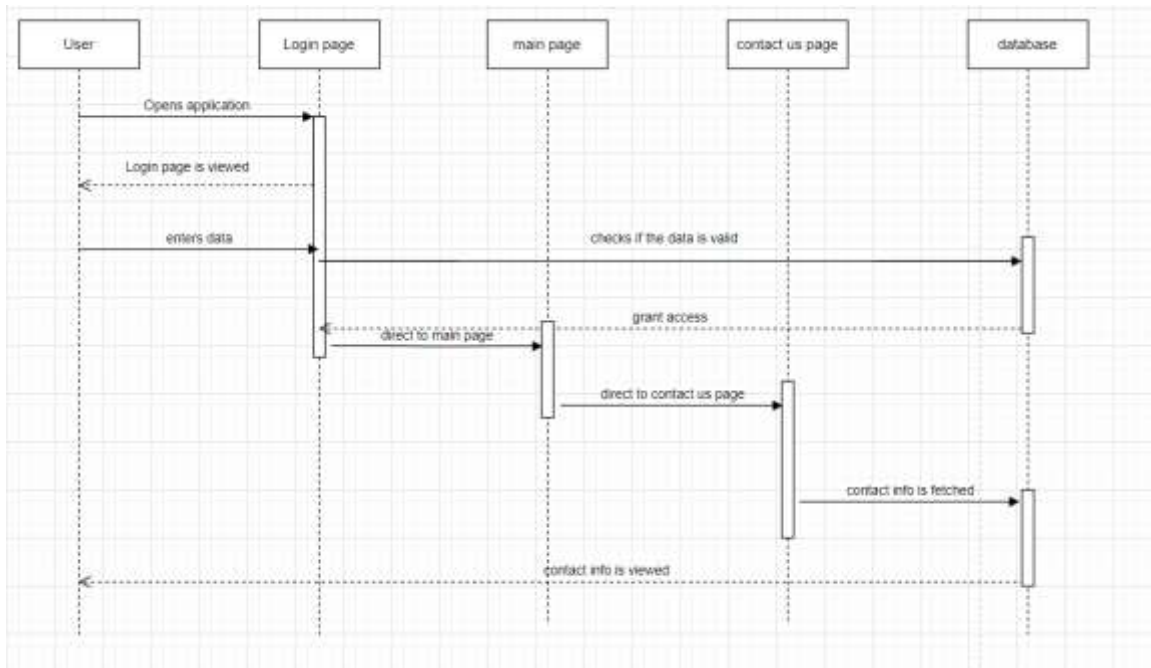


Figure 29 Sequence Diagram for view Contact details

### 4.3.7 Feedback:

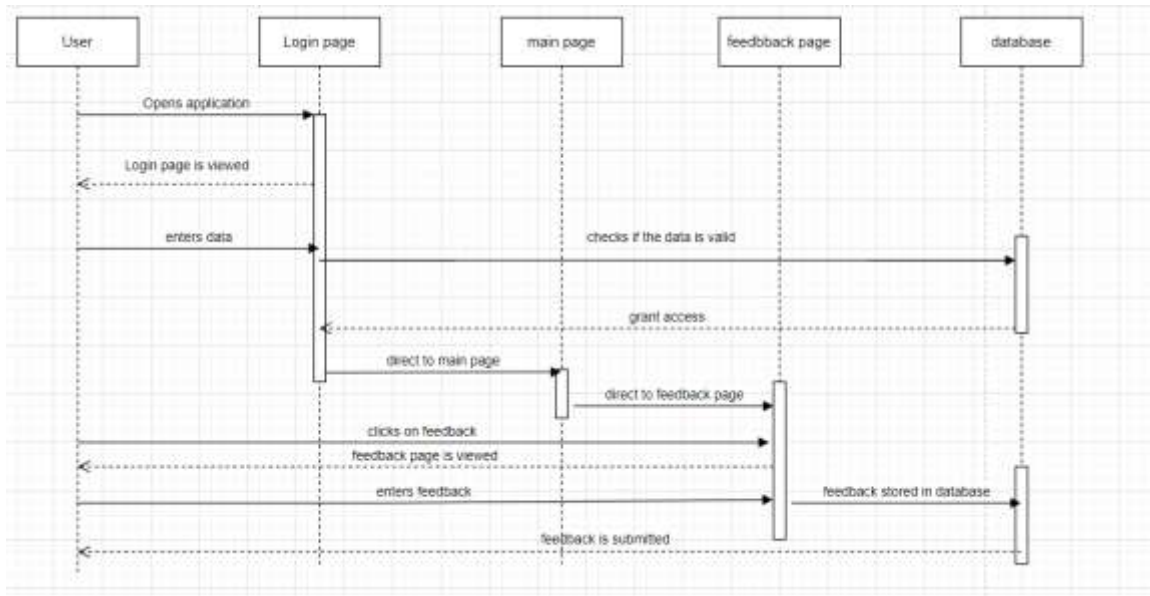


Figure 30 Sequence diagram for Feedback

#### 4.3.8 Chat:

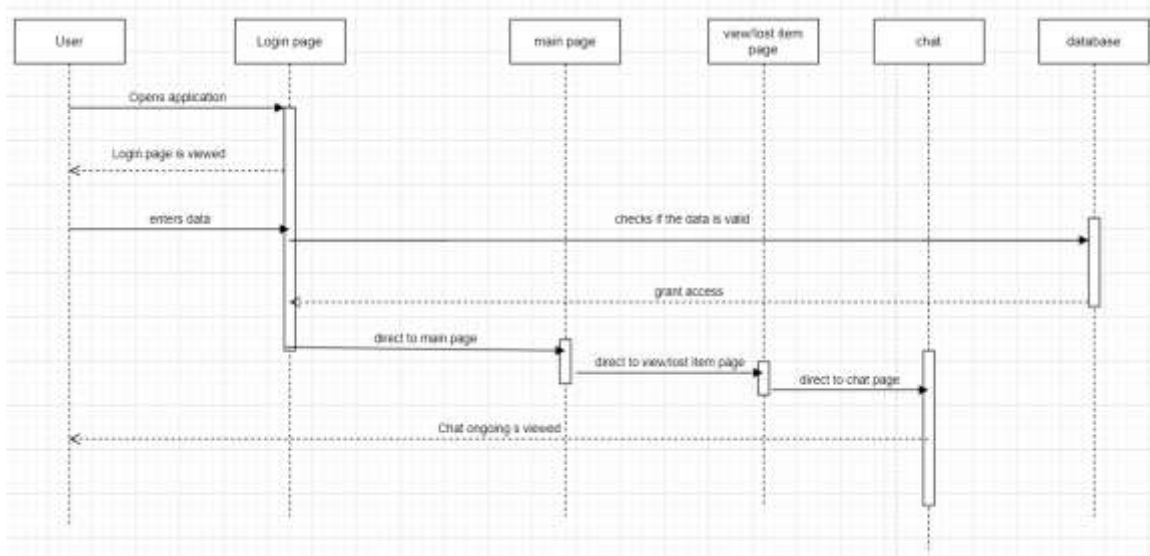


Figure 31 Sequence diagram for chat

#### 4.3.9 View homepage:

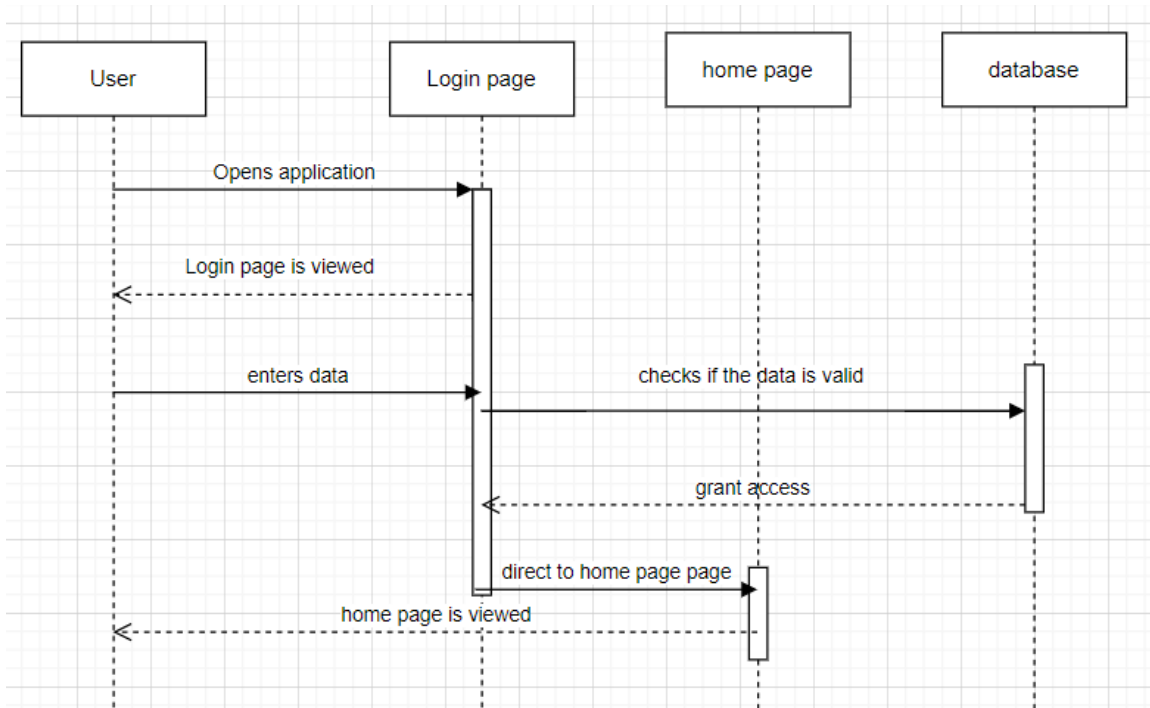


Figure 32 Sequence diagram for view homepage

#### 4.3.10 View lost items:

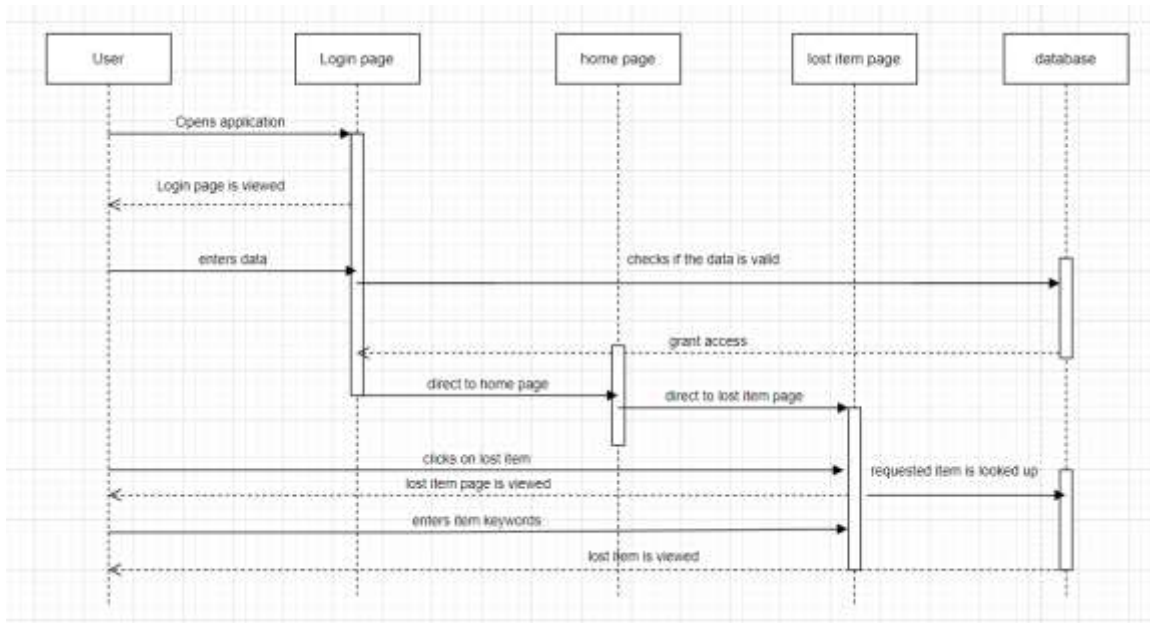


Figure 33 Sequence diagram for view lost items

#### 4.3.11 View found items:

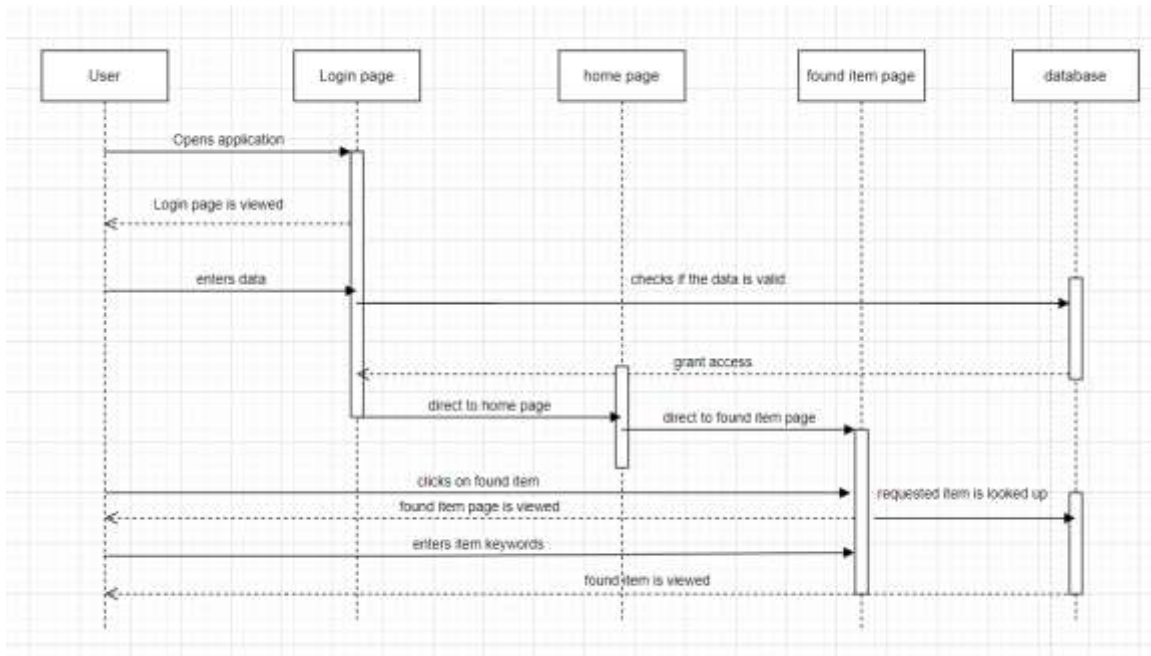


Figure 34 Sequence diagrams for view found items

#### 4.3.12 Update account info:

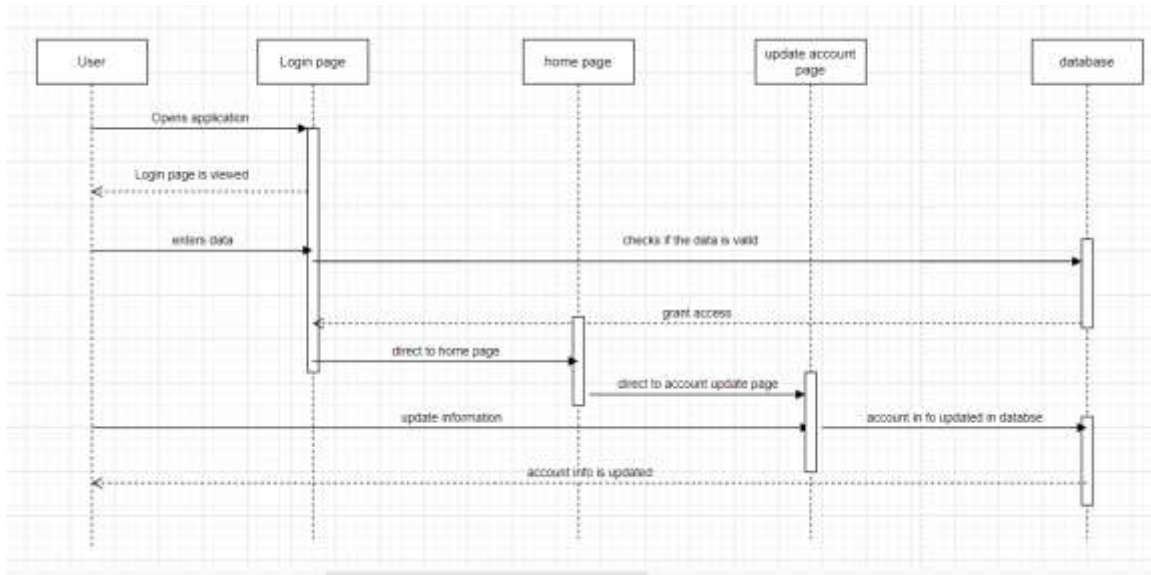


Figure 35 Sequence diagram for update account info

#### 4.3.13 Deactivate account:

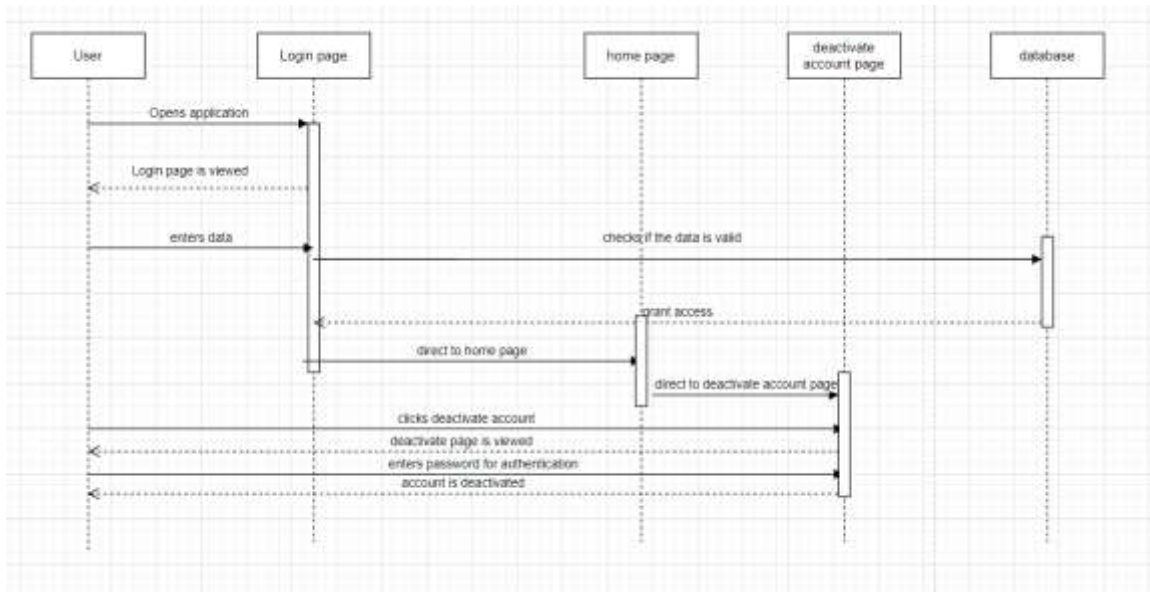


Figure 36 Sequence diagram for Deactivate account

#### 4.3.14 User guidance:

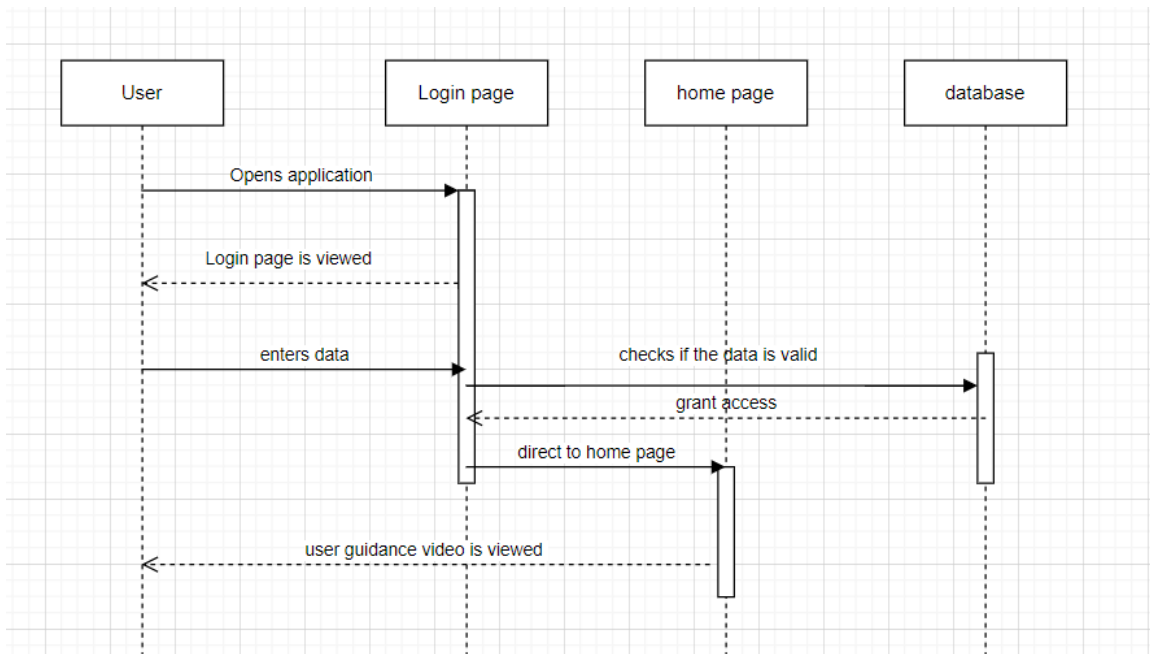


Figure 37 Sequence diagram for user guidance

#### 4.3.15 View history:

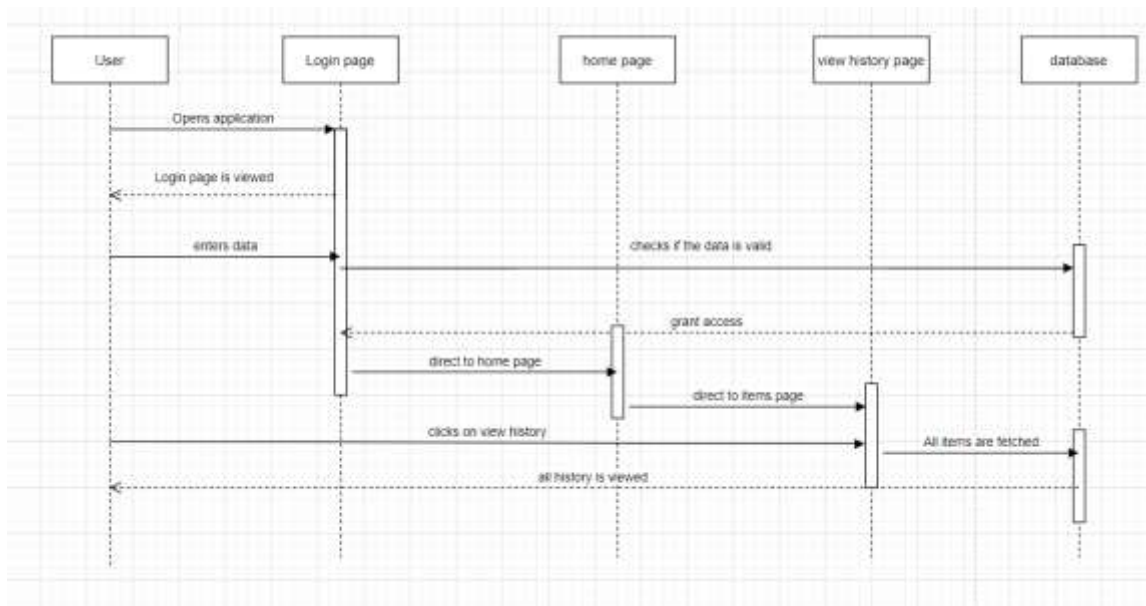


Figure 38 Sequence diagram for view history

#### 4.3.16 Search by Category

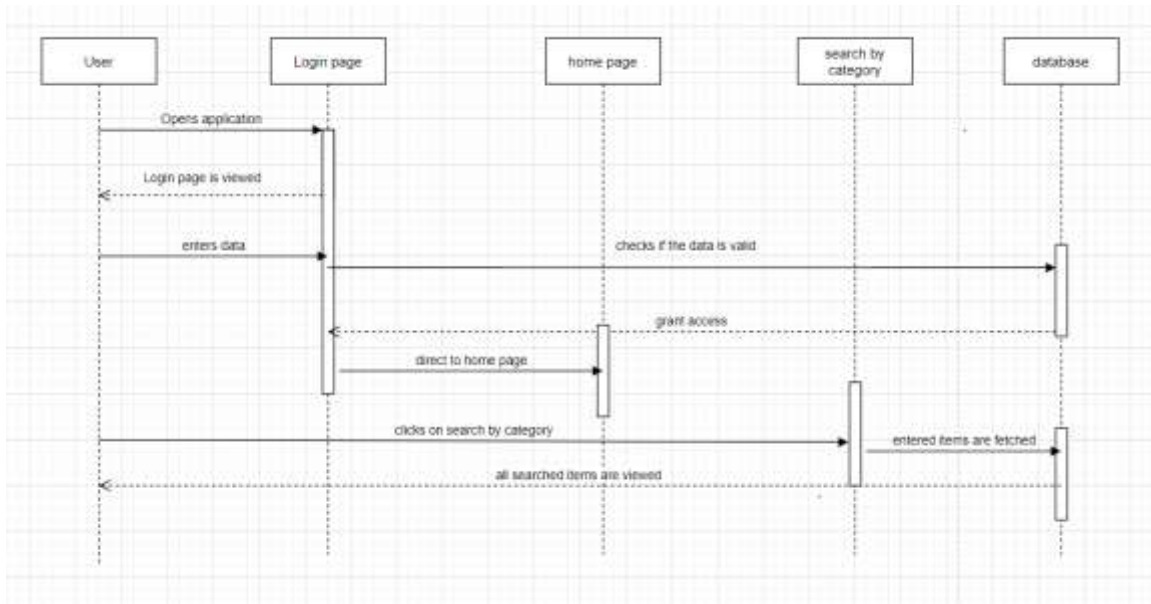


Figure 39 sequence diagram of search by category

### 4.3.17 View Statistics

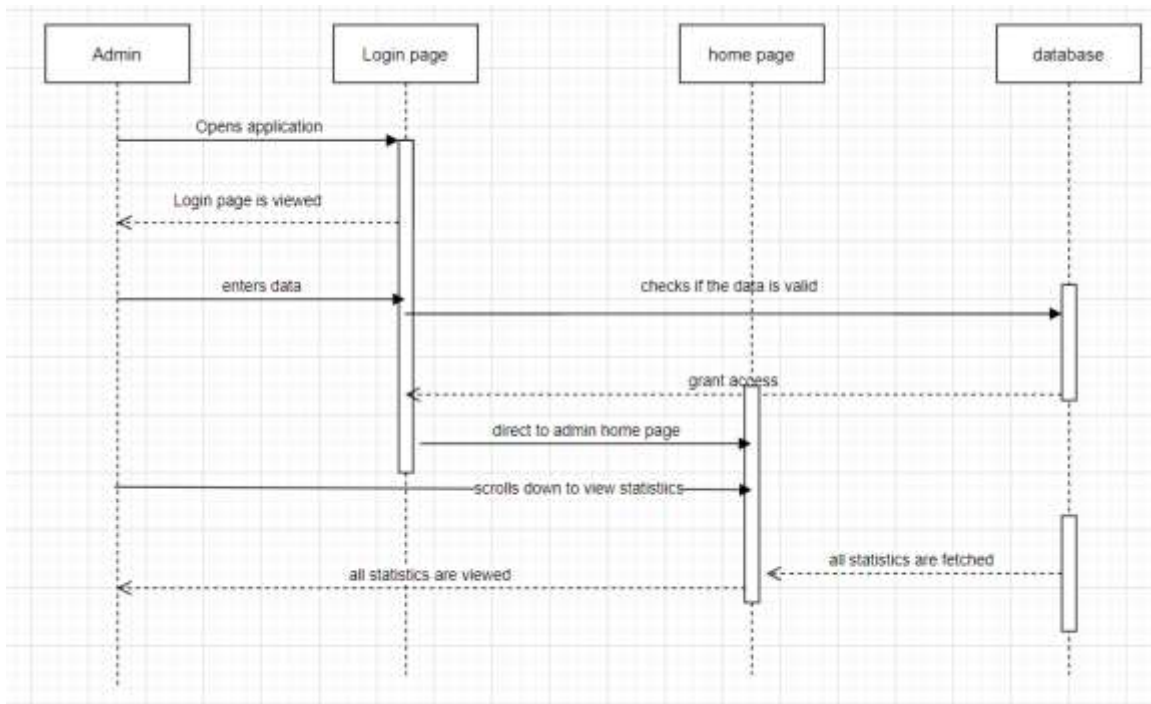


Figure 40 sequence diagram of view statistics

#### 4.4 SOFTWARE ARCHITECTURE

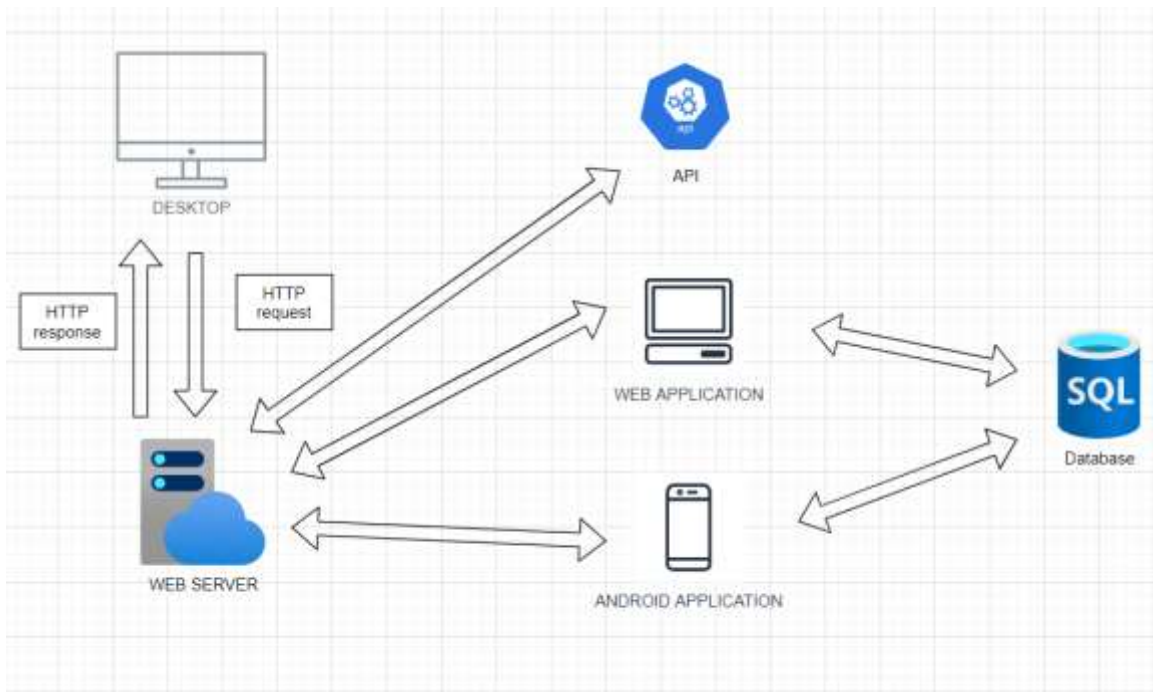


Figure 41 Software Architecture

## 4.5 CLASS DIAGRAM

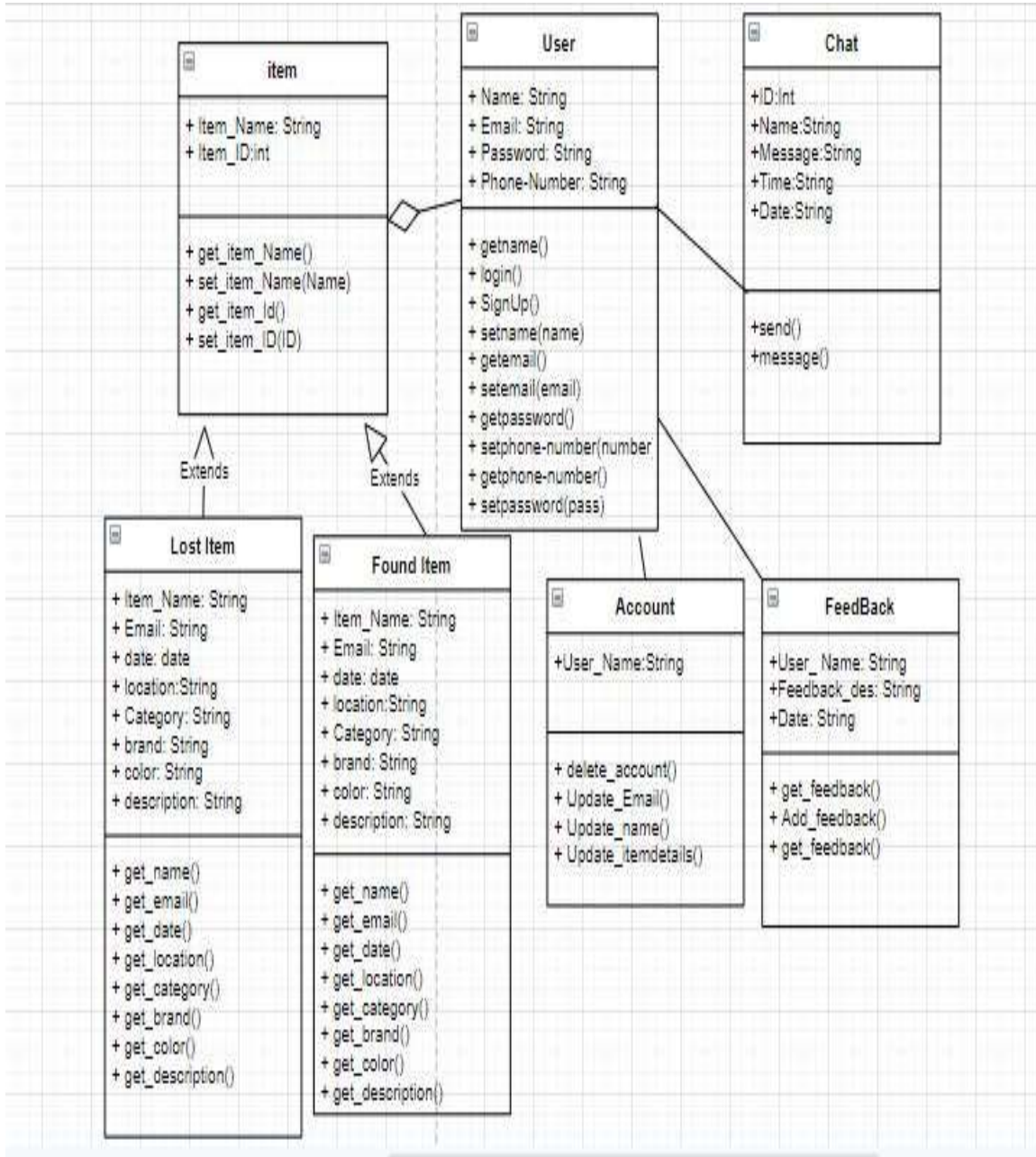


Figure 42 Class diagram

## 4.6 ENTITY RELATIONSHIP DIAGRAM

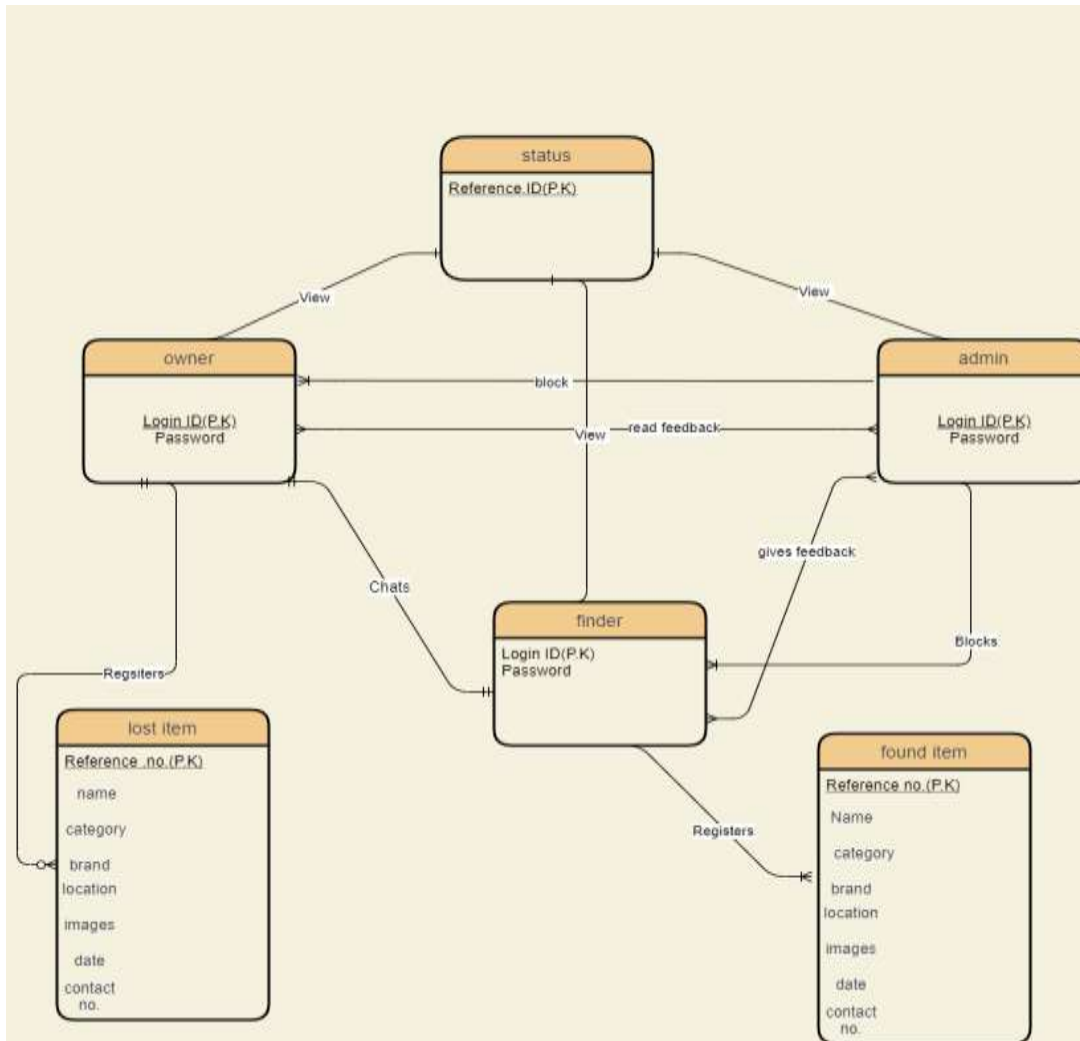


Figure 43Entity relationship diagram

## 4.7 COLLABORATION DIAGRAM

### 4.7.1 login

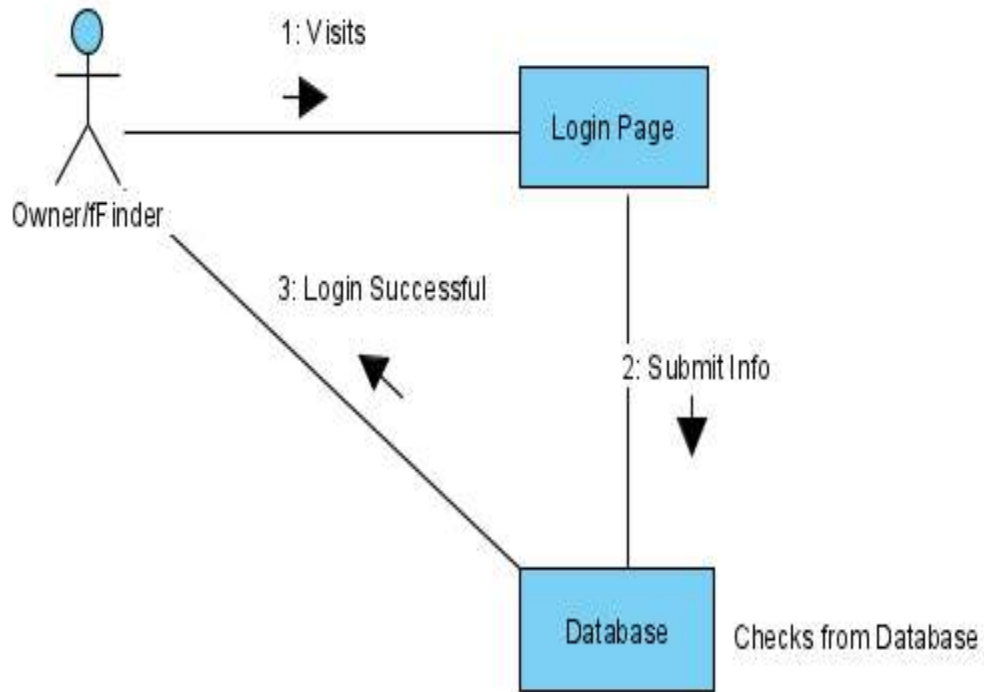


Figure 44 Collaboration diagram for login

#### 4.7.2 REGISTER

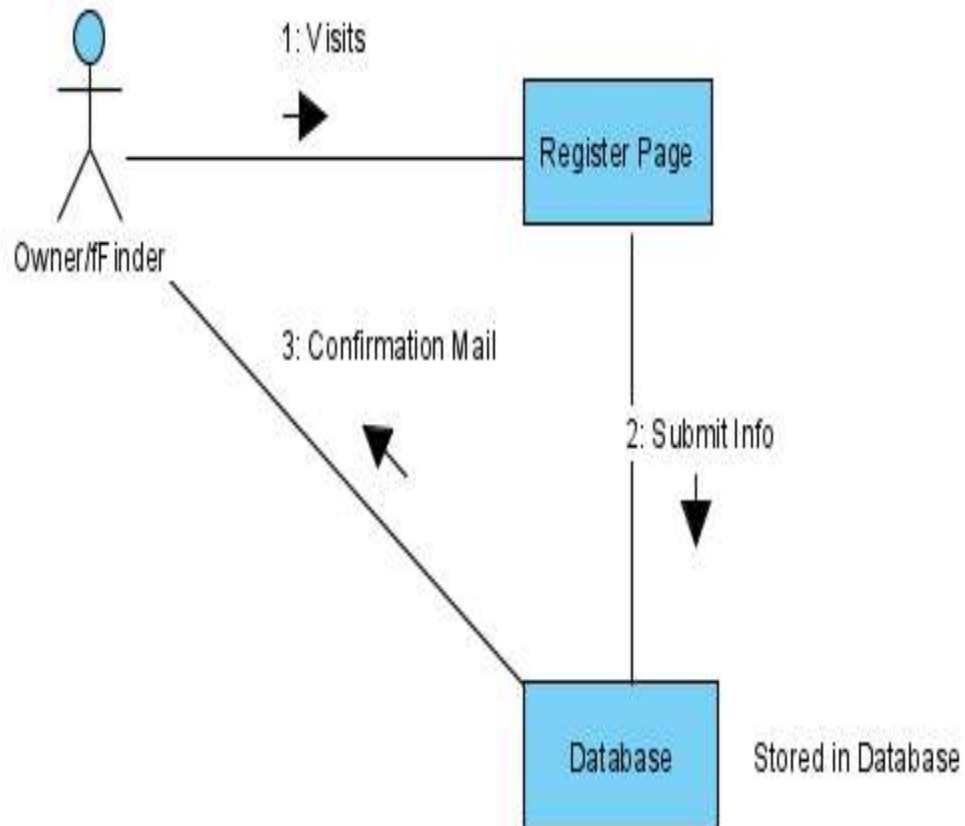


Figure 45 Collaboration diagram for registration

### 4.7.3 REPORT FOUND ITEM

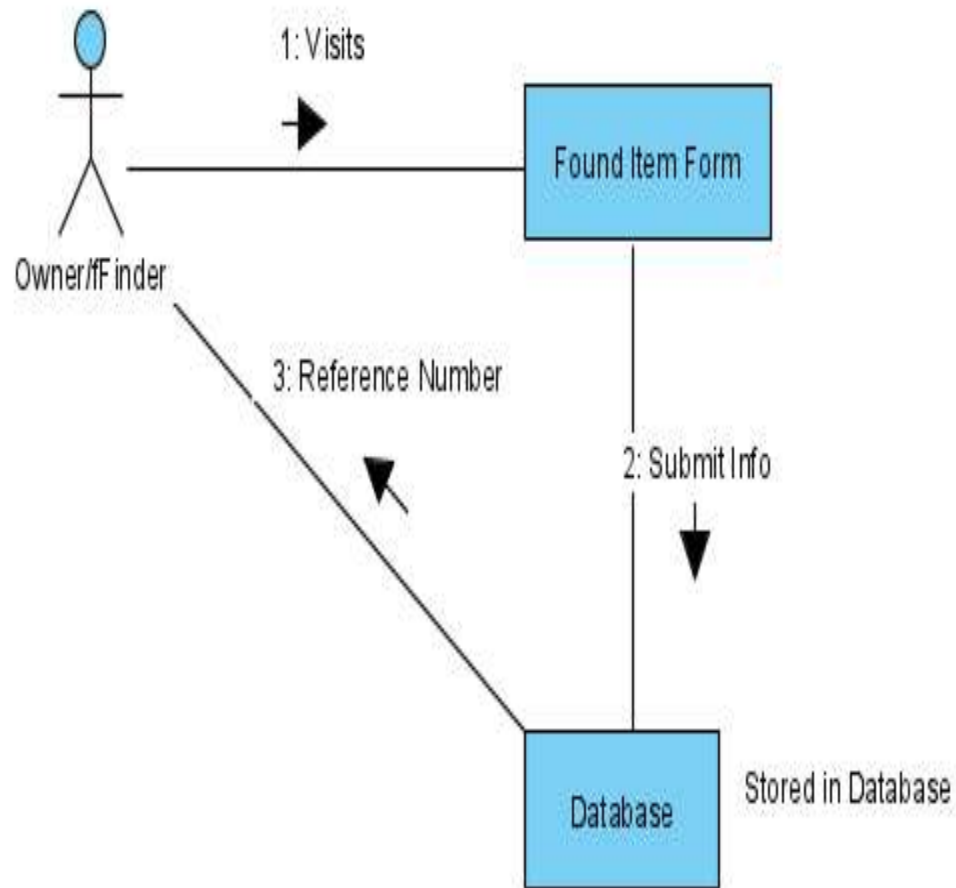


Figure 46 collaboration diagram for report found item

#### 4.7.4 REPORT LOST ITEM

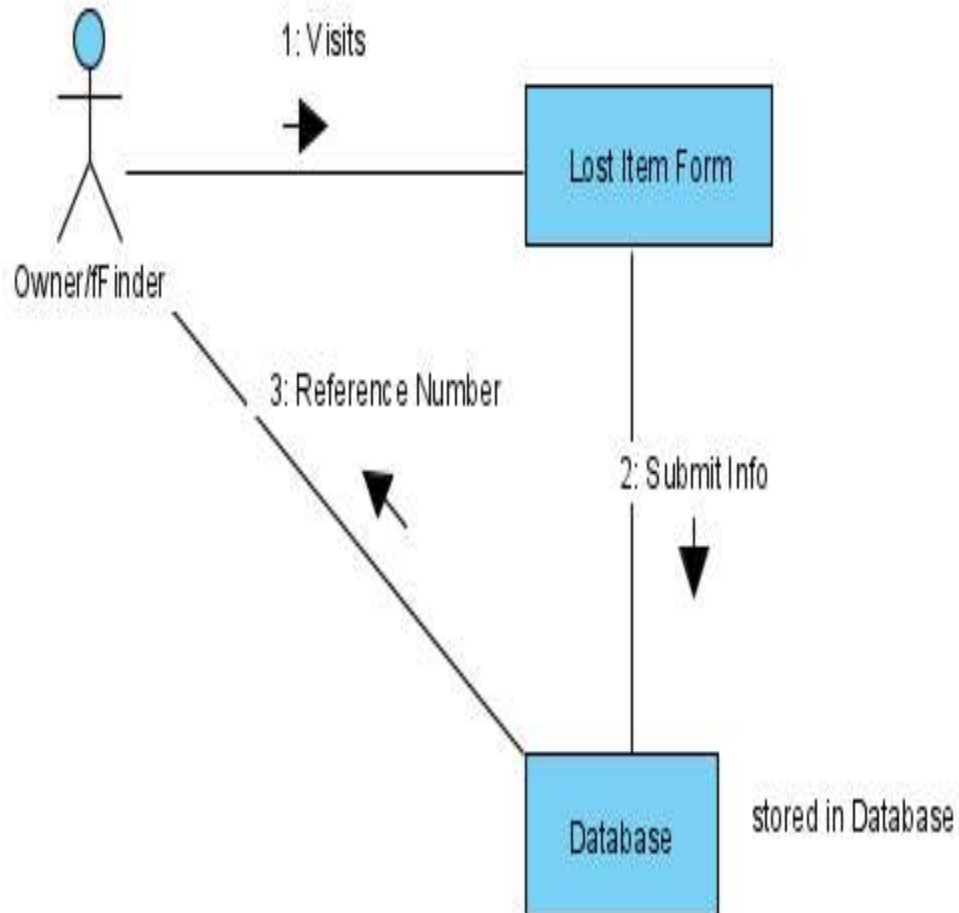


Figure 47 Collaboration diagram for report lost item

#### 4.7.5 VIEW FOUND ITEM

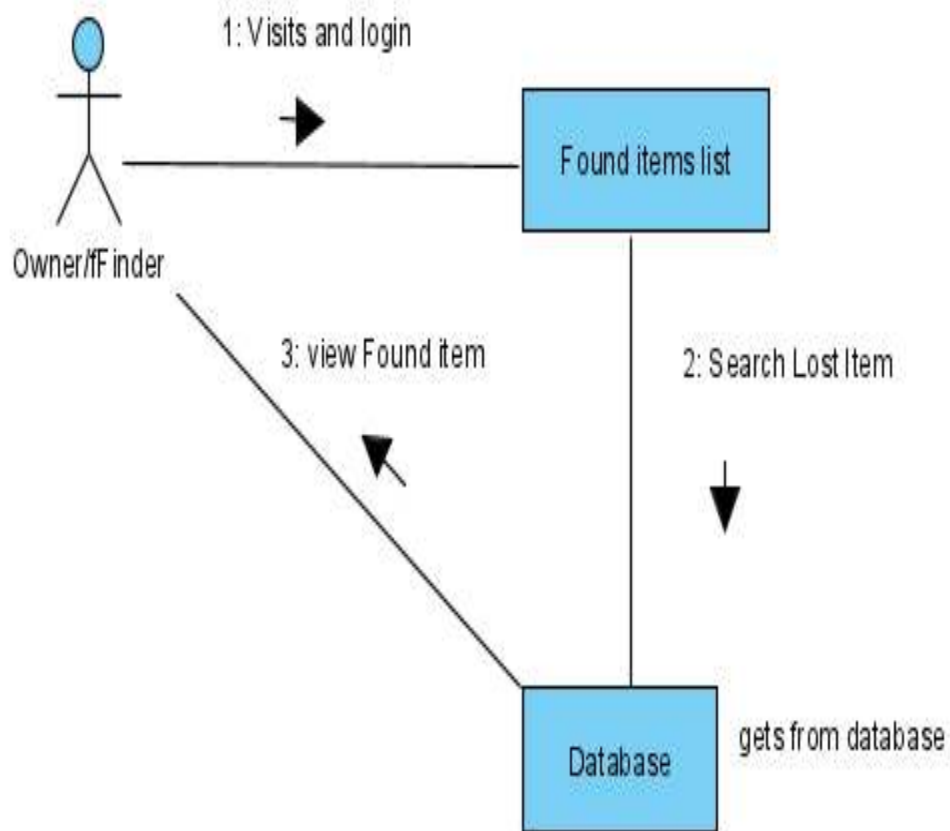


Figure 48 Collaboration diagram for View found items

#### 4.7.6 VIEW LOST ITEM

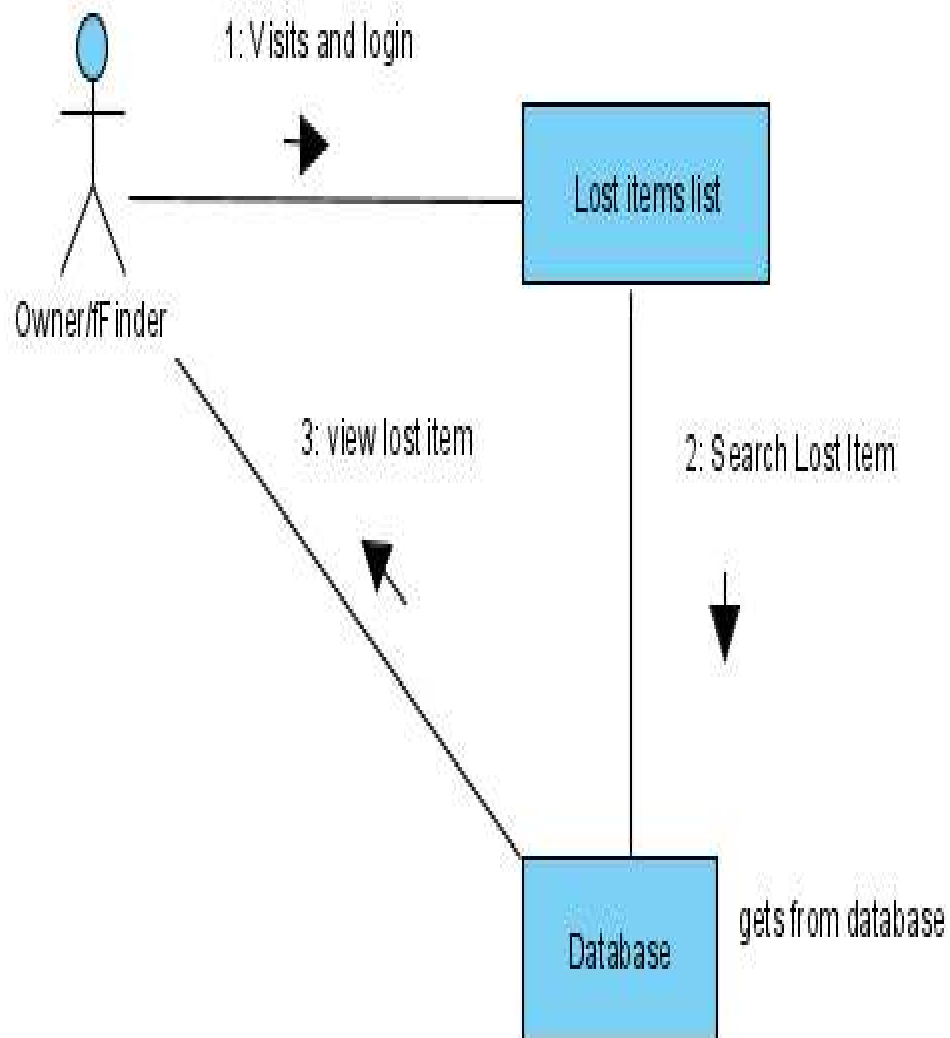


Figure 49 Collaboration diagram for View lost items

#### 4.7.7 VIEW ITEMS

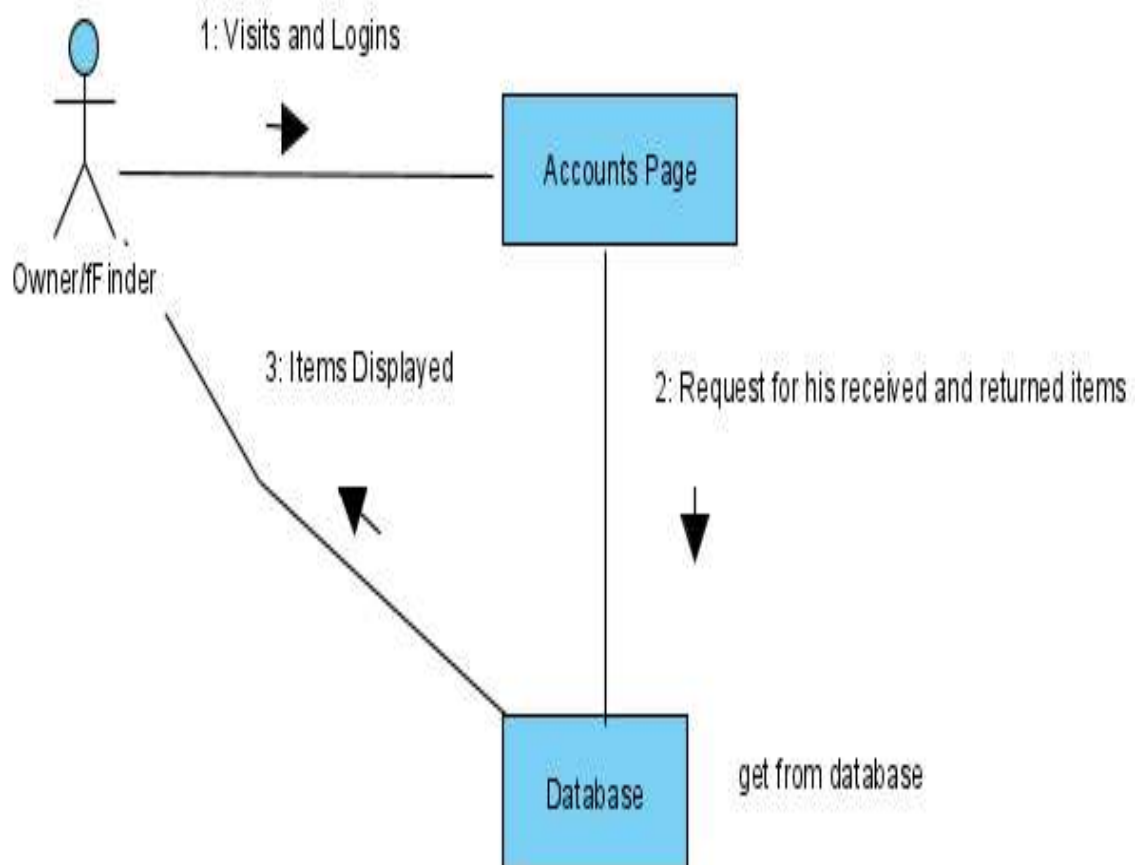


Figure 50 Collaboration diagram for View items

#### 4.7.8 CHAT

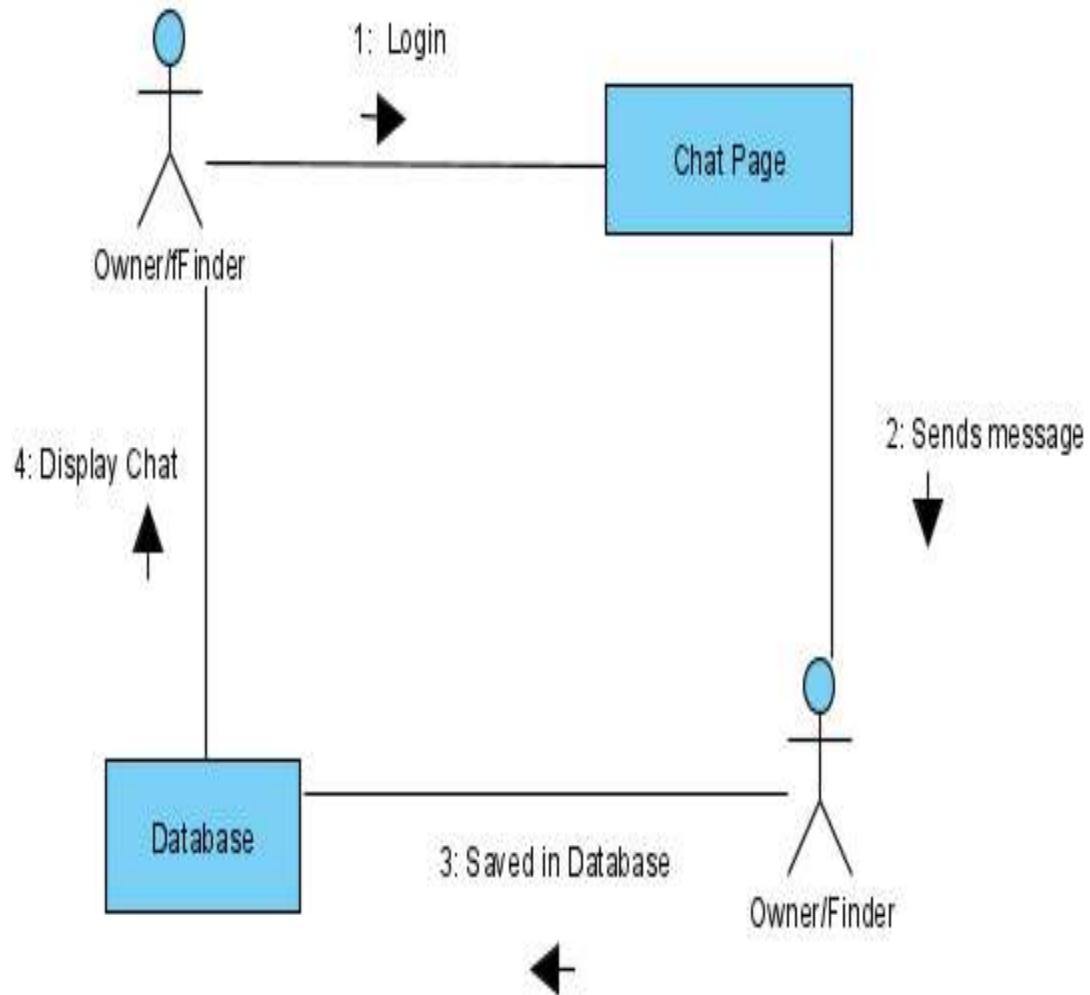


Figure 51 Collaboration diagram for Chat

#### 4.7.9 CONTACT DETAILS

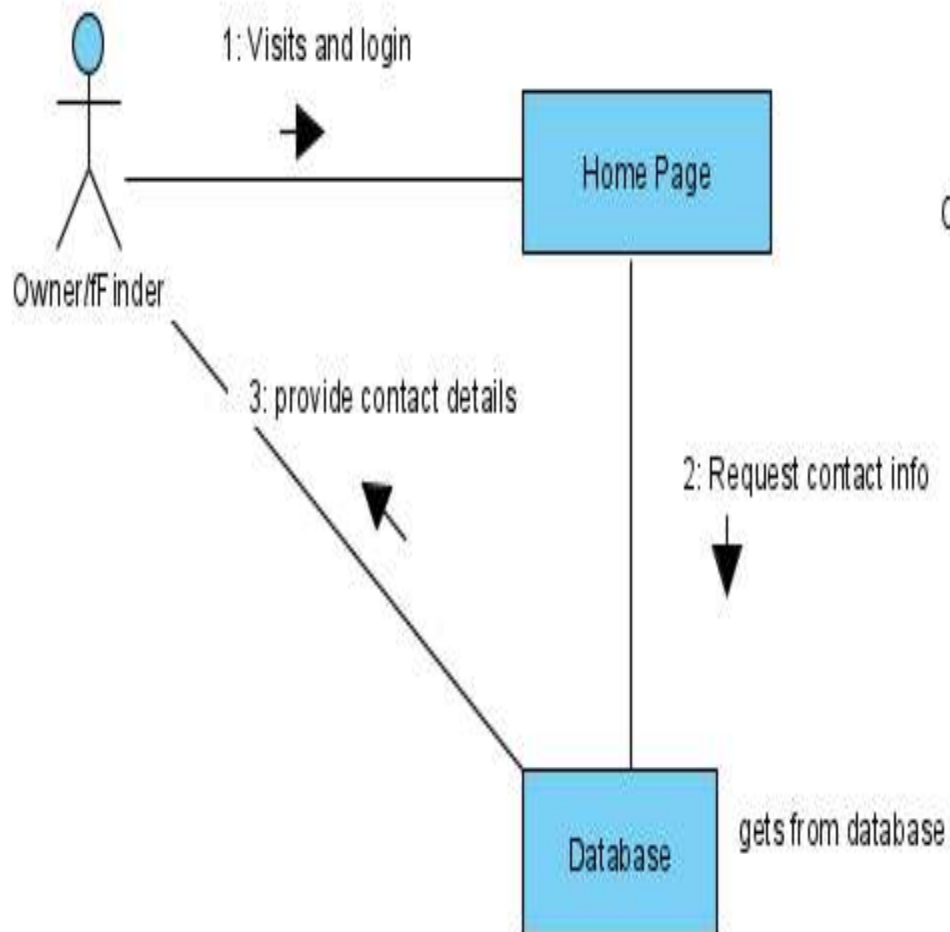


Figure 52 Collaboration diagram for Contact details

#### 4.7.10 VIEW STATUS

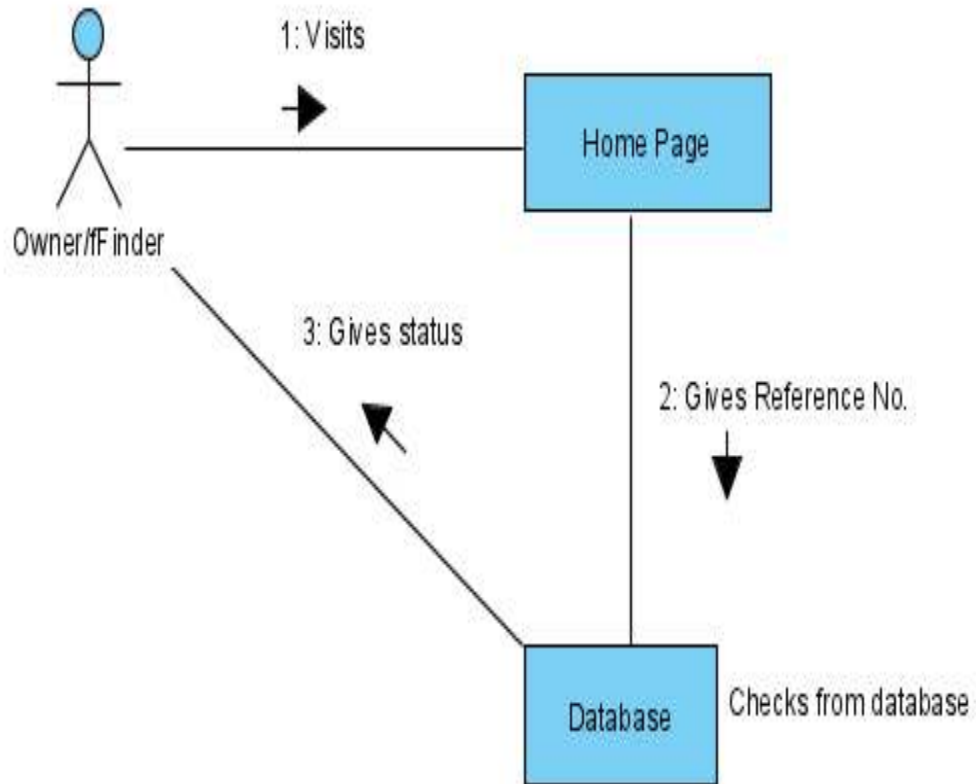


Figure 53 Collaboration diagram for View status

#### 4.7.11 DEACTIVATE ACCOUNT

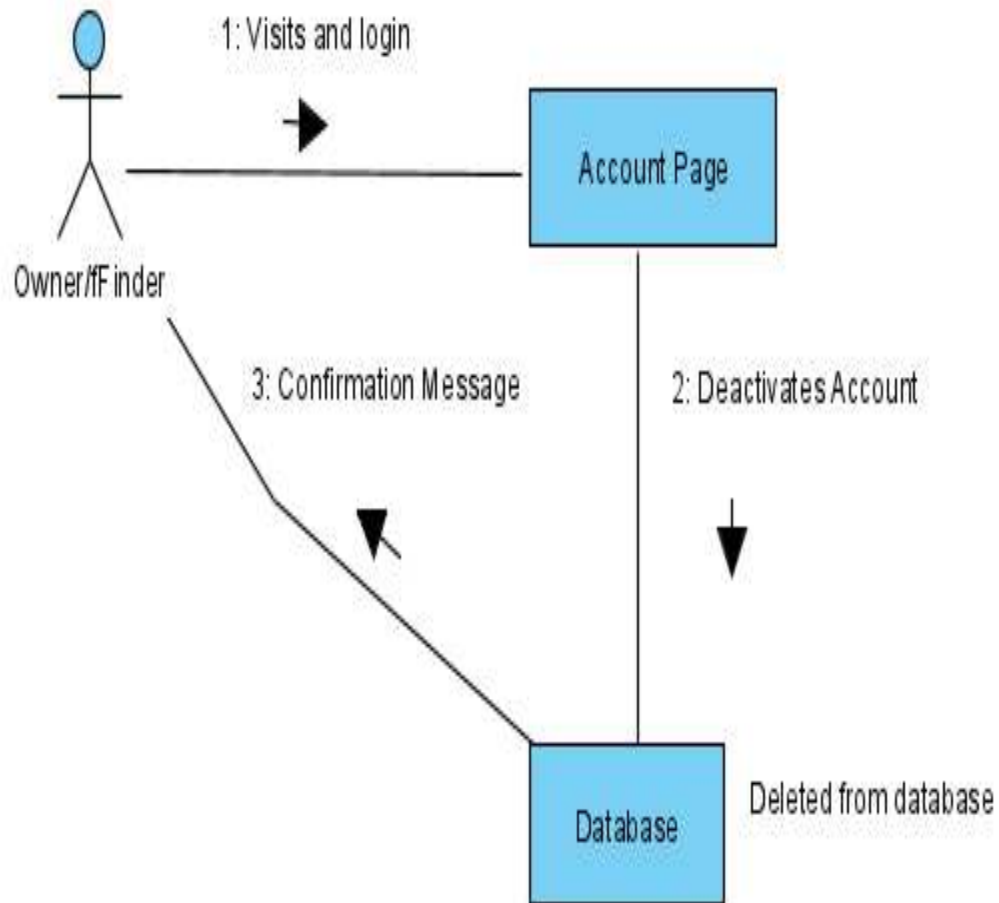


Figure 54 Collaboration diagram for deactivate account

#### 4.7.12 GUIDANCE

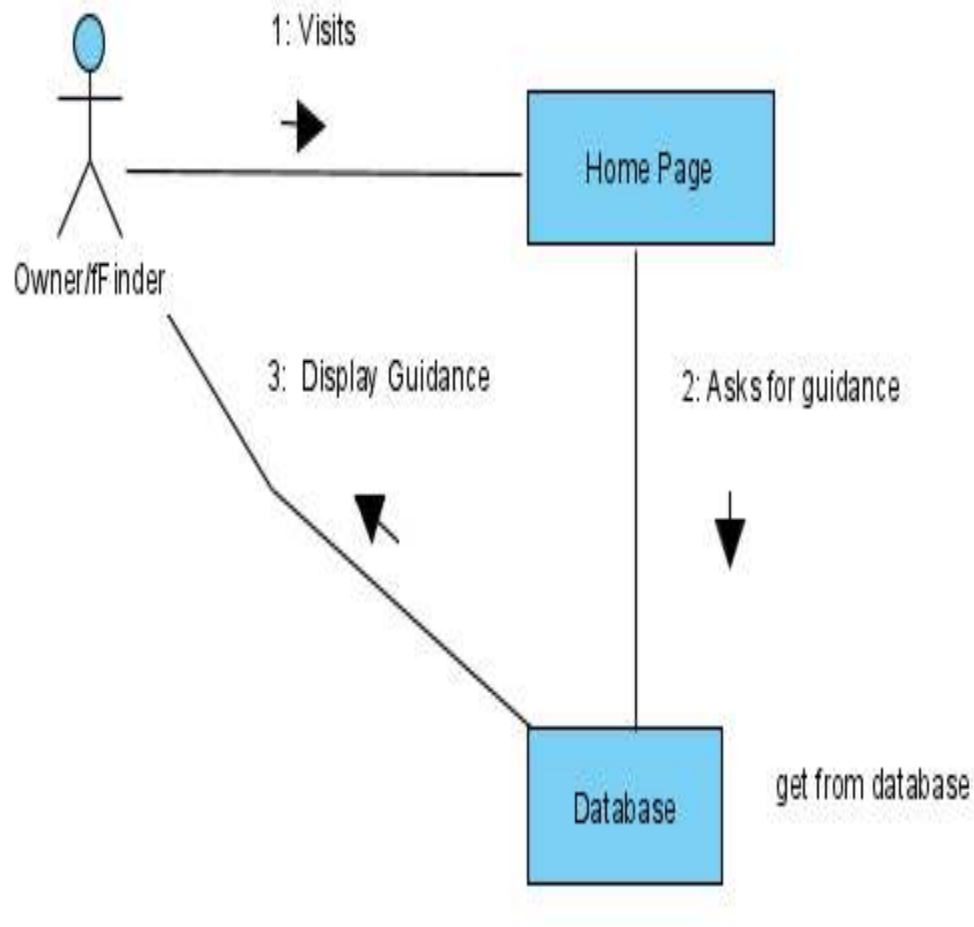


Figure 55 Collaboration diagram for Guidance

#### 4.7.13 HIGHLIGHTS

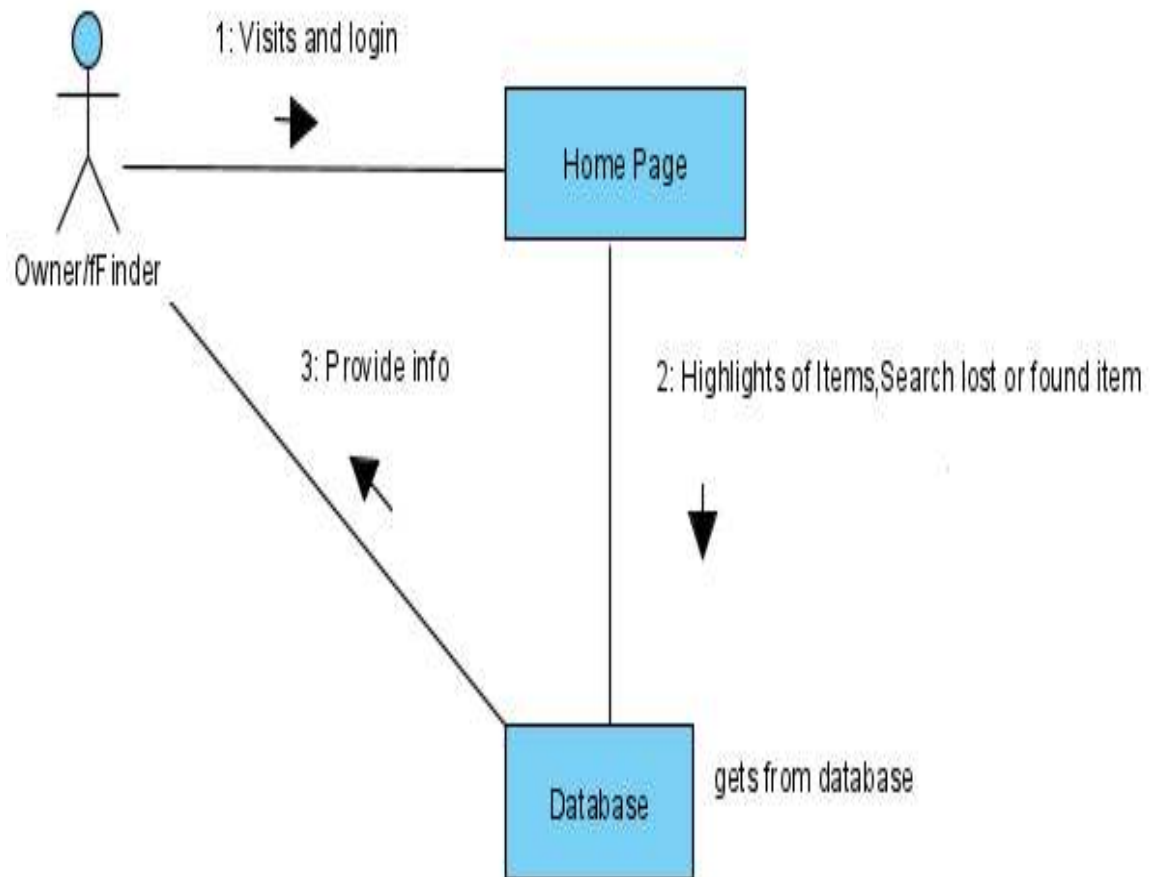


Figure 56 Collaboration diagrams for highlights

#### 4.7.14 UPDATE ACCOUNT

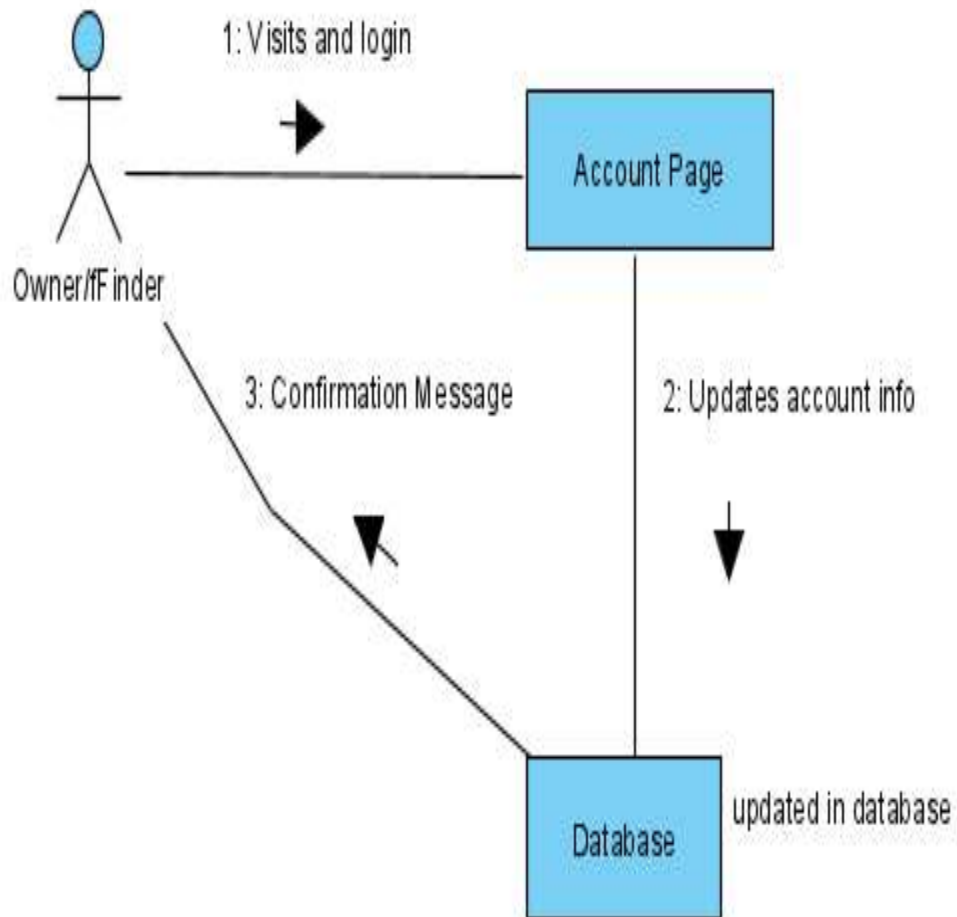


Figure 57 Collaboration diagram for update account info

#### 4.7.15 FEEDBACK

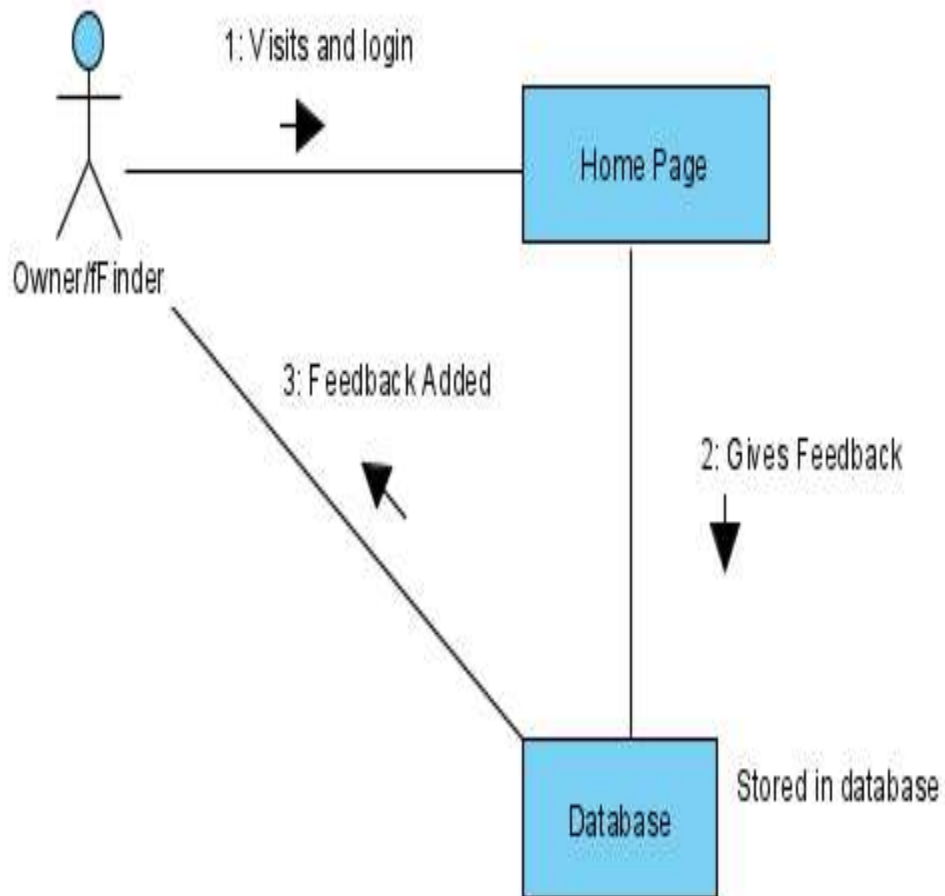


Figure 58 Collaboration diagram for Feedback

#### 4.7.16 View Statistics:

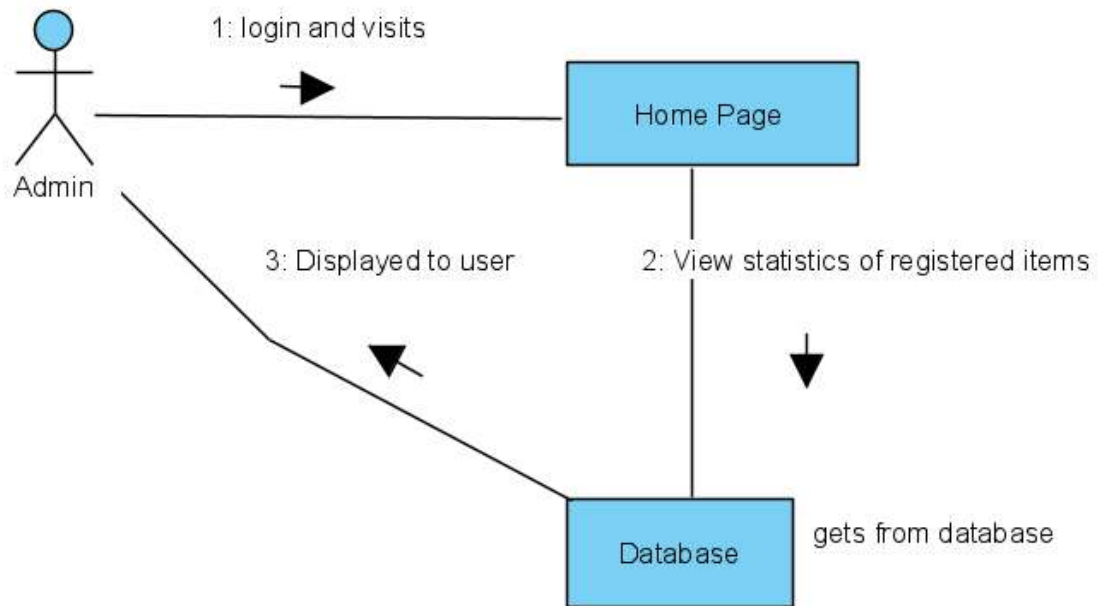


Figure 59 collaboration diagram for view statistics.

#### 4.7.17 View History:

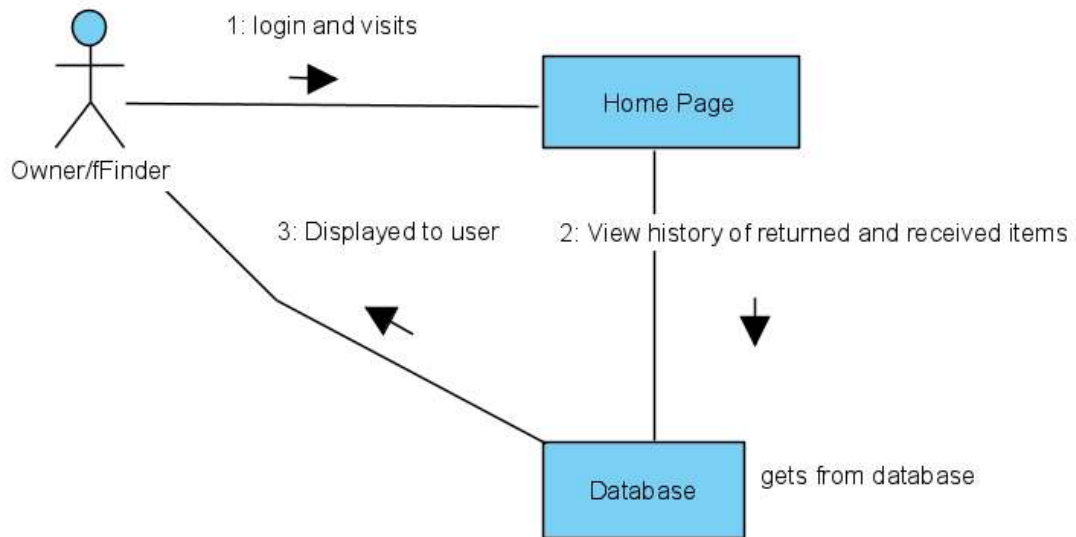


Figure 60 collaboration diagram for view history.

#### 4.7.18 Search with Category:

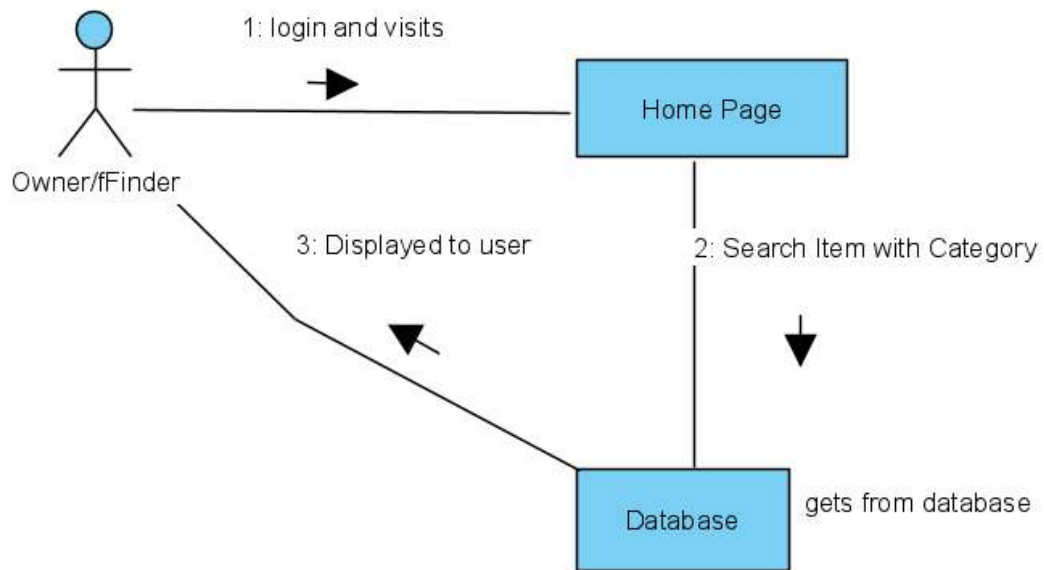


Figure 61 collaboration diagram for search by category.

## 4.8 NETWORK DIAGRAM

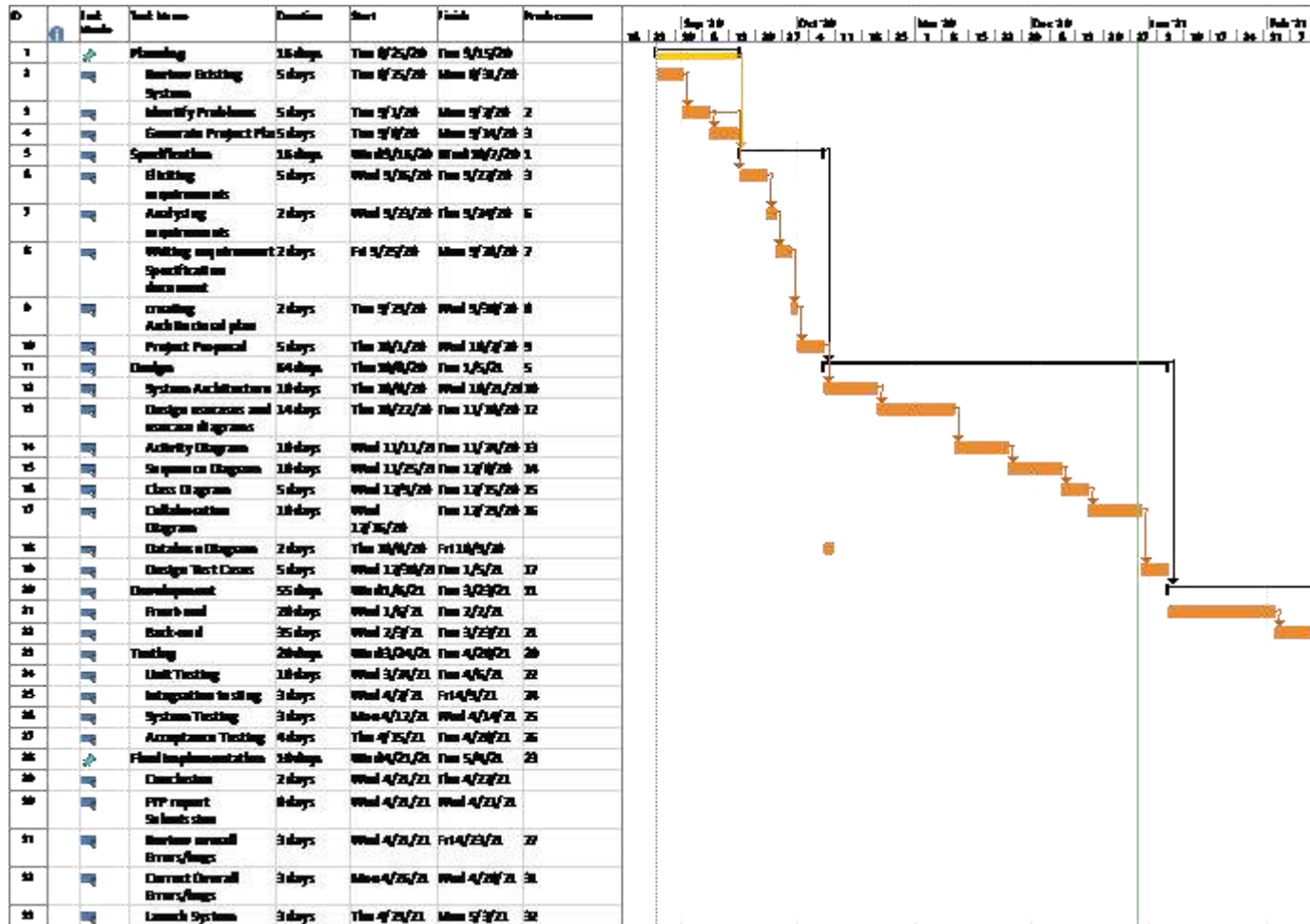


Figure 62 Network Diagram

## 5 CHAPTER 04 SYSTEM TESTING

### 5.1 TEST CASES

#### 5.1.1 TC01: REGISTER NEW USER

|                                                                                                                                                                                                        |                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case # 1</b><br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by:</b> MUHAMMAD FREED<br><b>Executed by :</b> MUHAMMAD FREED<br><b>Short description:</b> To register new user | <b>Test Case Name:</b> Register new user<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**  
Owner/Finder/Admin opens website or mobile application.

| Step | Action                                  | Expected Response | Pass/Fail | Comments |
|------|-----------------------------------------|-------------------|-----------|----------|
| 1.   | The user enter his/her name             | Ali               | Pass      |          |
| 2.   | The user enter his/her email            | Ali123@gmail.com  | Pass      |          |
| 3.   | The user enters his/her password.       | QWERTY            | Pass      |          |
| 4.   | The user enters his/her contact number. | 03226756899       | pass      |          |
| 5.   | The user enter his/her contact number   | 0322ygyuuj6       | Fail      |          |
| 6.   | The user enter his/her city name        | Lahore            | Pass      |          |
| 7.   | The user enter his/her email            | ali@@4526\$%      | Fail      |          |

**Post-Condition:**  
New user registered successfully.

*Table 19 Test case for register user*

### 5.1.2 TC02: LOGIN

|                                                                                                                                                                                                                           |                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case # 2</b><br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by :</b> MUHAMMAD FREED<br><b>Executed by :</b> MUHAMMAD FREED<br><b>Short description:</b> to allow the user to access the portal | <b>Test Case Name:</b> Login<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

Owner/Finder/Admin must be registered with the system.

| Step | Action                            | Expected Response           | Pass/Fail | Comments |
|------|-----------------------------------|-----------------------------|-----------|----------|
| 1.   | The user enter his/her email      | Freesalmans855.fs@gmail.com | Pass      |          |
| 2.   | The user enters his/her password. | QWERTY                      | Pass      |          |
| 3.   | The user enters his/her password. | 123                         | Fail      |          |
| 4.   | The user enter his/her email      | Freed21434tfgfyrygh.com     | Fail      |          |

**Post-Condition:**

The user successfully logged in.

*Table 20 Test case for login*

### 5.1.3 TC03: REPORT LOST ITEM

|                                                                                                                                                                                                                              |                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case # 3</b><br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by:</b> MUHAMMAD FREED<br><b>Executed by :</b> MUHAMMAD FREED<br><b>Short description</b> to allow the owner to report his lost item. | <b>Test Case Name</b> Report lost item<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

Owner must be logged in.

| Step | Action                                      | Expected Response                    | Pass/Fail | Comments |
|------|---------------------------------------------|--------------------------------------|-----------|----------|
| 1.   | Owner enters his/her item category.         | Watch                                | Pass      |          |
| 2.   | Owner enters his/her item colour.           | Silver                               | Pass      |          |
| 3.   | Owner enters his/her item brand.            | Rolex                                | Pass      |          |
| 4.   | Owner enters his/her item lost location.    | Canteen                              | Pass      |          |
| 5.   | Owner enters his/her lost item description. | My watch got misplaced near canteen. | Pass      |          |
| 6.   | Owner upload his/her lost item pictures     | image                                | Pass      |          |

**Post-Condition:**

Lost item registered successfully.

*Table 21 Test case for Report lost item*

#### 5.1.4 TC04: REPORT FOUND ITEM

|                                                                                                                                                                                                                                 |                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case # 4</b><br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by :</b> MUHAMMAD FREED<br><b>Executed by :</b> MUHAMMAD FREED<br><b>Short description</b> to allow the finder to report the found item. | <b>Test Case Name:</b> Report found item<br><b>Subsystem :</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

Finder must be logged in.

| Step | Action                                         | Expected Response                     | Pass/Fail | Comments |
|------|------------------------------------------------|---------------------------------------|-----------|----------|
| 1.   | Finder enters his/her item category.           | Wallet                                | Pass      |          |
| 2.   | Finder enters his/her item colour.             | Brown                                 | Pass      |          |
| 3.   | Finder enters his/her item brand.              | Gucci                                 | Pass      |          |
| 4.   | Finder enters his/her item lost location.      | D block comsats                       | Pass      |          |
| 5.   | Finder enter the description of the found item | I found a wallet with some documents. | Pass      |          |

**Post-Condition:**

Found item registered successfully.

*Table 22 Test case for Report found item*

### 5.1.5 TC05: VIEW STATUS

|                                                                                                                                                                                                                                                 |                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case # 5</b><br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by:</b> MUHAMMAD FREED<br><b>Executed by:</b> MUHAMMAD FREED<br><b>Short description</b> to allow the user to check the status of his/her reported item. | <b>Test Case Name:</b> View Status<br><b>Subsystem :</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

User must report any lost or found item.

| Step | Action                           | Expected Response | Pass/Fail | Comments |
|------|----------------------------------|-------------------|-----------|----------|
| 1.   | User enters the reference number | 134525            | Pass      |          |
| 2.   | User enters the reference number | Seqw312           | Fail      |          |

**Post-Condition:**

Status checked successfully.

*Table 23 Test case for View status*

### 5.1.6 TC06: VIEW CONTACT DETAILS

|                                                                                                                                                                                                                               |                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case # 6</b><br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by :</b> MUHAMMAD FREED<br><b>Executed by :</b> MUHAMMAD FREED<br><b>Short description:</b> to allow the user to view contact details. | <b>Test Case Name:</b> View Contact Details<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

User must log into the system.

| Step | Action                        | Expected Response      | Pass/Fail | Comments |
|------|-------------------------------|------------------------|-----------|----------|
| 1.   | User clicks on contact button | Contact info displayed | Pass      |          |

**Post-Condition:**

Contact details viewed successfully.

*Table 24 Test case for View contact details*

### 5.1.7 TC07: FEEDBACK

|                                                                                                                                                                                                                                          |                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case # 7</b><br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by:</b> MUHAMMAD FREED<br><b>Executed by :</b> MUHAMMAD FREED<br><b>Short description:</b> to allow the user to give feedback about the services. | <b>Test Case Name:</b> Feedback<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

User must be logged in to the system.

| Step | Action                                        | Expected Response | Pass/Fail | Comments |
|------|-----------------------------------------------|-------------------|-----------|----------|
| 1.   | User type the feedback in the feedback field. | Love this website | Pass      |          |

**Post-Condition:**

Feedback submitted successfully.

*Table 25 Test case for Feedback*

### 5.1.8 TC08: CHAT

|                                                                                                                                                                                                                                                                                                                 |                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case # 8</b><br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by:</b> MUHAMMAD FREED<br><b>Executed by :</b> MUHAMMAD FREED<br><b>Short description</b> to allow the owner to chat with the person who has found his lost item OR allow the finder to chat with the owner of the item. | <b>Test Case Name:</b> Chat<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

User must report any lost or found item.

| Step | Action                                               | Expected Response | Pass/Fail | Comments |
|------|------------------------------------------------------|-------------------|-----------|----------|
| 1.   | User types message and send it to the desired person | Hi                | Pass      |          |

**Post-Condition:**

Message sent successfully.

*Table 26 Test case for chat*

#### 5.1.9 TC09: VIEW HOME PAGE

|                                                                                                                                                                                                                          |                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case # 5</b><br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by:</b> MUHAMMAD FREED<br><b>Executed by:</b> MUHAMMAD FREED<br><b>Short description:</b> to allow the user to view the home page | <b>Test Case Name:</b> View Home Page<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

User must be logged in to the system.

| Step | Action                     | Expected Response   | Pass/Fail | Comments |
|------|----------------------------|---------------------|-----------|----------|
| 1.   | User clicks on home button | Home page displayed | Pass      |          |

**Post-Condition:**

Home page viewed successfully.

*Table 27 Test case for View home page*

#### 5.1.10 TC10: VIEW LOST ITEM

|                                                                                                                                                                                                                         |                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case # 5</b><br><b>System</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by</b> MUHAMMAD FREED<br><b>Executed by</b> MUHAMMAD FREED<br><b>Short description:</b> to allow the user to view the lost items. | <b>Test Case Name:</b> View Lost Items<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

User must logged in to the system.

| Step | Action                               | Expected Response    | Pass/Fail | Comments |
|------|--------------------------------------|----------------------|-----------|----------|
| 1.   | User clicks on view lost item button | Lost items displayed | Pass      |          |

**Post-Condition:**

Lost item viewed successfully.

*Table 28 Test case for View lost items*

#### 5.1.11 TC11: VIEW FOUND ITEMS

|                                                                                                                                                                                                                                                                   |                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case #:</b> 11<br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by:</b> ABDULLAH AHMAD<br><b>Executed by:</b> ABDULLAH AHMAD<br><b>Short description:</b> It allows the user to view all the view all the found items and their details. | <b>Test Case Name:</b> View Found Items<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

User must be login into the system.

| Step | Action                                | Expected Response         | Pass/Fail | Comments |
|------|---------------------------------------|---------------------------|-----------|----------|
| 1.   | User clicks on view found item button | Found item list displayed | Pass      |          |

**Post-Condition:**

The user will see all the found item.

*Table 29 Test case for View found items*

### 5.1.12 TC12: UPDATE ACCOUNT INFO

|                                                                                                                                                                                                                                   |                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case #:</b> 12<br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by:</b> ABDULLAH AHMAD<br><b>Executed by:</b> ABDULLAH AHMAD<br><b>Short description:</b> It allows the user to update his account info. | <b>Test Case Name:</b> Update Account Info<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

User must be login into the system.

| Step | Action                           | Expected Response                                          | Pass/Fail | Comments |
|------|----------------------------------|------------------------------------------------------------|-----------|----------|
| 1.   | User updates his name.           | Freed                                                      | Pass      |          |
| 2.   | The user updates his email.      | <a href="mailto:Freed123@gmail.com">Freed123@gmail.com</a> | Pass      |          |
| 3.   | The user update his phone number | 0322556825                                                 | Pass      |          |
| 4.   | User updates his email.          | Frewerwdsx134fhjghjgy.com                                  | Fail      |          |

**Post-Condition:**

The will successfully change his account info.

*Table 30 Test case for Update Account info*

### 5.1.13 TC13: DEACTIVATE ACCOUNT

|                                                                                                                                                                                                                              |                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case #:</b> 13<br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by:</b> ABDULLAH AHMAD<br><b>Executed by:</b> ABDULLAH AHMAD<br><b>Short description:</b> It allows the user to delete his account. | <b>Test Case Name:</b> Deactivate Account.<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

User must be login into the system.

| Step | Action                                        | Expected Response   | Pass/Fail | Comments |
|------|-----------------------------------------------|---------------------|-----------|----------|
| 1.   | The user clicks on deactivate account button. | Account deactivated | Pass      |          |

**Post-Condition:**

Account and items will be deleted.

*Table 31 Test case for Deactivate account*

#### 5.1.14 TC14: USER GUIDANCE

|                                                                                                                                                                                                                                      |                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case #:</b> 14<br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by:</b> ABDULLAH AHMAD<br><b>Executed by:</b> ABDULLAH AHMAD<br><b>Short description:</b> It allows the user to know how to use the system. | <b>Test Case Name:</b> User Guidance<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

The website must be opened.

| Step | Action                                | Expected Response | Pass/Fail | Comments |
|------|---------------------------------------|-------------------|-----------|----------|
| 1.   | The user clicks on how to use button. | Video is viewed   | Pass      |          |

**Post-Condition:**

The user will be able to know how to use the system.

*Table 32 Test case for User guidance*

#### 5.1.15 TC15: VIEW HISTORY

|                                                                                                                                                                                                                                           |                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case #:</b> 15<br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by:</b> ABDULLAH AHMAD<br><b>Executed by:</b> ABDULLAH AHMAD<br><b>Short description:</b> It allows the user to see returned and received items. | <b>Test Case Name:</b> View Items<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

User must be login into the system.

| Step | Action                                  | Expected Response             | Pass/Fail | Comments |
|------|-----------------------------------------|-------------------------------|-----------|----------|
| 1.   | The user clicks on view history button. | History of the user displayed | Pass      |          |

**Post-Condition:**

The user will see all the returned and received items.

*Table 33 Test case for View history*

#### 5.1.16 TC16: SEARCH BY CATEGORY

|                                                                                                                                                                                                                                             |                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case #:</b> 16<br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by:</b> ABDULLAH AHMAD<br><b>Executed by:</b> ABDULLAH AHMAD<br><b>Short description:</b> It allows the user to search item by typing the category | <b>Test Case Name:</b> search by category<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

User must be login into the system.

| Step | Action                                             | Expected Response | Pass/Fail | Comments |
|------|----------------------------------------------------|-------------------|-----------|----------|
| 1.   | The user search category by clicking on search bar | Watch             | Pass      |          |
| 2.   | The user search category by clicking on search bar | 1234              | Fail      |          |

**Post-Condition:**

The user will see all the results according to its searched category.

*Table 34 test case for search by category*

#### 5.1.17 TC17: VIEW STATISTICS

|                                                                                                                                                                                                                                                          |                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case #:</b> 17<br><b>System:</b> LOST AND FOUND MANAGEMENT SYSTEM<br><b>Designed by:</b> ABDULLAH AHMAD<br><b>Executed by:</b> ABDULLAH AHMAD<br><b>Short description:</b> It allows the admin to check the stats of the android app or website. | <b>Test Case Name:</b> View statistics<br><b>Subsystem:</b> Web and android development<br><b>Date:</b> 22/12/20<br><b>Execution date:</b> 22/12/20 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|

**Pre-Condition:**

User must be login into the system.

| Step | Action                       | Expected Response | Pass/Fail | Comments |
|------|------------------------------|-------------------|-----------|----------|
| 1.   | Admin clicks on users button | statistics        | Pass      |          |

**Post-Condition:**

The admin will see the statistics of the website.

*Table 35 testcase for view statistics*

## 6 CHAPTER CONCLUSION

### 6.1 Problems Faced and Lessons Learned:

Initially during documentation, we were confused for choosing correct language for the development of this application, but our supervisor helped us for choosing the appropriate one. So we opt for choosing angular for front-end out our web application and we have made our android application using flutter for the backend of both (web and android) we have choose node.js. Our website has two step verification for better security. We also faced connecting mongoose with node.js but we did that after doing RND. We did our all work in team; we divided our tasks and then it was very easy for us to accomplish any type of development. From this we learned that teamwork is necessary in every type of work which we are doing.

### 6.2 Project Summary:

Main purpose of Lost and Found System is tracking all the lost and found items. This application is designed for anyone who has lost his item anywhere or found someone else item. But this app is especially designed for the students and faculty members. It can be used anywhere for example in school, universities. In universities lost and found department is working manually because of that they are facing many problems, so we want to automate their system through this application.

Our application/web is user friendly so that it can be used by anyone. It will have GUI (Graphical User Interface). We have also provided guidance for how to use the site. In the end we have tested and validated our website for best results.

### 6.3 Future Work:

We will develop IOS app in the future so that iPhone users can easily access all the features of the application.

## References

[1] METHOD AND SYSTEM FOR RETRIEVAL OF LOST GOODS

<https://patentimages.storage.googleapis.com/41/16/cf/90c15ba9a32ccb/US6546088.pdf>

[2] LOST ITEM NOTIFICATION AND RECOVERY SYSTEM

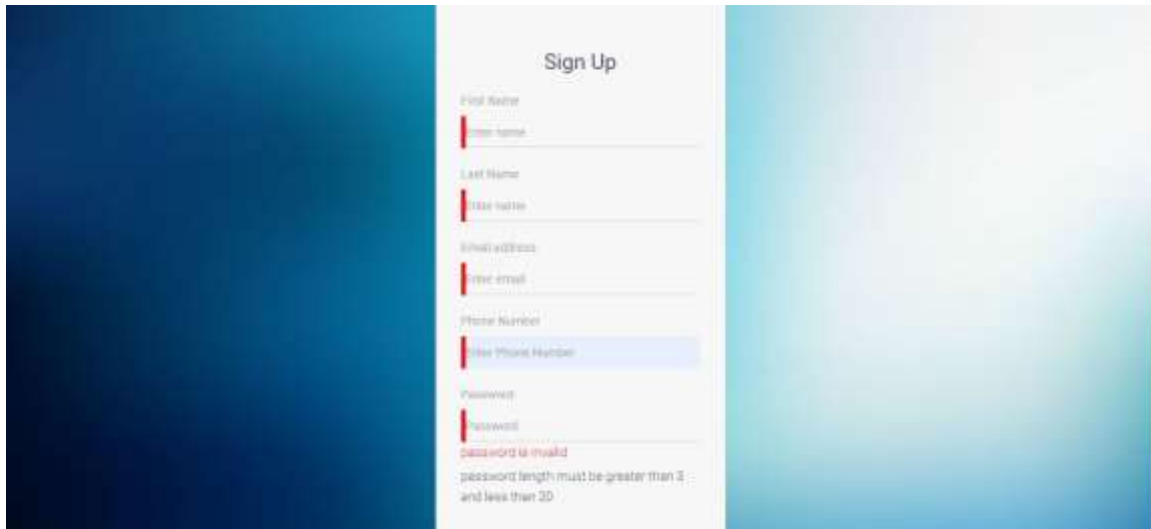
<https://patentimages.storage.googleapis.com/7a/d4/d3/8c606eeb77f89a/US20080079581A1.pdf>

[3] LOST AND FOUND WEB APPLICATION FOR CAL POLY POMONA STUDENTS

[http://dspace.calstate.edu/bitstream/handle/10211.3/194123/KaurHarkamal\\_Project2017.pdf?sequence=3](http://dspace.calstate.edu/bitstream/handle/10211.3/194123/KaurHarkamal_Project2017.pdf?sequence=3)

## 6.4 Web Prototypes:

### 6.4.1 Sign up screen



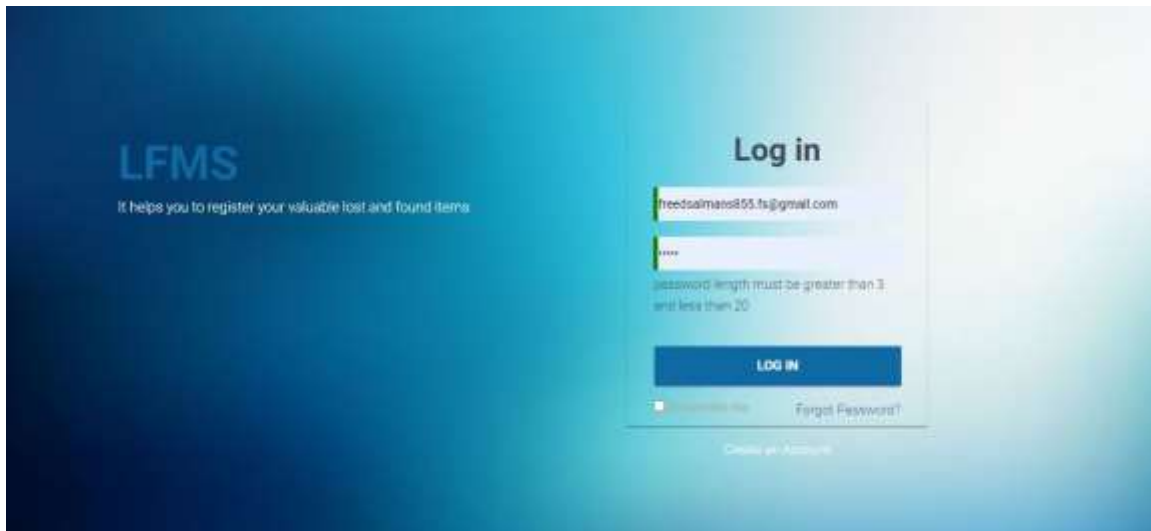
A web prototype of a sign-up screen. The background is a gradient of blue and green. On the left, there is a dark blue vertical bar. The sign-up form is centered and has a white background. It contains the following fields and labels:

- First Name**: Input field with a red vertical bar on the left.
- Last Name**: Input field with a red vertical bar on the left.
- Email address**: Input field with a red vertical bar on the left.
- Phone Number**: Input field with a red vertical bar on the left.
- Password**: Input field with a red vertical bar on the left.

Below the password field, there is a red error message: "password is invalid". Below that, there is a red error message: "password length must be greater than 3 and less than 20". At the bottom of the form, there is a "LOG IN" button.

Figure 63 sign up prototype

### 6.4.2 Login screen:



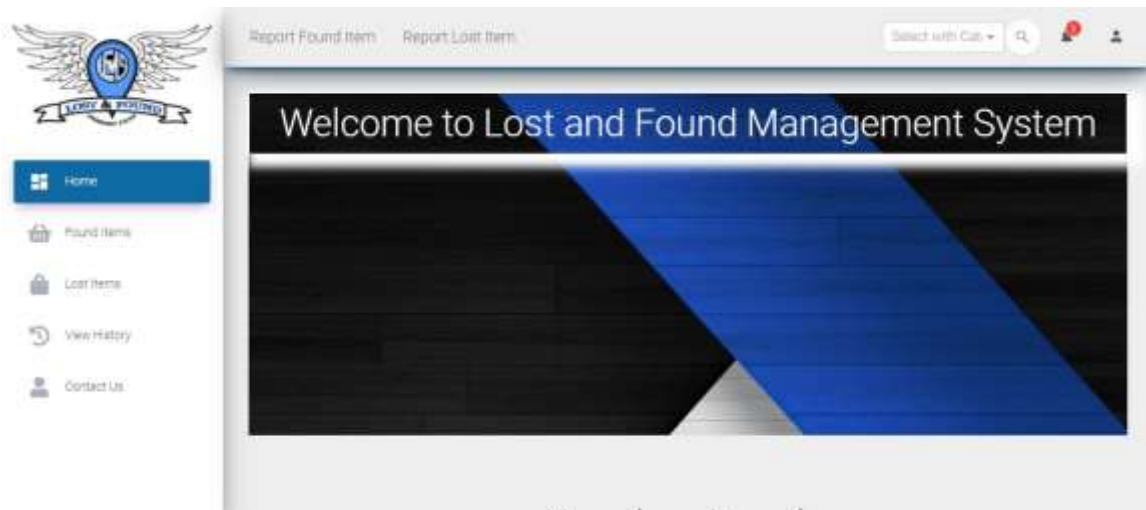
A web prototype of a login screen. The background is a gradient of blue and green. On the left, there is a dark blue vertical bar. The login form is centered and has a white background. It contains the following fields and labels:

- Log in**: Title of the form.
- Email address**: Input field with a red vertical bar on the left.
- Password**: Input field with a red vertical bar on the left.

Below the password field, there is a red error message: "password length must be greater than 3 and less than 20". Below that, there is a "LOG IN" button. At the bottom of the form, there is a "Forgot Password?" link and a "Create an Account" link.

Figure 64 login prototype

### 6.4.3 Home screen



*Figure 65 home prototype*

#### 6.4.4 Report lost item screen

The screenshot displays a web application interface for reporting a lost item. On the left, a sidebar contains a logo and navigation links: Home, Found Items, Lost Items, View History, and Contact Us. The main content area is titled 'Add Lost Item' and 'Item details'. It features several input fields: 'Item Name', 'Brand', 'Description', 'Phone Number', 'Category' (a dropdown menu), 'Color', 'Item Type' (a dropdown menu), and a 'Lost' checkbox. Below these is an 'Add Date' field with a date picker. A map section includes 'Enter Location' and 'Select Origin' buttons. At the bottom, there is a file upload section with the text 'Upload your image', a 'Choose File' button, a 'NO FILE CHOSEN' status, and a 'CLEAR FILE' button.

Figure 66 Report lost item prototype

#### 6.4.5 Report found item screen:

**Add Found Item**  
Item details

Item Name:

Brand:

Description:

Item Number:

Category:

Item Type:

Found:

Upload your image:  NO FILE CHOSEN

Enter Location:

**Home**

Found Items

Lost Items

View History

Contact Us

Figure 67 Report found item prototype

#### 6.4.6 Lost item screen:

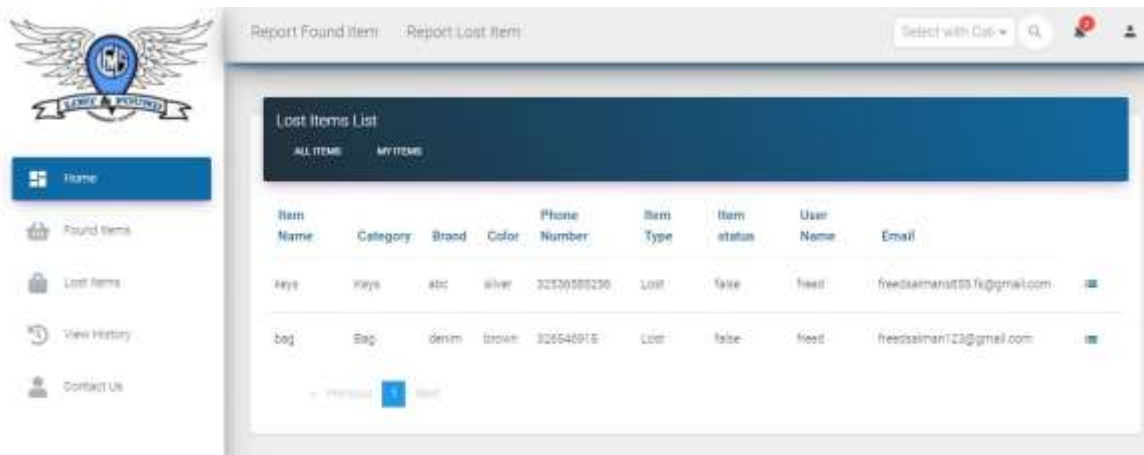


Figure 68 Lost item prototype

#### 6.4.7 Found item screen:

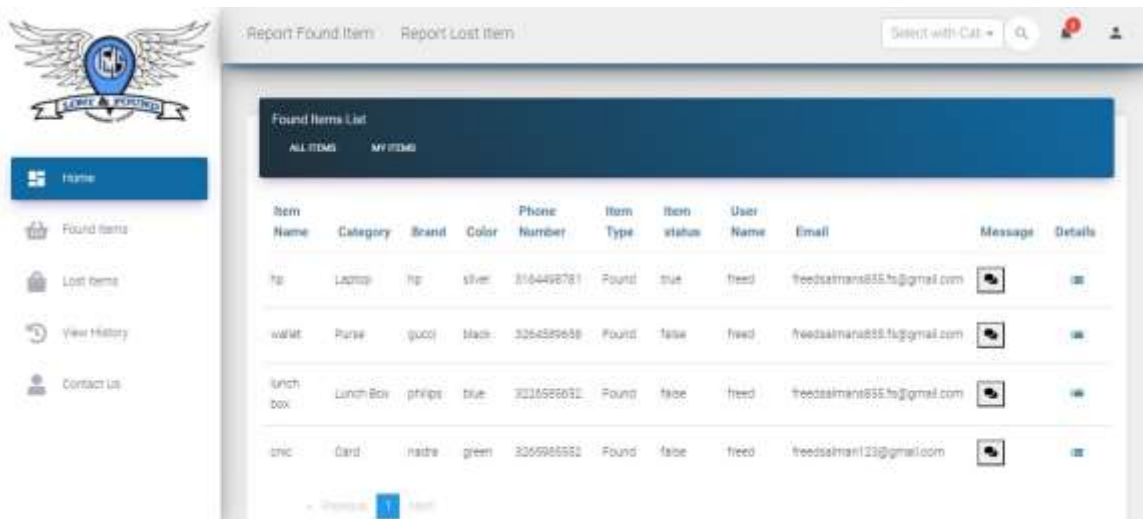


Figure 69 Found item prototype

#### 6.4.8 View status screen:

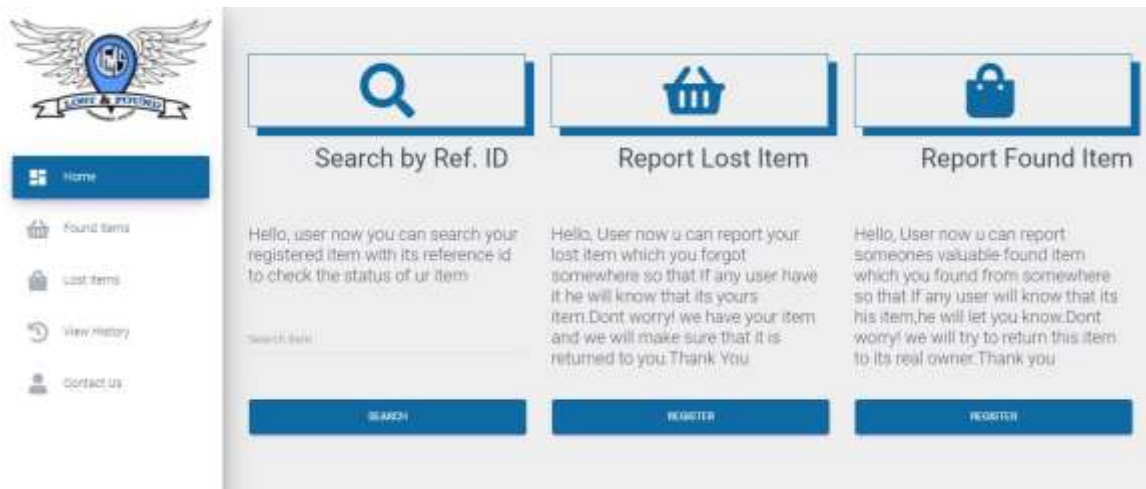
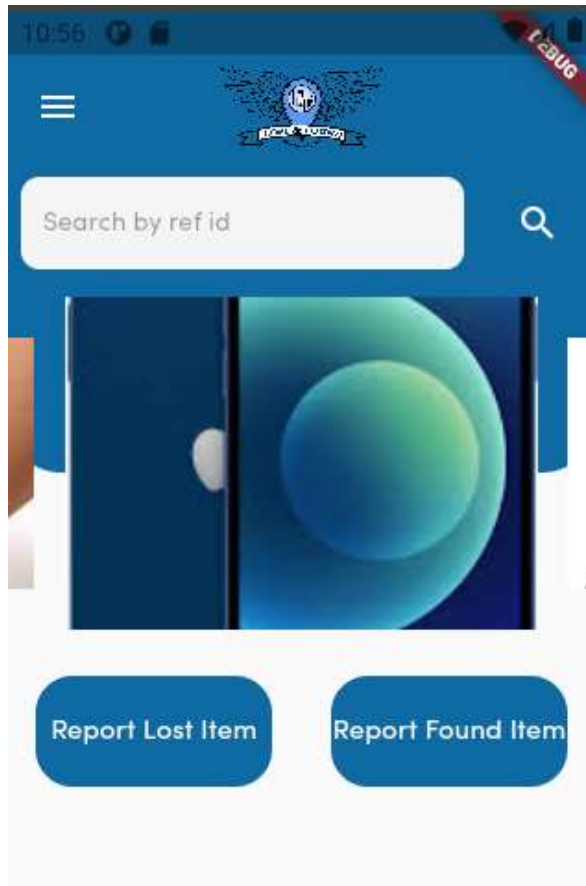


Figure 70 View status prototype

## 5.5 Mobile Prototypes

### 5.5.1 View Home Page



*Figure 72 Home page mobile prototype*

### 5.5.2 Menu Screen

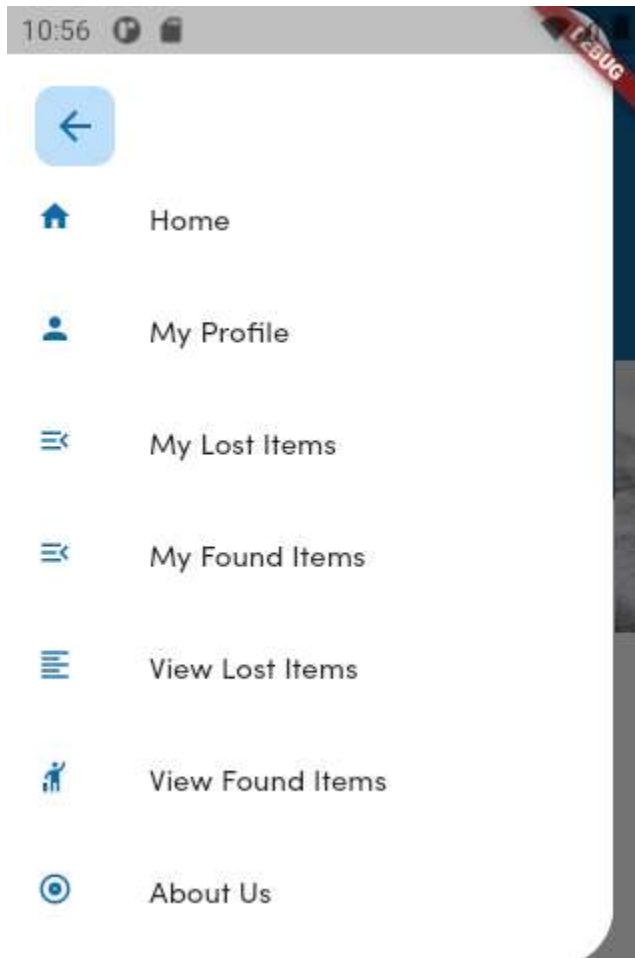
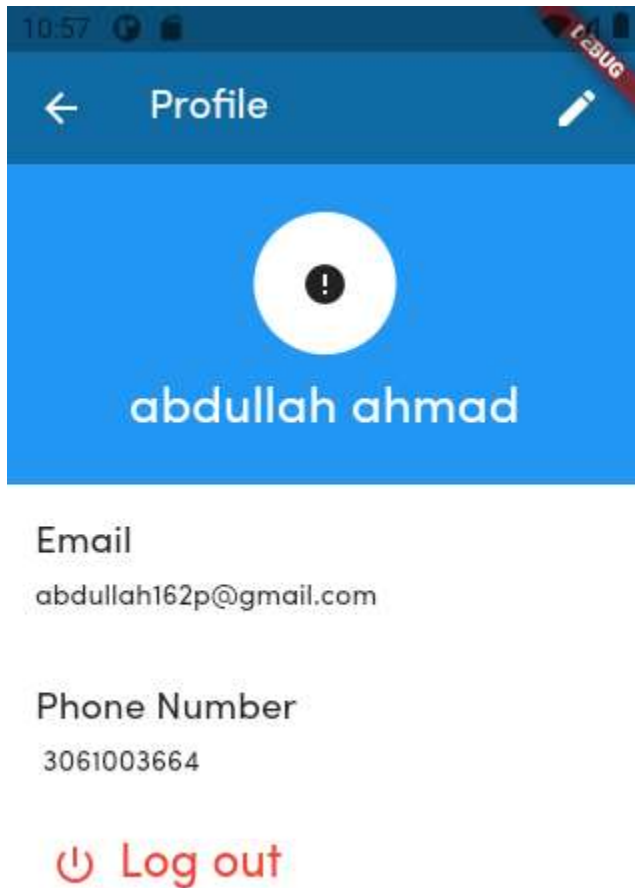


Figure 71 Menu screen mobile prototype

### 5.5.3 User Details Screen



*Figure 72 User Details Screen Mobile prototype*

#### 5.5.4 Item details screen



Figure 73 Item details screen Mobile prototype

### 5.5.5 My Lost items Screen

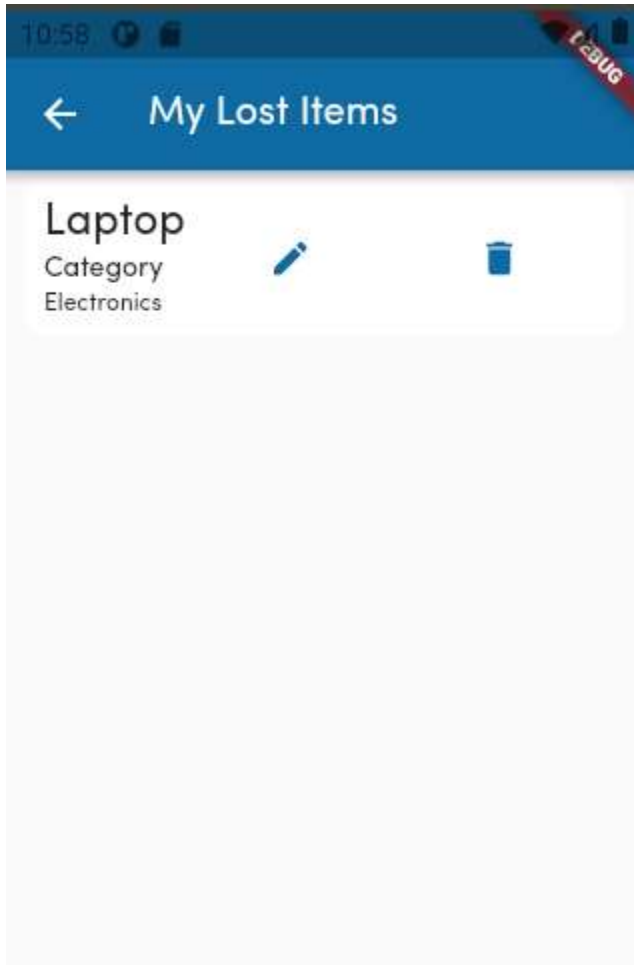


Figure 74 My Lost items Screen mobile prototype

### 5.5.6 My found items Screen

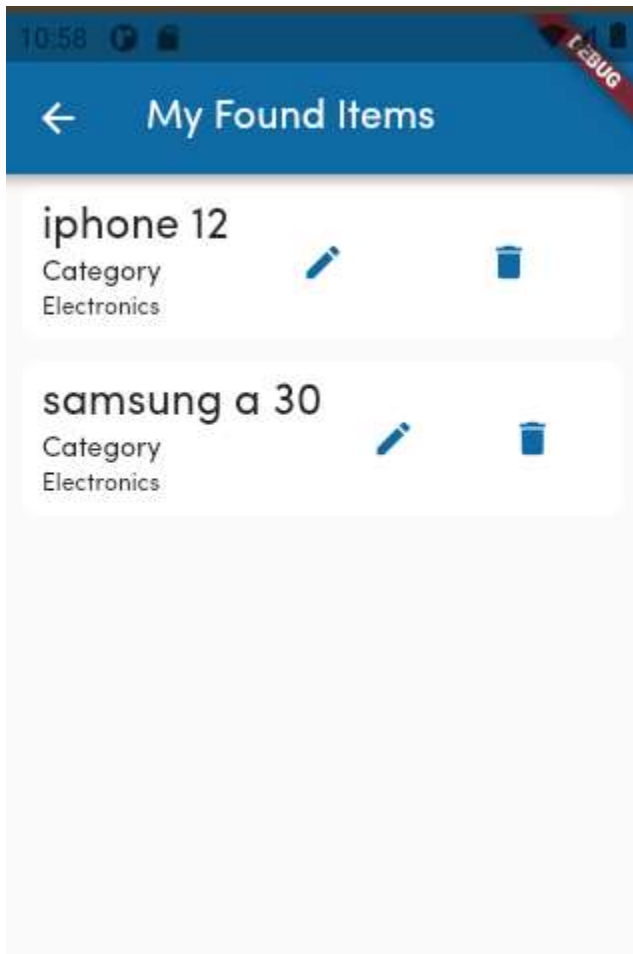


Figure 75 my found items Screen mobile prototype

### 5.5.7 All Lost items screen

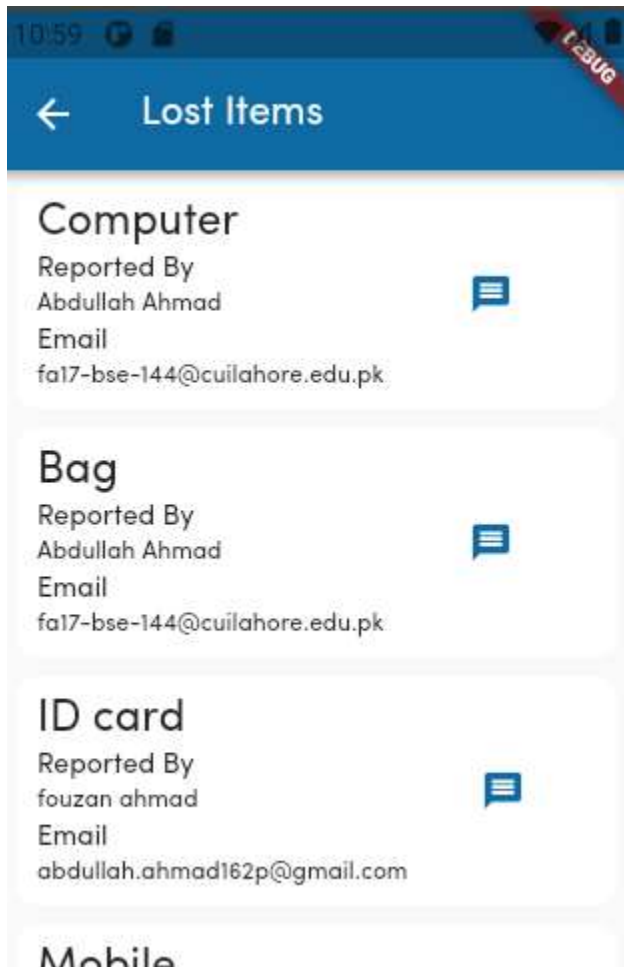


Figure 76 All Lost items screen mobile prototype

### 5.5.8 All found items screen

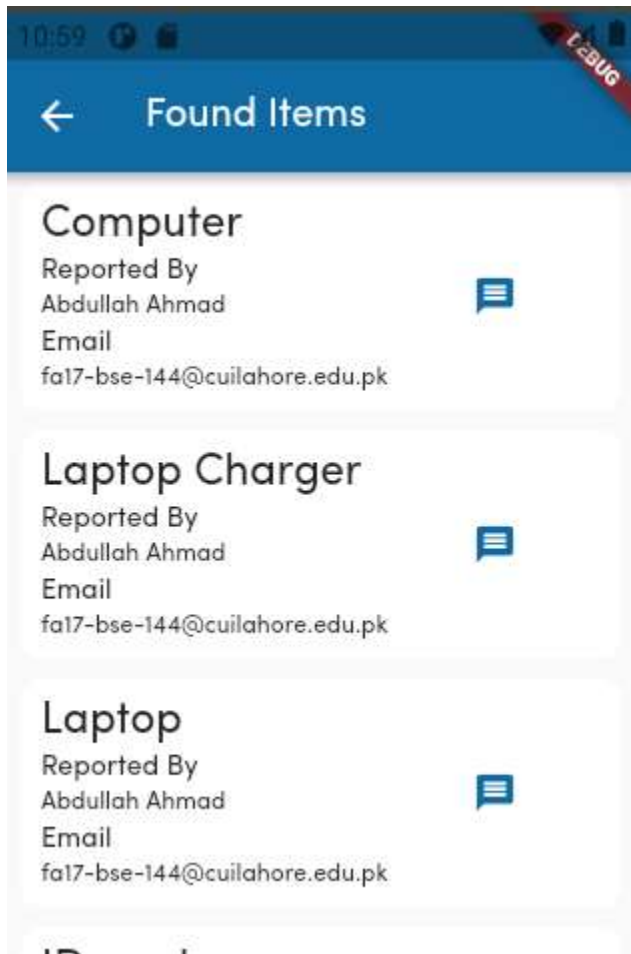


Figure 77 all found items screen mobile prototype

### 5.5.9 About us screen



Figure 78 about us screen mobile prototype

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